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These notes are based on the course taught by Prof. Victor Filipe Martins-Da-Rocha at FGV-EESP (Brazil). All remaining errors are ours.

Class 1

Logic

Statements

Definition 1. A claim, statement, or proposition is a sentence that is true (T) or false (F).

Negation

Definition 2. A claim, statement or proposition which is modified by the word “not” is called the negation of the original statement.

P	not(P)
T	F
F	T

Conjunction

The word “and” is used to join two statements which is called the conjunction of the two statements.

P	Q	P and Q
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

A statement formed by connecting two statements with the word “or” is called the disjunction of the two statements. “or” is used in the inclusive sense.

A	B	A or B
T	T	T
T	F	T
F	T	T
F	F	F

Conditional

From two statements P and Q, we may construct one of the form “If P, then Q”; this is called a conditional statement and denoted by “ $P \Rightarrow Q$ ”.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

We also write this as “P implies Q”, “P only if Q”, “P is a sufficient condition for Q”, and “Q is a necessary condition for P”.

Bi-conditional

From two statements P and Q, we may construct the following biconditional statement “ $P \iff Q$ ” defined as follows:

P	Q	$P \iff Q$
T	T	T
T	F	F
F	T	F
F	F	T

We use the statement “P if, and only if, Q”, “P and Q are equivalent”, “P is a necessary and sufficient condition for Q”, and “Q is a necessary and sufficient condition for P”.

Universal Quantifier

The phrase “for every x ” is called a universal quantifier. We regard “for every x ”, “for all x ” and “for each x ” as having the same meaning and symbol

$$\forall x$$

For any arbitrary statement $P(x)$ defined for any x , we have

P(a)	P(b)	P(c)	...	$\forall x, P(x)$
T	T	T	...	T
T	F	T	...	F
F	T	F	...	F
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Existential Quantifier

Definition 3. The phrase “there exists an x ” is called an existential quantifier.

We regard “there exists an x ”, “for some x ” and “for at least one x ” as having the same meaning, and symbol

$$\exists x$$

For any arbitrary statement $P(x)$ defined for any x , we have

P(a)	P(b)	P(c)	...	$\exists x, P(x)$
F	F	F	...	F
T	F	T	...	T
F	T	F	...	T
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Observe that the statement

$$\text{not}(\forall x, P(x))$$

coincides with the statement

$$\exists x, \text{ not}(P(x)).$$

Sets

Definition 4. A set is a collection of objects called elements. The notation $x \in S$ represents the statement

“ x is an element of S ”. The statement “ x is not an element of S ” is denoted by “ $x \notin S$ ”. We can define a finite set by listing its elements (between curly brackets).

Example:

$$X = \{2, 3, 5, 7, 11\}$$

is a set with five elements. Alternatively, we can define X as “the set whose elements are the first five prime numbers”. If there are no elements inside of a set X we say that the set is **empty** and denote it by $\emptyset$.

Universal Quantifiers

Fix an arbitrary set X and consider a statement $P(x)$ for each member of X . The statement is denoted by

$$\forall x \in X, P(x).$$

Alternatively we could have said: “for every $x \in X$, we have $P(x)$ ”, “for all $x \in X$, we have $P(x)$ ” and “for each $x \in X$, we have $P(x)$ ”.

Existential Quantifier

For simplicity, the statement “there exists an $x \in X$ such that $P(x)$ ” is denoted by

$$\exists x \in X, P(x)$$

Alternatively, we could have said: “there exists an $x \in X$ such that $P(x)$ ”, “for some $x \in X, P(x)$ ”, and “ $P(x)$ for at least one x ”. If there is no x satisfying the statement $P(x)$ we say that $\{x \in X : P(x) \text{ is true}\} = \emptyset$

More on Sets

Sets are determined entirely by their elements.

Definition 5. We say that the sets X, Y are equal, written $X = Y$, when

$$\forall x, x \in X \Leftrightarrow x \in Y.$$

The order in which the elements are listed or repetitions of the same element are irrelevant. In particular, we have,

$$\{1, 2\} = \{2, 1\}, \quad \{1, 1\} = \{1\}.$$

Definition 6. Fix two sets A and B . The statement “ B is a subset of A ”, denoted by $B \subseteq A$, is defined by

$$\forall x, \quad x \in B \implies x \in A$$

Observe that

$$A = B \iff [A \subseteq B \text{ and } B \subseteq A].$$

Note that we use “[]” to emphasize the conjunction statement. When $B \subseteq A$ is true, we say that B is a subset of A . If, in addition, we have that $B \neq A$, we say that B is a strict subset of A , or $B \subset A$.

Definition 7. The set S of objects satisfying property P is denoted by

$$S := \{x : P(x)\}$$

Example: Fix two sets A and B . We let

$$A \cap B := \{x : x \in A \text{ and } x \in B\} \text{ and } A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

We also define

$$A \setminus B := \{x : x \in A \text{ and } x \notin B\} \text{ and } B \setminus A := \{x : x \notin A \text{ and } x \in B\}.$$

Observe that

$$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$$

Example: Fix a non-empty set S and consider the set, denoted by 2^S , of all subsets of S :

$$2^S := \{T : T \subseteq S\}.$$

The set B that contains all elements of some set A satisfying property P is denoted by

$$B := \{x \in A : P(x)\}$$

Some important sets that we will use throughout the course are going to be the natural numbers $\mathbb{N}$, the rational numbers $\mathbb{Q}$, and the real numbers $\mathbb{R}$.

Families

Definition 8. A class or a family is a non-empty collection of sets (set of sets).

If $\mathcal{A}$ is a class, then $\mathcal{A} \neq \emptyset$ and any element $A \in \mathcal{A}$ is a set.

Definition 9. The intersection of all elements of the class $\mathcal{A}$ is defined by

$$\{x : \forall A \in \mathcal{A}, \quad x \in A\}$$

We also use the notation,

$$\bigcap \mathcal{A}, \quad \bigcap \{A : A \in \mathcal{A}\}, \quad \text{or} \quad \bigcap_{A \in \mathcal{A}} A$$

Definition 10. The union of all elements of the class $\mathcal{A}$ is defined by

$$\{x : \exists A \in \mathcal{A}, \quad x \in A\}.$$

We also use the notation,

$$\bigcup \mathcal{A}, \quad \bigcup \{A : A \in \mathcal{A}\}, \quad \text{or} \quad \bigcup_{A \in \mathcal{A}} A.$$

A family $\mathcal{A}$ can be indexed by a set I , i.e., $\mathcal{A} := \{A_i : i \in I\}$ where each A_i is a set. In that case, the union $\bigcup \mathcal{A}$ is also denoted by

$$\bigcup_{i \in I} A_i \text{ or, equivalently, } \{x : \exists i \in I, \quad x \in A_i\}.$$

The intersection $\bigcap \mathcal{A}$ is also denoted by

$$\bigcap_{i \in I} A_i \text{ or, equivalently, } \{x : \forall i \in I, \quad x \in A_i\}.$$

If $I = \{k, k+1, \dots, K-1, K\}$ for some integers $k < K$, then $\bigcup_{i \in I} A_i$ is also denoted by

$$\bigcup_{i=k}^K A_i$$

and $\bigcap_{i \in I} A_i$ is also denoted by

$$\bigcap_{i=k}^K A_i$$

Ordered Pairs

Definition 11. An ordered pair is an ordered list (a, b) of two objects a and b . The list is ordered in the sense that

$$(a, b) = (a', b') \Leftrightarrow [a = a' \text{ and } b = b']$$

Definition 12. The (Cartesian) product of two non-empty sets A and B , denoted $A \times B$ is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$, i.e.,

$$A \times B := \{(a, b) : a \in A \text{ and } b \in B\}.$$

Definition 13. For any $n \in \mathbb{N}$, an n -vector is an ordered list $(a_1, a_2, \dots, a_n)$.

Definition 14. We define the (Cartesian) product of n sets $A_1, \dots, A_n$ as follows

$$A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) : \forall i \in \{1, \dots, n\}, \quad a_i \in A_i\}$$

The product $A_1 \times \dots \times A_n$ is also denoted $\prod_{i=1}^n A_i$ or $\times_{i=1}^n A_i$. Similarly, we will denote $\times_{i=1}^n A$ as A^n .

Functions

By a function f that maps X into Y , denoted by

$$f : X \longrightarrow Y$$

We mean a relation such that, for every $x \in X$, the set $\{y \in Y : f(x) = y\}$ is a singleton. In particular, this means that:

- for every $x \in X$, there exists a $y \in Y$ such that $f(x) = y$;
- for every $y, z \in Y$ with $f(x) = y$ and $f(x) = z$ for some $x \in X$, we must have $y = z$.

The function $f : X \rightarrow Y$ can also be denoted by $f : x \mapsto f(x)$ when X and Y are implicit.

The set of all functions that map X into Y is denoted by Y^X .

Example: $\mathbb{R}^{\{1,2\}}$ is the set of all real-valued functions on $\{1, 2\}$. A function $f \in \mathbb{R}^{\{1,2\}}$ can be identified with the pair $(f(1), f(2))$, in other words, we can identify $\mathbb{R}^{\{1,2\}}$ with $\mathbb{R}^2$.

Example: Let $X \subseteq \mathbb{R}$ and $f : X \rightarrow \mathbb{R}$ be the absolute value, denoted by $f(x) = |x|$. This means that:

$$f(x) = |x| := \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x \leq 0. \end{cases}$$

Definition 15. The set X is called the domain of f and Y the codomain of f .

Definition 16. The image of f , denoted $Im(f)$, is defined by

$$\begin{aligned} f(X) &:= \{y \in Y : \exists x \in X, \quad y = f(x)\} \\ &= \{f(x) : x \in X\}. \end{aligned}$$

Definition 17. The image of f on X is a subset of the codomain Y , that is, $f(X) \subseteq Y$.

Definition 18. The graph of the function f is defined by

$$\begin{aligned} \text{gph}(f) &:= \{(x, y) \in X \times Y : y = f(x)\} \\ &= \{(x, f(x)) : x \in X\}. \end{aligned}$$

Definition 19. A function $f : X \rightarrow Y$ is said to be:

1. A **surjection** when $f(X) = Y$.
2. An **injection** if $x \neq x'$ implies $f(x) \neq f(x')$ for all $x, x' \in X$.
3. A **bijection** if it is both injective and surjective.

Example: $f \in Y^X$ is constant if there exists $y \in Y$ such that $f(x) = y$ for all $x \in X$.

Example: Let $S \subseteq X$; the indicator function of S in X , denoted by $\mathbb{1}_S$, is the function that maps X into $\{0, 1\}$ defined by

$$\forall x \in X, \quad \mathbb{1}_S(x) = \begin{cases} 1 & \text{if } x \in S \\ 0 & \text{if } x \notin S \end{cases}$$

Example: Let $Z \subseteq X \subseteq W$ and fix a function $f \in Y^X$. The restriction of f to Z , denoted by f_Z is the function in Y^Z defined by

$$\forall z \in Z, \quad f_Z(z) := f(z).$$

Definition 20. Given $f : X \rightarrow Z$ and $g : Z \rightarrow Y$, we define the composition of f and g as the function $g \circ f : X \rightarrow Y$ defined by

$$\forall x \in X, \quad (g \circ f)(x) := g(f(x)).$$

Definition 21. Given a function $f : X \rightarrow Y$, the image of a set $A \subseteq X$ under f is the set $f(A) \subseteq Y$ defined by

$$\begin{aligned} f(A) &:= \{y \in Y : \exists a \in A, \quad y = f(a)\} \\ &= \{f(a) : a \in A\}. \end{aligned}$$

Definition 22. The inverse image of a set $B \subseteq Y$ is the set $f^{-1}(B) \subseteq X$ defined by

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

Definition 23. A function $f : X \rightarrow Y$ is invertible if $f^{-1}(\{y\})$ is a singleton for all $y \in Y$. If f is invertible we can consider the function $g : Y \rightarrow X$ defined by the equation

$$f^{-1}(\{y\}) = \{g(y)\}.$$

The function g is denoted by f^{-1} and called the inverse of f . Abusing notations, we write $f^{-1}(y)$ for $f^{-1}(\{y\})$, i.e.,

$$f^{-1}(y) := \{x \in X : f(x) = y\}.$$

even if f is not invertible.

Proposition 1. A function $f \in Y^X$ is invertible if, and only if, it is a bijection.

Proof. We proof only ($\implies$). The other direction is an exercise from Problem Set 1. Assume that f is invertible. We first show that f is injective. Let $x \neq x'$ arbitrarily chosen in the domain X . We cannot have $f(x) = f(x')$ otherwise $f^{-1}(\{f(x)\}) = \{x, x'\}$ is not a singleton. We now show that f is surjective. Fix an arbitrary element y of the codomain Y . There must exist some $x \in X$ such that $f(x) = y$ otherwise $f^{-1}(\{y\}) = \emptyset$ is not a singleton. We have thus proved that if f is invertible then f is bijection. $\square$

Class 2

Introduction to Sequences

Let $\mathbb{N} := \{1, 2, \dots\}$ be the set of natural numbers.

Definition 24. A sequence in a non-empty set X is a function $f : \mathbb{N} \rightarrow X$. A sequence f is also denoted by $(x_1, x_2, \dots)$ where $x_i := f(i)$ for each $i \in \mathbb{N}$.

We also use the following notations

$$(x_n)_{n \in \mathbb{N}}, \quad \text{or} \quad (x_n)$$

Definition 25. $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing if $\varphi(n) < \varphi(m)$ for all $n < m$.

Definition 26. A sequence is injective if, and only if, $x_n \neq x_k$ for any $n \neq k$.

Definition 27. A subsequence of a sequence $f \in X^{\mathbb{N}}$ is a function $g \in X^{\mathbb{N}}$ of the form $g = f \circ \varphi$ where $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing.

If $(x_n)_{n \in \mathbb{N}}$ represents f , then $(x_{\varphi(n)})_{n \in \mathbb{N}}$ represents the subsequence g . We may use the notation $(x_{n_k})_{k \in \mathbb{N}}$ to represent the subsequence g where

$$\forall k \in \mathbb{N}, \quad n_k := \varphi(k).$$

Example: Take the sequence composed of all the natural numbers $(1, 2, 3, 4, 5, 6, 7, 8, \dots)$. Then the sequence $(1, 1, 1, 1, \dots)$ is not a subsequence of the original sequence.

Notation on $\mathbb{R}$

Fix any real numbers $a < b$.

Definition 28. The open interval (a, b) is defined by

$$(a, b) := \{t \in \mathbb{R} : a < t < b\}$$

Definition 29. The semi-open intervals are defined by

$$(a, b] := (a, b) \cup \{b\} \text{ and } [a, b) := (a, b) \cup \{a\}.$$

Definition 30. The closed interval $[a, b]$ is defined by

$$[a, b] := \{t \in \mathbb{R} : a \leq t \leq b\}.$$

Definition 31. We define ∞ as follows:

$$\forall x \in \mathbb{R}, \quad \infty > x.$$

This convention allows us to treat ∞ and $-\infty$ as numbers, defined such as:

$$\infty > -\infty \quad \text{and} \quad \infty > t > -\infty$$

Definition 32. The unbounded intervals are defined by

$$(a, \infty) := \{t \in \mathbb{R} : a < t\} \text{ and } [a, \infty) := (a, \infty) \cup \{a\}.$$

Similarly, we define $(-\infty, b)$ and $(-\infty, b]$.

We consider the following notations: $\mathbb{R}_+ := \{x \in \mathbb{R} : x \geq 0\}$ and $\mathbb{R}_{++} := \{x \in \mathbb{R} : x > 0\}$.

Definition 33. We define the extended real numbers such as:

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}.$$

Definition 34. A set $S \subseteq \mathbb{K}$ is said to be bounded from above if it has an upper bound, i.e.,

$$\exists a \in \mathbb{K}, \quad \forall s \in S, \quad a \geq s.$$

Definition 35. A set $S \subseteq \mathbb{K}$ is said to be bounded from below if it has a lower bound, i.e.,

$$\exists b \in \mathbb{K}, \quad \forall s \in S, \quad b \leq s.$$

Definition 36. $s^* \in \mathbb{K}$ is the supremum of $S \subseteq \mathbb{K}$ if it is an upper bound of S and $a \geq s^*$ for all upper bounds of S .

Definition 37. $i^* \in \mathbb{K}$ is the infimum of $S \subseteq \mathbb{K}$ if it is a lower bound of S and $b \leq i^*$ for all lower bounds of S .

We denote the supremum of S as $\sup S$ and the infimum of S as $\inf S$. The supremum of S is often called the least upper bound of S and the infimum of S , denoted by $\inf S$ is often called the greatest lower bound of S and is denoted $\inf S$.

Example: Let $A = (0, 1)$, then $\sup A \notin A$. Note that $\mathbb{R}_+ \subseteq \mathbb{R}$ is not bounded from above.

Remark: Every set in the extended reals is trivially bounded from above and below. We will follow the usual convention that implies that if $S \subseteq \overline{\mathbb{R}}$ we will allow ∞ and $-\infty$ to upper and lower bounds, respectively. Therefore, a major advantage of the $\overline{\mathbb{R}}$ is that every set in $\overline{\mathbb{R}}$ has both a sup and an inf.

Axiom 1 (The Completeness Axiom). Every non-empty subset S of $\mathbb{R}$ that is bounded from above has a supremum in $\mathbb{R}$.

Proposition 2. The following statements are true;

1. For any $a > 0$ and $b \in \mathbb{R}$, there exists an integer $n \in \mathbb{N}$ such that $b < na$.

2. For any $a, b \in \mathbb{R}$ such that $a < b$, there exists a rational number $q \in \mathbb{Q}$ such that $a < q < b$.

Proof. We first prove the first. Assume, by way of contradiction, that b is an upper bound of the set $A := \{na : n \in \mathbb{N}\}$. Since A is a set of real numbers, it admits a supremum that we denote by c . This means that $na \leq c$ for every $n \in \mathbb{N}$ and c is the lowest number satisfying this property. In particular, $c - a$ cannot be an upper bound of A . This means that there exists $\bar{n} \in \mathbb{N}$ such that $c - a < \bar{n}a$. We then get that $c < (\bar{n} + 1)a$. This contradicts the fact that c is the supremum of A since $(\bar{n} + 1)a \in A$.

$\mathbb{Q}$ is given by all the numbers that can be expressed as the quotient of two integers. We now prove the second statement. Choose $n \in \mathbb{N}$ such that $n > 1/(b - a)$. Such an integer exists by property 1 above. We then have $nb - na > 1$. This implies that there must be an integer $p \in \mathbb{N}$ in the interval (na, nb) . We then get that $a < p/n < b$. $\square$

Countable and Uncountable Sets

A set is countable if you can count its elements. Of course, if the set is finite, you can easily count its elements. If the set is infinite, being countable means that you are able to put the elements of the set in order just like natural numbers are in order. Yet in other words, it means you are able to put the elements of the set into a “standing line” where each one has a “waiting number”, but the “line” is allowed to continue to infinity.

Definition 38. A set X is called *countably infinite* if there exists a bijection f that maps X onto the set $\mathbb{N}$ of natural numbers.

Equivalently,

Definition 39. X is *countably infinite* if, and only if, X is infinite and there exists a injective sequence (x_n) in X such that

$$X = \{x_1, x_2, \dots\}$$

Recall that a sequence (x_n) in X is a function from $\mathbb{N}$ to X . Therefore, the sequence is injective if, and only if, $x_n \neq x_k$ for any $n \neq k$.

Definition 40. X is called *countable* if it is either finite or countably infinite.

Definition 41. X is called *uncountable* if it is not countable.

Proposition 3. The following properties are satisfied.

1. Every subset of a countable set is countable.
2. Every infinite set has a countably infinite subset.

Proof. 1. Let X be a countable set and $Y \subseteq X$ a non-empty subset. Since X is countable, there exists a bijection $f : X \rightarrow \mathbb{N}$. Denote by g the function from $\mathbb{N}$ to X satisfying $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_{\mathbb{N}}$. We have that $Y \subseteq g(\mathbb{N})$. We construct $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ by induction as follows:

$$\varphi(1) := \min\{n \in \mathbb{N} : g(n) \in Y\}$$

and, for each $k \in \mathbb{N} \setminus \{1\}$,

$$\varphi(k) := \min\{n \geq \varphi(k-1) + 1 : g(n) \in Y\}.$$

Intuitively, $\varphi(1)$ is the first index n such that $g(n) \in Y$, $\varphi(2)$ is the second index n such that $g(n) \in Y$. Observe that, for every $k \in \mathbb{N} \setminus \{1\}$, we have

$$\varphi(k) = \min\{n \in \mathbb{N} : g(n) \in Y \setminus \{g(\varphi(1)), \dots, g(\varphi(k-1))\}\}$$

This is a possible way to order the elements of Y . Indeed, observe that the function φ is injective since it is strictly increasing. Let $h := g \circ \varphi$. It is an injective function from $\mathbb{N}$ to Y since g and φ are injective. We now prove it is surjective. Fix an arbitrary $y \in Y$. Since $Y \subseteq g(\mathbb{N})$, there exists $n \in \mathbb{N}$ such that $y = g(n)$. If $n = 1$ then $\varphi(1) = 1$ (see the definition of $\varphi(1)$) and $y = h(1) \in h(\mathbb{N})$. Assume that $n > 1$. Let $T_y := \{k \in \{1, 2, \dots, n-1\} : g(k) \in Y\}$ be the set of all indexes k strictly lower than n such that $g(k)$ belongs to Y . Let m be the cardinal (number of elements) of T_y (if T_y is empty then $m = 0$). Observe that $h(m+1) = g(\varphi(m+1)) = y$. We have thus proved that h is surjective.

2. Let X be an infinite set. Pick an arbitrary element $x_1 \in X$. The set $X_1 := X \setminus \{x_1\}$ is still infinite. Pick an arbitrary element $x_2 \in X_1$. The set $X_2 := X \setminus \{x_1, x_2\}$ is still infinite. Pick an arbitrary element $x_3 \in X_2$. Iterating the above argument, we can construct $\{x_1, x_2, \dots\} \subseteq X$ such that $x_{n+1} \notin \{x_1, \dots, x_n\}$. Let $g : \mathbb{N} \rightarrow \{x_1, x_2, \dots\}$ be the function defined by $g(n) := x_n$. The function g is obviously surjective. We now prove it is injective. Let $k' \neq k$ be two distinct natural numbers. We can assume without any loss of generality that $k' > k$. Since

$$x_{k'} \notin \{x_1, \dots, x_k, \dots, x_{k'-1}\}$$

we must have $g(k') = x_{k'} \neq x_k = g(k)$. □

Useful results

Lemma 1. *Every natural number can be expressed in the form $n = 2^p q$, where p is a nonnegative integer and q is an odd natural number.*

Proof. We will prove this by strong induction. For the base case $n = 1$, just note that $n = 2^0 \cdot 1$. Now let $k \in \mathbb{N}$, and suppose that every natural number less than k can be written in the desired form. If k is odd, we just write $k = 2^0 k$. If k is even, then there is an integer l such that $k = 2l$, and l is positive because k is. Since $l < k$, the inductive hypothesis implies that there exists a nonnegative integer p and an odd natural number q such that $l = 2^p q$, and then $k = 2l = 2^{p+1} q$, which satisfies the conclusion. □

Proposition 4. *The set $\mathbb{N} \times \mathbb{N}$ is countably infinite.*

Proof. Define a function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by $f(m, n) = 2^{m-1}(2n-1)$. To show that f is injective, suppose $(m_1, n_1), (m_2, n_2)$ are elements of $\mathbb{N} \times \mathbb{N}$ such that $f(m_1, n_1) = f(m_2, n_2)$, which is to say

$$2^{m_1-1}(2n_1-1) = 2^{m_2-1}(2n_2-1) \quad (1)$$

We will first prove by contradiction that $m_1 = m_2$. Suppose not; then one is larger, and we may assume without loss of generality that $m_2 > m_1$. Multiplying both sides of (1) by 2^{1-m_1} , we obtain

$$2n_1 - 1 = 2^{m_2-m_1}(2n_2 - 1)$$

The fact that $m_2 > m_1$ implies that the right-hand side is even, while the left-hand side is odd; this is a contradiction, so we can conclude that $m_1 = m_2$. Then simple algebra shows that $n_1 = n_2$ as well, so $(m_1, n_1) = (m_2, n_2)$. To prove surjectivity, let $x \in \mathbb{N}$ be arbitrary. Lemma 1, shows we can write $x = 2^p q$ for some nonnegative integer p and some odd natural number q . The fact that q is odd means that $q = 2j + 1$ for some integer j , and that $q \geq 1$ means $j \geq 0$. Therefore, $(p + 1, j + 1) \in \mathbb{N} \times \mathbb{N}$, and

$$f(p + 1, j + 1) = 2^{(p+1)-1}(2(j + 1) - 1) = 2^p(2j + 1) = 2^p q = x.$$

Thus we have shown that f is bijective, so $\mathbb{N} \times \mathbb{N} \approx \mathbb{N}$. □

Proposition 5. *A countable union of countable sets is countable.*

Proof. Removing the redundant elements from some sets if necessary, we can assume without any loss of generality that $X_i \cap X_j = \emptyset$ for every $i \neq j$. We also assume that each X_i is countable infinite (the case where X_i is finite for some indexes i can be analyzed similarly). It then follows that each X_i can be written as follows: $X_i = \{x_1^i, x_2^i, \dots\}$ where $x_n^i \neq x_k^i$ for every $n \neq k$. Let $X := X_1 \cup X_2 \cup \dots$. We can easily construct an injection from X to $\mathbb{N}^2$. Indeed, if $x \in X$ then there exists a unique pair $(i_x, n_x) \in \mathbb{N}^2$ such that $x = x_{n_x}^{i_x}$ (this is where we use the assumption that $X_i \cap X_j \neq \emptyset$ and $x_n^i \neq x_k^i$ for every $i \neq j$ and $n \neq k$). Let $f : X \rightarrow \mathbb{N}^2$ be defined by $f(x) := (i_x, n_x)$. The function f is injective. Indeed, let $x \neq x' \in X$ be two distinct elements of X . If x and x' belong to different sets then $i_x \neq i_{x'}$. Assume that $i_x = i_{x'}$ and denote this index by i . x and x' belong to X^i . Since $x \neq x'$, we cannot have $n_x = n_{x'}$. Since $\mathbb{N}^2$ is countable, we are done. □

Lemma 2 (Cantor's Nested Intervals). *Let $I_n := [a_n, b_n]$ be a closed interval for each $n \in \mathbb{N}$. If $I_1 \supseteq I_2 \supseteq \dots$ then*

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$$

If, in addition, $\lim(b_n - a_n) = 0$, then the intersection is a singleton.

Corollary 1. $\mathbb{R}$ is uncountable.

Proof. Assume, by way of contradiction, that $[0, 1] = \{x_1, x_2, \dots\}$. We should find someone in $[0, 1]$ that does not belong to $\{x_1, x_2, \dots\}$. The real number x_1 does not belong to at least one of the three intervals $[0, 1/3]$, $[1/3, 2/3]$ and $[2/3, 1]$. Repeat the procedure and construct a sequence (I_n) of nested intervals $I_n = [a_n, b_n]$ such that $\lim(b_n - a_n) = 0$ and such that $\{x_1, \dots, x_n\} \cap I_n = \emptyset$. Apply the Cantor's Nested Interval Lemma to deduce that there exists $x \in [0, 1] \setminus \{x_1, x_2, \dots\}$. $\square$

The Cardinality Ordering

Definition 42. Let A and B be any two sets. We say that A is *cardinally larger than* B , denoted $A \succeq_{\text{card}} B$, if there exists an injection from B into A .

Definition 43. If we can find a bijection from A onto B , then we say that A and B are *cardinally equivalent* and denote this by $A \sim_{\text{card}} B$.

Class 3

Real Sequences

Definitions and Basic Properties

Definition 44. A real sequence is a function from $\mathbb{N} \rightarrow \mathbb{R}$ denoted by x_n

$$\mathbb{N} \rightarrow \mathbb{R}$$

$$n \mapsto x_n$$

We denote x_n as the image of the function when evaluated at n (the n -th element of a sequence). For us to represent the whole sequence write:

$$(x_1, x_2, x_3, \dots), \quad \text{or} \quad (x_n)_{n \in \mathbb{N}} \quad \text{or} \quad (x_n).$$

Definition 45. A sequence (x_n) is said to be increasing (decreasing) if $x_n \leq x_{n+1}, \forall n \geq 1$ ($x_n \geq x_{n+1}, \forall n \geq 1$). If the inequality among elements is strict we say the sequence is **strictly increasing (decreasing)**.

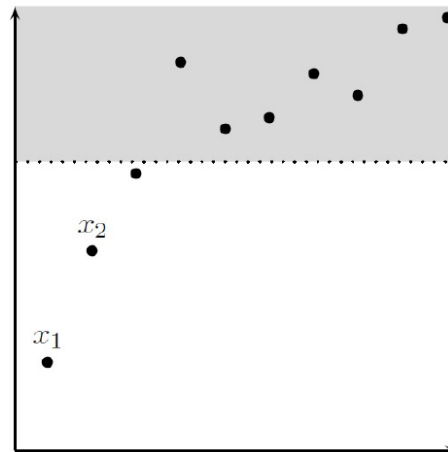
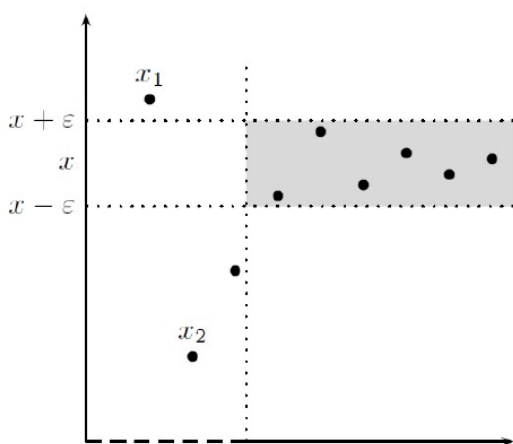
Definition 46. A sequence (x_n) converges to $x \in \mathbb{R}$, denoted by $\lim_{n \rightarrow \infty} x_n = x$ if,

$$\forall \epsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad |x_n - x| < \epsilon.$$

The idea of this definition is simple. After some point, this sequence gets arbitrarily close to its limit x . In a more precise way, only finitely many elements of the sequence (N of them) are outside of the interval $(x - \epsilon, x + \epsilon)$. At some point in this course (the Metric Spaces chapter, to be more precise), we will introduce some more structure to this definition re-stating it in terms of *open balls*.

Definition 47. A sequence (x_n) diverges if,

$$\forall y \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad x_n \geq y$$



Lemma 3 (Sandwich). Consider three real sequences (x_n) , (y_n) , and (z_n) such that, for each n large enough,

$$x_n \leq y_n \leq z_n$$

If $\exists \ell \in \overline{\mathbb{R}}$ such that $\lim x_n = \lim z_n = \ell$, then $\lim y_n$ exists and $\lim y_n = \ell$.

Proof. We prove the case $\ell \in \mathbb{R}$. Fix $\epsilon > 0$. Since both (x_n) and (z_n) converge to $\ell \in \mathbb{R}$ we have that:

$$\exists N_x \in \mathbb{N}, \quad \forall n \geq N_x, \quad |x_n - \ell| < \epsilon$$

$$\exists N_z \in \mathbb{N}, \quad \forall n \geq N_z, \quad |z_n - \ell| < \epsilon$$

Pick $N \equiv \max\{N_x, N_z\}$:

$$-\epsilon < x_n - \ell \leq y_n - \ell \leq z_n - \ell < \epsilon$$

$$\forall n \geq N, \quad |y_n - \ell| < \epsilon \implies \lim y_n = \ell$$

□

From a given sequence (x_n) one may want to analyze just a subset of the indices n , for example:

$$(x_2, x_4, x_6, \dots), \quad \text{or} \quad (x_1, x_3, x_5, \dots)$$

In a more general way, we can define a strictly increasing function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ that selects the indexes we want and makes **another** sequence out of the original (x_n) . We denote this new sequence as $(x_{\varphi(n)})$ and name it a **subsequence**.

Lemma 4. If a sequence (x_n) converges to $\ell \in \mathbb{R}$, then for every strictly increasing $\varphi : \mathbb{N} \rightarrow \mathbb{N}$, the subsequence $(x_{\varphi(n)})$ also converges to ℓ .

Proof. We start by showing that $\varphi(n) \geq n$. If we prove this, using the definition of convergence ($\dots \forall n \geq N$) we are able to say that the subsequence will also converge. First note that $\varphi(1) \in \mathbb{N}$, and thus it must be the case that $\varphi(1) \geq 1$. Since φ is strictly increasing: $\varphi(2) > \varphi(1) \geq 1$, also since $\varphi(2) \in \mathbb{N}$ we have that $\varphi(2) \geq 2$. Proceeding by induction we can show that $\varphi(n) \geq n$ for any finite n . Fix $\epsilon > 0$. Since (x_n) converges we have that $\exists N \in \mathbb{N}$ such that $\forall n \geq N, |x_n - \ell| < \epsilon$. But note that:

$$\forall n \geq N \implies \varphi(n) \geq N$$

Hence $|x_{\varphi(n)} - \ell| < \epsilon$.

□

Definition 48. A sequence (x_n) is said to be **bounded from above (below)** if

$$\exists K \in \mathbb{R}, \quad x_n \leq K, \quad \forall n \in \mathbb{N} \quad (\text{above})$$

$$\exists K \in \mathbb{R}, \quad x_n \geq K, \quad \forall n \in \mathbb{N} \quad (\text{below})$$

Definition 49. A sequence (x_n) is **bounded** if it is bounded from above and below

$$\exists K \in \mathbb{R}, \quad |x_n| \leq K, \quad \forall n \in \mathbb{N}$$

We are then able to conclude that $\sup\{|x_n| : n \in \mathbb{N}\} \leq K \in \mathbb{R}$.

Proposition 6. Every real sequence that converges in $\mathbb{R}$ must be bounded.

Proof. Let (x_n) be a sequence that converges to $x \in \mathbb{R}$. Fix $\epsilon > 0$.

$$\exists N \in \mathbb{N}, \quad \forall n \geq N, \quad x - \epsilon < x_n < x + \epsilon$$

This implies that:

$$\sup_{n \geq N} |x_n| \leq |x| + \epsilon$$

Since $N < \infty$:

$$\sup_{n \in \mathbb{N}} |x_n| \leq \max\{|x_1|, |x_2|, \dots, |x_{N-1}|, |x| + \epsilon\}$$

□

Proposition 7. Every increasing (decreasing) real sequence that is bounded from above (below) converges.

Proof. Let (x_n) be an increasing sequence that is bounded from above. Since the set $T := \{x_n : n \in \mathbb{N}\}$ is bounded from above it has a supremum $\ell \in \mathbb{R}$. Fix $\epsilon > 0$. Since ℓ is the supremum of T it is the **smallest** upper bound of T which means that $\ell - \epsilon$ is **not** an upper bound for T . This means there exists some $N \in \mathbb{N}$ such that $x_N > \ell - \epsilon$. Since (x_n) is an increasing sequence: $\forall n \geq N, x_n \geq x_N > \ell - \epsilon$. By definition of ℓ it must be the case that $x_n \leq \ell < \ell + \epsilon$ which means that $\forall n \geq N, |x_n - \ell| < \epsilon$. □

Definition 50. A sequence (x_n) is said to be *monotonic* if it is either increasing or decreasing.

Proposition 8. Every real sequence has a monotonic subsequence.

Proof. For each $k \in \mathbb{N}$ define the *tail* of the sequence (x_n) as:

$$T_k \equiv \{x_n : n \geq k\} = \{x_k, x_{k+1}, x_{k+2}, \dots\}$$

We start by the case that k is such that T_k does **not** admit a maximum. Pose $\varphi(1) \equiv k$. Since T_k does not have a maximum, there must exist $n_2 > k$ such that $x_{n_2} > x_k$. Pose $\varphi(2) \equiv n_2$. Note that since $\varphi(2) > \varphi(1)$ we have that $T_{\varphi(2)} \subseteq T_{\varphi(1)}$. Since $T_{\varphi(1)}$ does not admit a maximum, $T_{\varphi(2)}$ also does not admit a maximum. Hence, there must exist $n_3 > \varphi(2)$ such that $x_{n_3} > x_{\varphi(2)}$. Pose $\varphi(3) \equiv n_3$. Following this procedure we can construct a

strictly increasing function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that $x_{\varphi(n+1)} > x_{\varphi(n)}$ which means that $(x_{\varphi(n)})$ is a strictly increasing subsequence of (x_n) .

We now assume T_k admits a maximum for each k . Since T_1 admits a maximum, there must exist $n_1 \geq 1$ such that x_{n_1} attains the maximum of T_1 . Pose $\varphi(1) \equiv n_1$. Since T_k admits a maximum for each k , we know that $T_{\varphi(1)+1}$ admits a maximum. Let $n_2 \geq \varphi(1) + 1$ be such that x_{n_2} attains the maximum of $T_{\varphi(1)+1}$. Pose $\varphi(2) \equiv n_2$. Note that since $T_{\varphi(2)} \subseteq T_{\varphi(1)}$ it must be the case that $x_{\varphi(1)} \geq x_{\varphi(2)}$. Proceeding inductively we can construct $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that $\varphi(n+1) > \varphi(n)$ and $(x_{\varphi(n)})$ is a decreasing subsequence of (x_n) . $\square$

Having stated the last definitions and propositions we naturally get to an essential result.

Theorem 1 (Bolzano-Weierstrass). *Every bounded real sequence has a convergent subsequence.*

The proof of the above Theorem relies on previously proved properties of real sequences and is left as an exercise. This is a good example of how results are built out of smaller building blocks and as time goes on and you are more familiar with definitions and some basic properties, you can use them to derive new (and more general) mathematical statements.

Lemma 5. *If $S \neq \emptyset$ and $S \subseteq \mathbb{R}$, then there is an increasing sequence $(x_n) \in S^{\mathbb{N}}$ such that $x_n \rightarrow \sup S$.*¹

Proof. Let $\ell := \sup S$ and assume the case $\ell < \infty$. If $\ell = \max S$ the proof is trivial², so we assume it is not. By the definition of a supremum, we know that $\ell - 1$ cannot be an upper bound of S . Therefore there exists $x_1 \in S$ such that $x_1 \geq \ell - 1$. Let $\epsilon_1 := (\ell - x_1)/2$ and note that since ℓ is **not** a maximum of S we must have that $x_1 < \ell$ which means that $\epsilon_1 \in (0, 1/2)$. In particular, (again) by the definition of the supremum we have that $\ell - \epsilon_1$ cannot be an upper bound for S , and by the same argument $\exists x_2 \in S$ such that $x_2 \geq \ell - \epsilon_1$. Note that by the definition of ϵ_1 we have that $x_2 > x_1$. Again, pose $\epsilon_2 := (\ell - x_2)/2$ and note that $\epsilon_2 \in (0, 1/2^2)$. We can proceed with the same argument inductively to construct a strictly increasing sequence (x_n) such that $x_{n+1} - \ell \leq 1/2^n$ and $x_n \leq \ell$ (since ℓ is not a maximum). We thus conclude that (x_n) converges to ℓ . $\square$

With the above property, we started linking some specific objects in a set with real sequences (in the above case, the sup and inf of a set). The next lemma is an additional step in that direction, but now with an arbitrary element of the real numbers.

Lemma 6. *For any real number x and any real sequence (x_n) we have $\lim x_n = x \iff$ Every subsequence of (x_n) has itself a subsequence that converges to x .*

¹ There is also an equivalent Lemma for the increasing/inf case.

² Just set $x_n := \ell$ for each $n \in \mathbb{N}$

Proof. $[\implies]$. This proof relies exclusively on using one of the previous Lemmas... which one?

$[\impliedby]$. Assume every subsequence of the real sequence (x_n) has itself a subsequence that converges to x . We shall prove $x_n \rightarrow x$. Suppose it is **not** the case, that is:

$$\exists \epsilon > 0, \quad \forall N \in \mathbb{N}, \quad \exists n \geq N, \quad |x_n - x| \geq \epsilon$$

Since the above property holds $\forall N \in \mathbb{N}$, we can start by setting $N = 1$ and getting the existence of:

$$\exists n_1 \geq 1, \quad |x_{n_1} - x| \geq \epsilon$$

By setting $N = n_1 + 1$ we get that:

$$\exists n_2 \geq n_1 + 1, \quad |x_{n_2} - x| \geq \epsilon$$

Proceeding inductively we can construct (x_{n_k}) such that:

$$\forall k \geq 1, \quad |x_{n_k} - x| \geq \epsilon$$

Define $\varphi(k) := n_k$ for $k \in \mathbb{N}$ and we get that the subsequence $(x_{\varphi(n)})$ does **not** converge. However, by assumption, every subsequence of (x_n) has itself a convergent subsequence that converges to x . Since it is also the case for $(x_{\varphi(n)})$ we know that $\exists \psi : \mathbb{N} \rightarrow \mathbb{N}$ such that $x_{\varphi(\psi(n))} \rightarrow x$. This is already a contradiction since we just showed that $(x_{\varphi(n)})$ does not converge to x . $\square$

Cauchy-Sequences

This is going to be an important class of sequences for the following chapters since it is widely used in the context of the convergence of value functions. We are going to state a definition in which the relevant distance is not between an element of a sequence and a real number, but between two elements of the sequence.

Definition 51. A real sequence (x_n) is said to be a **Cauchy sequence** if

$$\forall \epsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n, p \geq N, \quad |x_p - x_n| < \epsilon$$

Intuitively, any convergent sequence will have its terms closer and closer. We now show an important result stating that in $\mathbb{R}$ any sequence for which the distance between its terms shrinks will converge.

Proposition 9. Every real Cauchy sequence converges.

Proof. We first show that (x_n) is bounded. Choosing $\epsilon := 1$, we get the existence of $N \in \mathbb{N}$ such that

$$\forall n \geq N, \quad |x_n| - |x_N| \leq |x_n - x_N| < 1 \implies |x_n| \leq 1 + |x_N|$$

Fix now two arbitrary increasing sequences $\varphi, \varphi' : \mathbb{N} \rightarrow \mathbb{N}$. The two subsequences $(x_{\varphi(n)})$ and $(x_{\varphi'(n)})$ are both bounded. By Bolzano-Weierstrass theorem $\exists \psi, \psi' : \mathbb{N} \rightarrow \mathbb{N}$ such that $(x_{\varphi(\psi(n))})$ and $(x_{\varphi'(\psi'(n))})$ converge to ℓ and ℓ' . We already know that if every subsequence of (x_n) has itself a subsequence that converges to ℓ it must be the case that $x_n \rightarrow \ell$, therefore we shall prove that $\ell = \ell'$. Assume by contradiction $\ell \neq \ell'$. Take $\epsilon := |\ell - \ell'|/2 > 0$. And since (x_n) is a Cauchy sequence:

$$\exists N \in \mathbb{N}, \quad \forall n, p \geq N, \quad |x_n - x_p| < \epsilon$$

We then deduce that:

$$\forall k \geq N, \quad |x_{\varphi(\psi(k))} - x_{\varphi'(\psi'(k))}| < \epsilon$$

Passing to the limit when $k \rightarrow \infty$, we get that

$$0 < |\ell - \ell'| < |\ell - \ell'|/2$$

Which is a contradiction.

□

Class 4

Problem Set 1

Exercise 1. Prove that if $f \in Y^X$ is a bijection, then it is invertible.

Proof. Assume that f is a bijection. We want to prove that f is invertible. Fix y an arbitrary element of the codomain Y . The set $f^{-1}(\{y\})$ cannot be empty since f is surjective. This means that $f^{-1}(\{y\})$ has at least one element. Actually, it cannot have more than one element. Indeed, assume by contradiction³, that $f^{-1}(\{y\})$ contains at least two distinct elements $x \neq x'$. This means that $f(x) = f(x')$ and contradicts the fact that f is injective. $\square$

Exercise 2. Prove that a set X is countable if, and only if, there exists an injection from X to a countable set.

Proof. Since $\mathbb{N}$ is countable, we only have to prove the sufficiency part ("if"). Let X be a set and assume that there exists an injection $h : X \rightarrow Z$ where Z is a countable set. By definition, there exists a bijection $f : Z \rightarrow \mathbb{N}$. Let $g := f \circ h$. The function $g : X \rightarrow \mathbb{N}$ is injective since h and f are injective. Therefore, the function $\bar{g} : X \rightarrow g(X)$ defined by $\bar{g}(x) := g(x)$ is a bijection from X onto $g(X)$. If $g(X)$ is finite, then X is also finite. If $g(X)$ is infinite, then there exists a strictly increasing function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that $\varphi(\mathbb{N}) = g(X)$. Indeed, it is sufficient to pose

$$\varphi(1) := \min\{n \in \mathbb{N} : n \in g(X)\}$$

and for every $k \in \mathbb{N}$

$$\begin{aligned} \varphi(k+1) &:= \min\{n \geq \varphi(k) + 1 : n \in g(X)\} \\ &= \min\{n \in \mathbb{N} : n \in g(X) \setminus \varphi(\{1, 2, \dots, k\})\}. \end{aligned}$$

Since $\bar{g}$ is bijective, there exists a bijection, denoted by $\bar{g}^{-1}$, from $g(X)$ onto X , such that $\bar{g} \circ \bar{g}^{-1} = \text{Id}_{g(X)}$ and $\bar{g}^{-1} \circ \bar{g} = \text{Id}_X$. The function $\bar{g}^{-1} \circ \varphi$ is then a bijection from $\mathbb{N}$ onto X . This is sufficient to deduce that X is countably infinite. $\square$

Exercise 3. Prove that a set X is countable if, and only if, there exists a surjection from a countable set to X .

Proof. Since $\mathbb{N}$ is countable, we only have to prove the sufficiency part ("if"). Let X be a set and assume that there exists a surjection $h : Z \rightarrow X$ where Z is a countable set. By definition, there exists a bijection $f : \mathbb{N} \rightarrow Z$. Let $g := h \circ f$. The function $g : \mathbb{N} \rightarrow X$ is surjective since h and f are surjective. For every $x \in X$, the set

$$g^{-1}(\{x\}) := \{n \in \mathbb{N} : g(n) = x\}$$

³ Note that we are finishing the proof by using the equivalence between $A \implies B = \text{not}(A) \implies \text{not}(B)$.

is non-empty. We then consider a function $k : X \rightarrow \mathbb{N}$ defined by: $k(x)$ is one (arbitrary) element of $g^{-1}(\{x\})$. If we prove that the function $k : X \rightarrow \mathbb{N}$ is an injection, then we get the desired result by **exercise 2**. Let $x \neq x'$ two distinct elements of X . Since

$$x = g(k(x)) \text{ and } x' = g(k(x'))$$

we cannot have $k(x) = k(x')$. ■

□

Exercise 4. Prove the following Lemma

Lemma 7 (Cantor's Nested Intervals). Let $I_n := [a_n, b_n]$ be a closed interval for each $n \in \mathbb{N}$. If $I_1 \supseteq I_2 \supseteq \dots$ then

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$$

If, in addition, $\lim(b_n - a_n) = 0$, then the intersection is a singleton.

Proof. Since $I_{n+1} \subseteq I_n$, we must have $a_{n+1} \geq a_n$ and $b_{n+1} \leq b_n$. The sequence (a_n) is increasing and bounded from above by b_1 . This implies that there exists $a \in \mathbb{R}$ such that (a_n) converges to a . Moreover, we have $a = \sup\{a_n : n \in \mathbb{N}\}$. Similarly, we get the existence of some real number $b \in \mathbb{R}$ such that (b_n) converges to b and $b = \inf\{b_n : n \in \mathbb{N}\}$. Observe that $[a, b] \subseteq [a_n, b_n] = I_n$ for each n , i.e.,

$$[a, b] \subseteq \bigcap_{n \in \mathbb{N}} I_n$$

If $\lim(b_n - a_n) = 0$ then $a = \lim a_n = \lim b_n = b$ and we get that $[a, b] = \{a\} = \{b\}$. □

Exercise 5. Consider two real sequences (x_n) , and (y_n) such that:

$$\exists N \in \mathbb{N}, \quad \forall n \geq N, \quad x_n \leq y_n.$$

Prove that if both sequences converge in $\overline{\mathbb{R}}$ then $\lim x_n \leq \lim y_n$.

Proof. Let $\ell(x) \equiv \lim x_n$ and $\ell(y) \equiv \lim y_n$. Fix $\epsilon > 0$ arbitrary. Since both sequences converge there must exist $N_x, N_y \in \mathbb{N}$ such that $x_n \geq \ell(x) - \epsilon$ for $n \geq N_x$ and $y_n \geq \ell(y) - \epsilon$ for $n \geq N_y$. It follows that:

$$\forall n \geq \max\{N_x, N_y, N\}, \quad \ell(x) - \epsilon \leq x_n \leq y_n \leq \ell(y) + \epsilon$$

We have thus proved that:

$$\forall \epsilon > 0, \quad \ell(x) - \epsilon \leq \ell(y) + \epsilon$$

Since ϵ is arbitrary we deduce that $\ell(x) \leq \ell(y)$. If not, choose $\epsilon := [\ell(x) - \ell(y)]/3$ and get a contradiction. □

Exercise 6. Prove that every bounded real sequence has a convergent subsequence.
(Hint. This is the Bolzano-Weierstrass Theorem just use some results proved in the chapter to get the result)

Proof. Let (x_n) be a bounded real sequence. By Proposition (8): $\exists \varphi : \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing such that $(x_{\varphi(n)})$ is monotone. Since it is also bounded, by Proposition (7) it converges in $\mathbb{R}$. $\square$

Fix a real sequence (x_n) and denote by $(y_n), (z_n) \in \overline{\mathbb{R}}^{\mathbb{N}}$ the following sequences:

$$y_n := \sup\{x_n, x_{n+1}, \dots\}$$

$$z_n := \inf\{x_n, x_{n+1}, \dots\}$$

Definition 52. We call the limits: $\lim y_n = \limsup x_n$ (limit superior) and $\lim z_n = \liminf x_n$ (limit inferior).

Exercise 7. Prove that for any $\ell \in \mathbb{R}$ and $(x_n) \in \mathbb{R}^{\mathbb{N}}$, we have

$$\limsup x_n = \ell$$

if, and only if,

1. $\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \text{ such that } x_n < \ell + \epsilon$
2. $\forall \epsilon > 0, \forall N \in \mathbb{N}, \exists n > N, \text{ such that } x_n > \ell - \epsilon$

Proof. [$\Leftarrow$]. Suppose (1) and (2) hold and let $\epsilon > 0$. Let $T_n := \{x_n, x_{n+1}, \dots\}$ be the tail starting at n and define the sequence $y_n := \sup T_n$. By (1), $\ell + \epsilon$ is an upper bound for T_N , which means that $y_N \leq \ell + \epsilon$. Since (y_n) is a decreasing sequence we have that $y_n \leq \ell + \epsilon$. By (2) we know that $\ell - \epsilon$ is **not** an upper bound of T_k for any $k \in \mathbb{N}$ which implies that $y_k > \ell - \epsilon$.

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \ell - \epsilon < y_n \leq \ell + \epsilon$$

We thus conclude that $y_n \rightarrow \ell$.

[$\Rightarrow$]. Assume that $y_n \rightarrow \ell$. Since (y_n) is a decreasing sequence for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $\ell \leq y_n < \ell + \epsilon$ for all $n \geq N$. Since y_n is an upper bound of T_N :

$$\forall n \geq N, x_n \leq y_N \leq \ell + \epsilon$$

We have just shown that (1) holds. Now assume by contradiction that (2) does not hold: $\exists \epsilon > 0, \exists N \in \mathbb{N}$, such that $\ell - \epsilon$ is an upper bound of T_N . This implies that $y_N \leq \ell - \epsilon$, which is a contradiction. $\square$

Class 5

Real Functions

Definition 53. A real function on a non-empty set T is an element of $\mathbb{R}^T$,

$$f : T \rightarrow \mathbb{R}$$

More intuitively, a function is a map between elements of an arbitrary set T to the real numbers.

Definition 54. A function is said to be increasing (decreasing) if:

$$\forall x, y \in T, \quad x \geq y \implies f(x) \geq f(y) \quad (f(x) \leq f(y))$$

In the definition of a **strictly** increasing (decreasing) the inequalities are strict.

Example: Let $X \subseteq \mathbb{R}$ and $f : X \rightarrow \mathbb{R}$ be the absolute value, denoted by $f(x) = |x|$. This means that:

$$f(x) = |x| := \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x \leq 0. \end{cases}$$

Definition 55. We say that a function is bounded if $\exists M \in \mathbb{R}$ such that:

$$\forall t \in T, \quad |f(t)| \leq M.$$

Limits

Definition 56. Consider a function on the extended real numbers $f : \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}}$

1. Let $x \in \mathbb{R}$ and $\ell \in \mathbb{R}$, we say that $\lim_{t \rightarrow x} f(t) = \ell$ if, and only if,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies |f(t) - \ell| < \varepsilon$$

2. Let $\ell \in \mathbb{R}$, we say that $\lim_{t \rightarrow \infty} f(t) = \ell$ if, and only if,

$$\forall \varepsilon > 0, \quad \exists \tau \in \mathbb{R}, \quad \forall t \in T, \quad t > \tau \implies |f(t) - \ell| < \varepsilon$$

3. Let $\ell \in \mathbb{R}$, we say that $\lim_{t \rightarrow -\infty} f(t) = \ell$ if, and only if,

$$\forall \varepsilon > 0, \quad \exists \tau \in \mathbb{R}, \quad \forall t \in T, \quad t < \tau \implies |f(t) - \ell| < \varepsilon$$

4. Let $x \in \mathbb{R}$, we say that $\lim_{t \rightarrow x} f(t) = \infty$ if, and only if,

$$\forall y \in \mathbb{R}, \quad \exists \delta > 0, \quad \forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies f(t) > y$$

5. Let $x \in \mathbb{R}$, we say that $\lim_{t \rightarrow x} f(t) = -\infty$ if, and only if,

$$\forall y \in \mathbb{R}, \quad \exists \delta > 0, \quad \forall t \in T \setminus \{x\}, \quad |t - x| < \delta \Rightarrow f(t) < y$$

6. We say that $\lim_{t \rightarrow \infty} f(t) = \infty$ if, and only if,

$$\forall y \in \mathbb{R}, \quad \exists \tau \in \mathbb{R}, \quad \forall t \in T, \quad t > \tau \Rightarrow f(t) > y$$

7. We say that $\lim_{t \rightarrow -\infty} f(t) = \infty$ if, and only if,

$$\forall y \in \mathbb{R}, \quad \exists \tau \in \mathbb{R}, \quad \forall t \in T, \quad t < \tau \Rightarrow f(t) > y$$

8. We say that $\lim_{t \rightarrow \infty} f(t) = -\infty$ if, and only if,

$$\forall y \in \mathbb{R}, \quad \exists \tau \in \mathbb{R}, \quad \forall t \in T, \quad t > \tau \Rightarrow f(t) < y$$

9. We say that $\lim_{t \rightarrow -\infty} f(t) = -\infty$ if, and only if,

$$\forall y \in \mathbb{R}, \quad \exists \tau \in \mathbb{R}, \quad \forall t \in T, \quad t < \tau \Rightarrow f(t) < y$$

Continuity at a Point

There are multiple ways to define the continuity of a function at a point,

Definition 57. Let $S \subset \mathbb{R}$, $c \in S$ and let $f : S \rightarrow \mathbb{R}$. The function f is continuous at c , if and only if,

$$\lim_{x \rightarrow c} f(x) = f(c)$$

where $\lim_{x \rightarrow c} f(x) = f(c)$ means that

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in S \setminus \{c\}, \quad |x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon.$$

Definition 58. Let $S \subset \mathbb{R}$, $c \in S$ and let $f : S \rightarrow \mathbb{R}$. The function f is continuous at c , if and only if,

$$\forall (x_n) \in \left\{ (y_n) \in S^{\mathbb{N}} : \lim_{n \rightarrow \infty} y_n = c \right\} \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = f(c).$$

Definition 59. Fix a subset $S \subseteq T$; if f is continuous at every $x \in S$ then f is said to be continuous on S . If $S = T$ then we say that f is continuous. The set of all continuous functions on S is denoted by $\mathbf{C}(S)$.

Proposition 10 (Composition). Let S and T be two non-empty subsets of $\mathbb{R}$. Take $f \in \mathbb{R}^T$ and $g \in \mathbb{R}^S$ with $f(T) \subseteq S$. If f is continuous at $x \in T$ and g is continuous at $f(x)$, then $g \circ f$ is continuous at x .

Proof. Fix $\varepsilon > 0$. Since g is continuous at $f(x)$, $\exists \alpha > 0$ st,

$$\forall y \in S \setminus \{f(x)\}, \quad |y - f(x)| < \alpha \implies |g(y) - g(f(x))| < \varepsilon.$$

Given $\alpha > 0$, since f is continuous at x , $\exists \delta > 0$ st,

$$\forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies |f(t) - f(x)| < \alpha.$$

Combining the two properties above, we get that

$$\forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies |g(f(t)) - g(f(x))| < \varepsilon,$$

which is the definition of $g \circ f$ is continuous at x . □

Proposition 11. *If f is continuous at x , then $|f|$ is also continuous at x .*

Theorem 2 (Intermediate Value Theorem). *If $f \in C[a, b]$ then,*

$$\forall x, x' \in [a, b], \quad \forall y \in \mathbb{R}, \quad f(x) < y < f(x') \implies \exists \hat{x} \in [a, b], \quad y = f(\hat{x})$$

Proof. Let $x, x' \in [a, b]$. Interchanging x and x' if necessary, we can assume wlog that $f(x) < f(x')$ and fix an arbitrary real number $y \in (f(x), f(x'))$. We split the proof in two cases: If $x < x'$. Let T be the set of all $t \in [x, x']$ such that $f(t) \leq y$, i.e.,

$$T := \{t \in [x, x'] : f(t) \leq y\}.$$

The set T is non-empty since it contains x and is bounded from above (by x'). Therefore, there exists $\hat{x} \in [x, x']$ such that $\hat{x} = \sup(T)$. By Lemma (5):

If $S \neq \emptyset$ and $S \subseteq \mathbb{R}$, then there is an increasing sequence $(x_n) \in S^{\mathbb{N}}$ such that $x_n \rightarrow \sup S$,

there exists a sequence (t_n) in T such that $\lim t_n = \hat{x}$. Since $f(t_n) \leq y$ for each n , passing to the limit and using the continuity of f at $\hat{x}$, we deduce that $f(\hat{x}) \leq y$. Observe that we must have $\hat{x} < x'$, since we have $f(\hat{x}) \leq y < f(x')$ and $x' \notin T$. Therefore, for $\varepsilon > 0$ small enough, $\hat{x} + \varepsilon$ still belongs to $[x, x']$. Since $\hat{x}$ is an upper bound of T , we cannot have $\hat{x} + \varepsilon \in T$. This implies that $y < f(\hat{x} + \varepsilon)$. Since this is true for any small enough value of ε , we can let $\varepsilon := 1/n$ for n large enough. Passing to the limit and using the continuity of f , we get that

$$y \leq \lim f(\hat{x} + 1/n) = f(\hat{x}).$$

We have thus proved that $y = f(\hat{x})$. If $x' < x$. Reproduce a similar argument by considering the set $T' := \{t \in [x', x] : f(t) \leq y\}$ and taking the infimum of T' . □

Definition 60. *f is said to be uniformly continuous on $S \subseteq T \subseteq \mathbb{R}$ if*

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x, t \in S, \quad |x - t| < \delta \implies |f(x) - f(t)| < \varepsilon.$$

Theorem 3 (Heine). Let $[a, b] \subseteq \mathbb{R}$ be a non-empty subset of $\mathbb{R}$. The function f is continuous on $[a, b]$ if, and only if, it is uniformly continuous on $[a, b]$.

Proof. We only have to prove the $[\implies]$ part. Let f be a function continuous on $[a, b]$ and assume, by way of contradiction, that f is not uniformly continuous. This means that

$$\exists \varepsilon > 0, \quad \forall \delta > 0, \quad \exists x, t \in [a, b], \quad |x - t| < \delta \quad \text{and} \quad |f(x) - f(t)| \geq \varepsilon.$$

Since the above property is true for any arbitrary $\delta > 0$, we can take $\delta := 1/n$ and deduce the existence of $x_n, t_n \in [a, b]$ such that

$$|x_n - t_n| < 1/n \quad \text{and} \quad |f(x_n) - f(t_n)| \geq \varepsilon.$$

Since the sequence (x_n) is bounded (recall $x_n \in [a, b]$), we can use the theorem

Every bounded real sequence has a convergent subsequence.

to deduce the existence of a strictly increasing function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ and some real number $x \in \mathbb{R}$ such that $(x_{\varphi(n)})$ converges to x . Since $x_n \in [a, b]$ for each n , it follows from the (3) that $x \in [a, b]$. Observe that the sequence $(t_{\varphi(n)})$ also converges to x since the sequence $(x_n - t_n)$ converges to 0. Since f and $|\cdot|$ are continuous functions, we get that

$$\varepsilon \leq \lim \left| f(x_{\varphi(n)}) - f(t_{\varphi(n)}) \right| = \left| \lim f(x_{\varphi(n)}) - \lim f(t_{\varphi(n)}) \right| = |f(x) - f(x)| = 0.$$

This contradicts the fact that $\varepsilon > 0$. □

Level Curves

Definition 61. The level curve $k \in \text{Im}(f)$ of a function f is given by,

$$\{(x, y) \in \mathbb{R}^2 : f(x, y) = k\}.$$

Concave and Convex Functions

Definition 62. A set $D \subseteq \mathbb{R}^n$ is said to be a **convex set** iff:

$$\forall x, y \in D, \quad \lambda \in [0, 1], \quad \lambda x + (1 - \lambda)y \in D$$

Definition 63. Let D be a convex set. $f \in \mathbb{R}^D$ is said to be concave if

$$\forall x, y \in D, \quad \lambda \in [0, 1], \quad f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$$

The definition of strictly concave functions uses strict inequalities. The definition of (strictly) convex functions just switches the inequalities sides.

Lemma 8. Fix $[a, b] \in \mathbb{R}$ and let $f \in \mathbb{R}^{[a, b]}$ be a concave function, then

$$\inf f([a, b]) > -\infty$$

Lemma 9. Let $f \in \mathbb{R}^I$ where I is an interval containing $a \leq b \leq c$. If f is concave, then

$$\frac{f(c) - f(b)}{c - b} \leq \frac{f(c) - f(a)}{c - a} \leq \frac{f(b) - f(a)}{b - a}$$

Definition 64. A function f on $D \subseteq \mathbb{R}^n$ is said to be quasi-concave if

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$$

The definition of a **strictly quasi-concave** function is the same with a strict inequality.

Definition 65. Let $f \in D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$. We define the upper contour set as follows:

$$U(f, \alpha) := \{x \in D : f(x) \geq \alpha\}$$

Proposition 12. Let $D \subseteq \mathbb{R}^n$ be a convex set. A function $f \in \mathbb{R}^D$ is quasi concave if, and only if, all of its upper contour sets are convex.

Proof. ($\implies$). Suppose that f is quasi-concave. Fix $\alpha \in \mathbb{R}$ such that $U(f, \alpha)$ is non-empty. Fix $x, y \in U(f, \alpha)$ and $\lambda \in [0, 1]$

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\} \geq \alpha$$

Therefore $\lambda x + (1 - \lambda)y \in U(f, \alpha)$ which means that the upper contour is convex.

($\impliedby$). Now suppose $U(f, \alpha)$ is a convex set for any $\alpha \in \mathbb{R}$. Then $\forall x, y \in U(f, \alpha)$ and $\lambda \in [0, 1]$ we have that $\lambda x + (1 - \lambda)y \in U(f, \alpha)$. Since this is true for any α , set $\alpha = \min\{f(x), f(y)\}$. Hence,

$$f(\lambda x + (1 - \lambda)y) \geq \alpha = \min\{f(x), f(y)\}$$

We then conclude that f is quasi-concave. □

Class 6

Metric Spaces

After getting some notions of sets, sequences, and functions we start to deal with more abstract spaces and more general properties. Let X be a non-empty set.

Definition 66. A function $d : X \times X \rightarrow \mathbb{R}_+$ that satisfies the following properties is called a **metric** on X , $\forall x, y \in X$

1. $d(x, y) = 0$ if, and only if, $x = y$;
2. $d(x, y) = d(y, x)$;
3. $d(x, y) \leq d(x, z) + d(y, z)$

Remark. Let $d(x, y) = |x - y|$, Pick $z = 0$ in the above definition. Which property (3) sounds like for you?

Example: Let $x, y \in X$ and let

$$d(x, y) := \begin{cases} 1, & \text{if } x \neq y \\ 0, & \text{if } x = y \end{cases}$$

Definition 67. A **metric space** is a pair (X, d) , that is, a non-empty set endowed with a metric function.

Example: Let Fix $n \in \mathbb{N}$ and $p \in [1, \infty)$. For any $x, y \in \mathbb{R}^n$. Define the following metric:

$$d_p(x, y) \equiv \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

Proving that d_p satisfies all the properties of a metric is left as an exercise. In particular, the property (3) in the definition of a metric (triangle inequality) when applied to d_p is called Minkowski's Inequality.

Definition 68. The space $(\mathbb{R}^n, d_2)$ is called the **n -dimensional Euclidean space**.

Example: Let ℓ^∞ denote the space of bounded real sequences and let $(\mathbf{B}(T))$ denote the space of bounded real functions from an arbitrary set T . We can define a distance between any two elements in the above sets by defining the *sup-metric* for each of those cases:

$$d_\infty((x_n), (y_n)) \equiv \sup\{|x_n - y_n| : n \in \mathbb{N}\}$$

$$D_\infty(f, g) \equiv \sup\{|f(x) - g(x)| : x \in T\}$$

Example: The pairs (ℓ^∞, d_∞) and $(\mathbf{B}(T), D_\infty)$ are metric spaces.

Topology

Let (X, d) be a metric space. We now use our definition of metric to formalize the notion of distance between elements of the set X .

Definition 69. For any $x \in X$ and $\epsilon > 0$ the **open ball centered at x with radius ϵ** is defined as follows:

$$B_X(x, \epsilon) \equiv \{y \in X : d(x, y) < \epsilon\}.$$

[Link: Different metrics induce different shapes of balls.](#)

The above definition allows us to say that elements of a set X are arbitrarily close (for a given metric) to each other when they are both elements of the open ball.

Definition 70. A subset $S \subseteq X$ is said to be **open in X** if

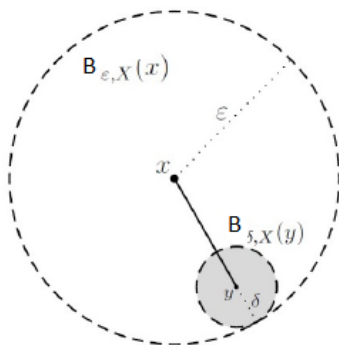
$$\forall x \in S, \quad \exists \epsilon > 0, \quad B_X(x, \epsilon) \subseteq S.$$

Lemma 10. The open ball $B_X(x, \epsilon)$ is open.

Proof. Fix $y \in B_X(x, \epsilon)$. We want to find $\delta > 0$ such that $B_X(y, \delta) \subseteq B_X(x, \epsilon)$. Let $\delta := \epsilon - d(x, y)$. By the triangle inequality:

$$\forall z \in B_X(y, \delta), \quad d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + \delta$$

□



Definition 71. A subset S of X is said to be **closed in X** , if $X \setminus S$ is open in X .

Remark: Sometimes the underlying metric space X will be implicit and we will just say that a set is *closed* (instead of *closed in X*)

Remark: We can define the **complement of S in X** or simply the **complement of S** (when X is implicit) as $S^c := X \setminus S$.

Definition 72. A **topology** associated to a metric d on X is the class of all open subsets of X , we denoted by $\mathcal{O}_X$.

Remind what a class was.

Proposition 13. Let $\mathcal{C} \subseteq \mathcal{O}_X$ be a class of open sets, then

$$\left[\bigcup_{S \in \mathcal{C}} S \right] \in \mathcal{O}_X$$

Proof. Define Y as follows:

$$Y := \bigcup \mathcal{C} = \bigcup_{C \in \mathcal{C}} C$$

Fix $y \in Y$. By definition:

$$\exists C \in \mathcal{C}, \quad y \in C$$

Since C is an open set.

$$\exists \epsilon > 0, \quad B_X(y, \epsilon) \subseteq C$$

But since $C \subseteq Y$, then $B_X(y, \epsilon) \subseteq Y$. Since this holds for any $y \in Y$ we must have that $Y \in \mathcal{O}_X$. $\square$

Remark: A direct consequence of the above proposition is that the intersection of any collection of closed sets is closed. However, it is **not** the case that an intersection of open sets is open (in the problem set you are invited to provide a counter-example of this claim).

Definition 73. The largest (in an inclusion sense) open set in X that is a subset of S is called **interior of S in X** and denoted by $\text{int}_X(S)$ (for short $\text{int}(S)$ when X is implicit). More formally:

$$\text{int}(S) := \bigcup \{V \in \mathcal{O}_X : V \subseteq S\}.$$

Definition 74. Let $\mathcal{C}_X$ denote the class of all closed sets in X .

Definition 75. The smallest (in an inclusion sense) closed set in X that contains S is called **closure of S in X** and denoted by $\text{cl}_X(S)$ (for short $\text{cl}(S)$ when X is implicit). More formally:

$$\text{cl}(S) := \bigcap \{C \in \mathcal{C}_X : S \subseteq C\}.$$

Definition 76. The boundary of S relative to X , denoted by $bd_X(S)$ (or $bd(S)$ for short, when X is implicit)

$$bd_X(S) := cl_X(S) \setminus int_X(S).$$

Proposition 14. Let $S \subseteq X$. S is open in $X \iff int(S) = S$.

Proof. ($\Leftarrow$). By definition, we know that $int(S)$ is an open set. If $int(S) = S$, then it must be the case that S is also open.

($\Rightarrow$). It is always the case that $int(S) \subseteq S$. By definition of the interior of S , we know it is the largest (for inclusion) open set contained in S , therefore it is also the case that $S \subseteq int(S)$. $\square$

Proposition 15. $x \in int(S) \iff \exists \epsilon > 0$ such that $B(x, \epsilon) \subset S$.

Proof. ($\Rightarrow$). Let $x \in int(S)$. Since $int(S)$ is open by definition we know $\exists \epsilon > 0$ such that $B(x, \epsilon) \subseteq int(S)$. Also, it is always the case that $int(S) \subset S$ ⁴. Hence we conclude that $B(x, \epsilon) \subset S$.

($\Leftarrow$). Assume there exists $\epsilon > 0$ such that $B(x, \epsilon) \subset S$. Since the open ball $B(x, \epsilon)$ is open by definition it must be the case that $B(x, \epsilon) \subset int(S)$, therefore $x \in int(S)$. $\square$

Limits

We re-write our notion of limit/convergence in terms of open balls now:

Definition 77. (x_n) converges to x if, and only if,

$$\forall \epsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad x_n \in B_X(x, \epsilon).$$

Proposition 16. A set S is closed in a metric space $(X, d) \iff$ Every sequence in S that converges in X has its limit point in S :

Proof. ($\Leftarrow$). Assume that every sequence that converges in X has its limit point in S . We shall prove that $X \setminus S$ is open. Assume by contradiction it is not the case, so there must exist $x \in X \setminus S$ such that

$$\nexists \epsilon > 0, \quad \text{such that } B_X(x, \epsilon) \subseteq X \setminus S$$

$$\forall \epsilon > 0, \quad B_X(x, \epsilon) \cap S \neq \emptyset$$

So for any $n \in \mathbb{N}$ we can choose $\epsilon = 1/n$ and since the intersection is non-empty, there exists $s_n \in B_X(x, 1/n) \cap S$. Note that $(s_n) \in S^{\mathbb{N}}$ is a sequence in S and it converges to x since

⁴ This inclusion holds with equality only if S is an open set.

$d(s_n, x) < 1/n$. But by assumption, all sequences that converge in X have their limit point in S , so $x \in S$ which is a contradiction since we fixed $x \in X \setminus S$.

($\implies$). We now assume S is closed and fix $(s_n) \in S^{\mathbb{N}}$ such that $s_n \rightarrow x \in X$. We want to show that $x \in S$. Suppose it is not the case, that is, $x \in X \setminus S$. By assumption, S is closed and then $X \setminus S$ is open. This means that:

$$\exists \epsilon > 0, \quad \text{such that } B_X(x, \epsilon) \subseteq X \setminus S$$

Since (s_n) converges to x :

$$\exists N \in \mathbb{N}, \quad \forall n \geq N, \quad s_n \in B_X(x, \epsilon)$$

But then it must be the case that $s_N \in B_X(x, \epsilon) \subseteq X \setminus S \implies s_N \in X \setminus S$ but this is a contradiction since we assumed $(s_n) \in S^{\mathbb{N}}$. $\square$

This is a very important characterization of closed sets. You will often use convergent sequences to prove a set is closed instead of using open balls.

Class 7

Compactness

Before stating formally one of the main concepts of a metric space, we start with some auxiliary definitions.

Definition 78. A subset S of X is said to be **bounded** if there exists $M > 0$ and $x \in S$ such that $S \subseteq B_X(x, M)$.

Definition 79. Let $\mathcal{C}$ be a class of open subsets of X . $\mathcal{C}$ is said to be an **open covering** of S if

$$S \subseteq \bigcup \{V : V \in \mathcal{C}\}.$$

Definition 80. A metric space X is said to be **compact** if every open cover of X has a finite subset that also covers X .

We can understand compactness as some sort of generalization of **finiteness**.

Example: $\{x_1, \dots, x_k\}$ for some $k \in \mathbb{N}$. Take $\epsilon := \max\{d(x_1, x_j) : j = 1, \dots, k\}$ and we have $\{x_1, \dots, x_k\} \subseteq B(x_1, \epsilon)$. But how to proceed if we have an infinite set? We cannot take $\epsilon > \sup\{d(x_1, x_j) : j = 1, \dots, k\}$ since this sup can be too big...

We now state an important result without proving it. Even though the proof is not very long, this result will be more important for us to have an alternative (and simpler) way to characterize compact sets without worrying about open coverings (which are very abstract objects).

Remark: Let $x, y \in \mathbb{R}^n$, we say that $x \leq y$ if $x_i \leq y_i$ for all $i = 1, \dots, n$.

Definition 81. Let $[x, y]$ be the set $\{z \in \mathbb{R}^n : x_i \leq z_i \leq y_i, \forall i = 1, \dots, n\}$.

Theorem 4 (Heine-Borel). For any $x \leq y \in \mathbb{R}^n$, the cube $[x, y]$ is compact.

Proposition 17. Any closed subset of a compact metric space is compact.

Proof. Let $F \subseteq X$ be a closed subset of the compact metric space X . Let $\mathcal{C}$ be an open covering of F , that is:

$$F \subseteq \bigcup \{V : V \in \mathcal{C}\}$$

If $\mathcal{C}$ is an open covering of F and F^c is also an open set we know that $\mathcal{C} \cup \{F^c\}$ is an open covering of X , which is compact. Therefore, there exist a finite family $\mathcal{D} \subseteq \mathcal{C}$ such that $\mathcal{D} \cup \{F^c\}$ covers X . We can then deduce that $\mathcal{D}$ covers F .

Draw the figure you have in your notes in the whiteboard! □

Proposition 18. *Any compact subset of a metric space X is closed and bounded*

Proof. Let K be a compact subset of a metric space X . We start showing that K is bounded. Let $\mathcal{C}$ be the class of open subsets of X defined as follows

$$\mathcal{C} := \{B(x, 1) : x \in K\}.$$

Note that this is an open covering of K . Therefore, since K is compact there exists a finite subset $F \subseteq K$ such that

$$K \subseteq \bigcup_{x \in F} B(x, 1)$$

Choose an arbitrary $\bar{x} \in F$ and define

$$M := \max\{d(x, \bar{x}) + 1 : x \in F\}$$

This definition of M ensures that $K \subseteq B(\bar{x}, M)$. We now proceed to the second step, which is proving that K is closed. Let $x \in K^c$ and define the following class

$$\mathcal{C} := \{B(k, d(x, k)/2) : k \in K\}$$

This is an open covering of K . Therefore, there exists a finite subset $F \subseteq K$ such that:

$$K \subseteq \bigcup_{f \in F} B(f, d(x, f)/2)$$

Let

$$\epsilon := \min\{d(x, f)/2 : f \in F\}.$$

Note that since $x \in K^c$ and $F \subseteq K$ we must have that $\epsilon > 0$. Moreover, by construction:

$$\forall f \in F, \quad B(x, \epsilon) \cap B(f, d(x, f)/2) = \emptyset.$$

Finally, note that

$$B(x, \epsilon) \subseteq \bigcap_{f \in F} B(f, d(x, f)/2)^c = \left[\bigcup_{f \in F} B(f, d(x, f)/2) \right]^c$$

But since the union of open balls covers K :

$$K \subseteq \bigcup_{f \in F} B(f, d(x, f)/2) \implies \left[\bigcup_{f \in F} B(f, d(x, f)/2) \right]^c \subseteq K^c$$

We thus conclude that

$$B(x, \epsilon) \subseteq K^c$$

□

Theorem 5. A subset of $\mathbb{R}^n$ (with the Euclidean metric) is compact if, and only if, it is closed and bounded.

Proof. We only prove the ($\Leftarrow$) part (the other direction is a problem set exercise). If K is bounded, there exists $k \in K$ and $M > 0$ such that:

$$K \subseteq \bigcap_{i=1}^n [k_i - M, k_i + M] = [x, y]$$

Where $x := (k_1 - M, k_2 - M, \dots, k_n - M)$ and $y := (k_1 + M, \dots, k_n + M)$. By Theorem (4) we already know that $[x, y]$ is compact so K is a closed subset of a compact set, by Proposition (17) we just proved, we know that K must also be compact. $\square$

The above results provide us with a simpler characterization of compact sets. In case we are dealing with finite vectors in the reals ($\mathbb{R}^n$) we can just substitute the notion of compactness with open coverages with way simpler notions of closed and bounded intervals.

We now state an important theorem that characterizes a compact set in terms of convergent sequences.

Theorem 6. A subset S of a metric space X is compact if, and only if, every sequence in S has a subsequence that converges to a point in S .

Proof. We only prove the ($\Rightarrow$). (For the other direction, refer to Efe Ok's book).

Let (x_n) be a sequence in S and assume by contradiction that there does not exist a converging subsequence.

Claim. The set $T := \{x_1, x_2, \dots, x_n, \dots\}$ is closed.

Proof. Let (y_n) be an arbitrary sequence in T and assume that it converges in X to some point ℓ . We shall prove that $\ell \in T$. For each $n \in \mathbb{N}$ there exists $k_n \in \mathbb{N}$ such that $y_n = x_{k_n}$. Observe that the function $n \mapsto k_n$ need not be injective. We claim that the set $\{x_{k_n} : n \in \mathbb{N}\}$ has finitely many elements. Otherwise, we can construct a subsequence of (y_n) that is also a subsequence of (x_n) . Indeed, let $\varphi(1) := \min\{k_n : n \in \mathbb{N}\}$ and $\theta(1)$ be such that $\varphi(1) = k_{\theta(1)}$. Let $\varphi(2) := \min\{k_n : k_n \geq \varphi(1) + 1 \text{ and } n \geq \theta(1) + 1\}$. This latter set must be non-empty by the assumption that $\{x_{k_n} : n \in \mathbb{N}\}$ takes infinitely many values. Proceeding by induction we can construct strictly increasing $\varphi, \theta \in \mathbb{N}^{\mathbb{N}}$ such that $\varphi(n) = k_{\theta(n)}$ for all $n \in \mathbb{N}$. This would imply that $y_{\theta(n)} = x_{\varphi(n)}$ is a subsequence of both y_n and x_n . This is a contradiction since there does not exist a converging subsequence of x_n . $\blacksquare$

Claim. $\forall n \in \mathbb{N}, \exists \epsilon_n > 0$ such that $B(x_n, \epsilon_n) \cap T = \{x_n\}$.

Proof. Fix $n \in \mathbb{N}$ and assume by contradiction

$$\forall \epsilon > 0, \quad B(x_n, \epsilon) \cap T \neq \{x_n\}$$

Choosing $\epsilon = 1$ we get the existence of $y_1 \in T$ with $y_1 \neq x_n$. Choosing $\epsilon = 1/2$, we get the existence of $y_2 \in T$ such that $y_2 \neq x_n$. By induction, we can construct a sequence (y_k)

that converges to x_n . By the previous claim, this sequence must take finitely many values. This is impossible since it converges to x_n and never coincides with x_n . ■
The set T is a closed subset of S . It is then compact. Observe that

$$\mathcal{C} := \{B(x_n, \epsilon_n) : n \in \mathbb{N}\}$$

is an open covering of T . Therefore, there must exist a finite sub-covering of T . This implies that at least one term of the sequence must be repeated infinitely often. In that case, we can extract a constant subsequence equal to this term, a contradiction. □

Completeness

We start by re-stating the definition of a Cauchy-sequence in terms of distances:

Definition 82. Let (X, d) be a metric space. A sequence $(x_n) \in X^{\mathbb{N}}$ is called a **Cauchy sequence** if:

$$\forall \epsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n, p \geq N, \quad d(x_n, x_p) < \epsilon.$$

Definition 83. A metric space X is said to be **complete** if every Cauchy sequence in X converges to a point in X .

Lemma 11. Let $T \neq \emptyset$ and $\mathbf{B}(T)$ be the space of bounded real functions from T to $\mathbb{R}$. $\mathbf{B}(T)$ is complete.

Proof. Let $f_n \in \mathbf{B}(T)^{\mathbb{N}}$ be a Cauchy sequence⁵

$$\forall \epsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n, p \geq N, \quad d_{\infty}(f_n, f_p) < \epsilon, \quad \sup\{|f_n(x) - f_p(x)| : x \in T\} < \epsilon$$

Fix $t \in T$. Since $|f_n(t) - f_p(t)| < \epsilon$ we get that $(f_n(t))$ is a Cauchy sequence. Since $\mathbb{R}$ is complete $(f_n(t))$ admits a limit. Let $f(t) := \lim f_n(t)$.

Claim. f is bounded⁶.

Proof. Let $p = N$.

$$|f_n(t) - f_N(t)| < \epsilon \implies \forall n \geq N, \quad f_N(t) - \epsilon \leq f_n(t) \leq f_N(t) + \epsilon$$

By the triangle inequality:

$$\epsilon > |f_n(t) - f_N(t)| \geq |f_n(t)| - |f_N(t)| \implies |f_n(t)| \leq |f_N(t)| + \epsilon$$

Passing to the limit when $n \rightarrow \infty$ and using the fact that $|\cdot|$ is continuous we get that:

$$|f(t)| \leq \epsilon + |f_N(t)|$$

⁵ This is a sequence of bounded functions from T to $\mathbb{R}$. Since this is a very abstract object we will use the fact that the real numbers are a complete metric space and work with the **images** of the functions.

⁶ We are being very picky here... We picked a sequence on bounded functions and passed it to the limit for a given $t \in T$. Now we want to show that this limiting function is also bounded.

$$\sup_{t \in T} |f(t)| \leq \epsilon + \sup_{t \in T} |f_N(t)| \quad \blacksquare$$

Claim. $f_n \rightarrow f$ in the sup metric.

Proof. Since we have that $|f_n(t) - f_p(t)| < \epsilon$:

$$\forall n, p \geq N, \quad \forall t \in T, \quad |f_n(t) - f_p(t)| \geq d_\infty(f_n, f_p) < \epsilon$$

Passing to the limit when $p \rightarrow \infty$ we have:

$$\forall n \geq N, \quad \forall t \in T, \quad |f_n(t) - f(t)| < \epsilon$$

$$\forall n \geq N, \quad d_\infty(f_n, f) := \sup_{t \in T} |f_n(t) - f(t)| < \epsilon \quad \blacksquare$$

In the final step of the above proof we just used the fact that since a property is valid for **any** $t \in T$, it must be valid for the sup. □

Class 8

Continuity

Definition 84. $f : X \rightarrow Y$ is **continuous at x** if, and only if,

$$\forall \epsilon > 0, \quad \exists \delta > 0, \quad f(B_X(x, \delta)) \subseteq B_Y(f(x), \epsilon)$$

or equivalently,

$$\forall \epsilon > 0, \quad \exists \delta > 0, \quad B_X(x, \delta) \subseteq f^{-1}(B_Y(f(x), \epsilon))$$

Proposition 19. A function $f : X \rightarrow Y$ is continuous at x if, and only, if

$$\forall V \in \mathcal{O}_Y, \quad f^{-1}(V) \in \mathcal{O}_X.$$

Proof. ($\implies$). Assume f is continuous and let V be an open subset of Y . We shall prove that $f^{-1}(V)$ is open. Fix $x \in f^{-1}(V)$. It must be the case that there exists $y \in V$ such that $y = f(x)$. Since V is open, $\exists \epsilon > 0$ such that $B_Y(y, \epsilon) \subseteq V$. Since f is continuous, $\exists \delta > 0$ such that $B_X(x, \delta) \subseteq f^{-1}(B_Y(f(x), \epsilon))$. We just proved that an open ball centered in an arbitrary element of the counter-domain is a subset of the pre-image $f^{-1}(V)$. Since x was arbitrary, we got the desired result.

Proof of ($\impliedby$). Assume that $f^{-1}(V)$ is open in X for any $V \in Y$. Fix $x \in X$ and $\epsilon > 0$. By definition $B_Y(f(x), \epsilon) \in \mathcal{O}_Y$. Since $f^{-1}(V) \in \mathcal{O}_X$ we must have that $f^{-1}(B_Y(f(x), \epsilon)) \in \mathcal{O}_X$. Note that $x \in f^{-1}(B_Y(f(x), \epsilon))$ and $f^{-1}(B_Y(f(x), \epsilon))$ is open $\exists \delta > 0$ such that $B_X(x, \delta) \subseteq f^{-1}(B_Y(f(x), \epsilon))$. $\square$

Theorem 7. Let $f : X \rightarrow Y$ be a continuous function. If S is a compact subset of X , then $f(S)$ is a compact subset of Y .

Proof. Let $\mathcal{C}$ be an open covering of $f(S)$. Denote by $\mathcal{D}$ the following class:

$$\mathcal{D} := \{f^{-1}(V) : V \in \mathcal{C}\}$$

Note that $\mathcal{D}$ is a class of open sets (since f is continuous) that covers S . Since S is compact, there must exist a finite $\mathcal{D}_f \subseteq \mathcal{D}$ that covers S . Since each element of $\mathcal{D}$ is the pre-image of some element in $\mathcal{C}$, we can write $\mathcal{D}_f$ such as:

$$\mathcal{D}_f := \{f^{-1}(V) : V \in \mathcal{C}_f\}$$

For some finite $\mathcal{C}_f \subseteq \mathcal{C}$. Since:

$$S \subseteq \bigcup_{C \in \mathcal{C}_f} f^{-1}(C)$$

We conclude:

$$f(S) \subseteq \bigcup_{C \in \mathcal{C}_f} C = \bigcup \mathcal{C}_f$$

$\square$

We just showed that when a function is said to be continuous sets in the domain and in the image preserve some topological properties (pre-image of open/closed sets will also be open/closed). We now introduce a characterization of a continuous function using sequences.

Proposition 20. $f : X \rightarrow Y$ is continuous if, and only if,

$$\forall x \in X, \quad (x_n) \in X^{\mathbb{N}}, \quad \lim x_n = x \implies \lim f(x_n) = f(x).$$

Proof. ($\implies$). Fix $x \in X$ and $(x_n) \in (X)^{\mathbb{N}}$ such that $\lim x_n = x$. We claim that $\lim f(x_n) = f(x)$. Fix $\epsilon > 0$. Since f is a continuous function we must have that $\exists \delta > 0$ such that $B_X(x, \delta) \subseteq f^{-1}(B_Y(f(x), \epsilon))$. Since $x_n \rightarrow x$ we know that there exists $N \in \mathbb{N}$ such that $\forall n \geq N, x_n \in B_X(x, \delta)$. This implies that $f(x_n) \in B_Y(f(x), \epsilon)$ by continuity. We have thus proved that $\lim f(x_n) = f(x)$.

Proof of ($\impliedby$). Assume by contradiction that $\exists x \in X$ such that f is **not** continuous at x .

$$\exists \epsilon > 0, \quad \forall \delta > 0, \quad f(B_X(x, \delta)) \not\subseteq B_Y(f(x), \epsilon)$$

Take $\delta := 1/n$ and we get the existence of $x_n \in B_X(x, 1/n)$. Such that $f(x_n) \notin B_Y(f(x), \epsilon)$. However, $x_n \rightarrow x$ and by assumption $f(x_n) \rightarrow f(x)$, which is a contradiction. $\square$

Theorem 8 (Weierstrass Theorem). *If X is a compact metric space and $f : X \rightarrow \mathbb{R}$ is continuous, then there exists $x, y \in X$ such that*

$$f(x) = \sup f(X) \quad \text{and} \quad f(y) = \inf f(X)$$

Proof. Note that since $f(X) \subseteq \mathbb{R}$ and $f(X) \neq \emptyset$, by Lemma (5) we know that $\exists (y_n) \in [f(X)]^{\mathbb{N}}$ such that $\lim y_n = \sup f(X)$. For each $n \in \mathbb{N}$, there exists $x_n \in X$ such that $y_n = f(x_n)$. Since X is compact, there exists $\psi \in \mathbb{N}^{\mathbb{N}}$ strictly increasing such that $\lim x_{\psi(n)} = x$. Since f is continuous, we can thus conclude that $\lim f(x_{\psi(n)}) = f(x)$:

$$\sup f(X) = \lim y_n = \lim y_{\psi(n)} = \lim f(x_{\psi(n)}) = f(x)$$

$\square$

This is a very important result. As we will see in the optimization problem (and during your first year) most economic models are written as optimization problems. Typically, some version of the Weierstrass theorem will ensure the existence of a solution. We now provide some examples to illustrate the importance of the assumptions of the theorem.

Example. Let $\mathcal{D} := \mathbb{R}_{++} \subseteq \mathbb{R}$ and define $f : \mathcal{D} \rightarrow \mathbb{R}$ as $f(x) = x$. Even though f is a continuous function the domain is clearly not compact since the positive real numbers are unbounded. Hence, the supremum is not well defined in $\mathbb{R}$.

Example. Let $\mathcal{D} := (0, 1) \subseteq \mathbb{R}$. Note that f is continuous and $\mathcal{D}$ is an open interval, so it is not compact. The supremum is not an element of $\mathcal{D}$: $\forall x \in \mathcal{D}$, there exists $x^* \in \mathcal{D}$ such that $f(x^*) > f(x)$.

Example. Now consider $\mathcal{D} := [0, 1] \subseteq \mathbb{R}$ a compact set, and define $f(x) = x\mathbb{1}_{x \in (0, 1)}$. This function is not a continuous function in one point of the domain.

Fixed Points

Definition 85. For a given function $f \in \mathbb{R}^X$ a fixed point x^* is such that $f(x^*) = x^*$.

In more general spaces, if a map maps some point to itself, this point is called a fixed point. This is a very important object and is frequently used in numerous applications in economic models.

In this section, we start with some simpler versions of fixed-point theorems and provide useful examples building on top of it.

Brouwer Fixed Point Here we state and prove a simple version of Brouwer's fixed point in the reals.

Theorem 9 (Baby Brouwer Fixed Point). Let $[a, b] \subseteq \mathbb{R}$ be a closed interval and $f : [a, b] \rightarrow [a, b]$ be a continuous function. Then, f has a fixed point.

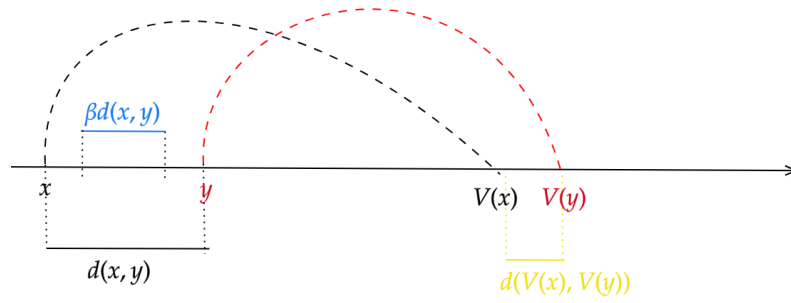
Proof. Define g as $g(x) := f(x) - x$. By continuity of f , g is continuous. Note that since $f(a) \in [a, b]$ we must have that $g(a) \geq 0$. If $g(a) = 0$ we already have a fixed point, so assume it is not the case: $g(a) > 0$. For the same argument, we must have that $g(b) < 0$. Since g is continuous, $\exists x^* \in [a, b]$ such that $g(x^*) = 0$, or $f(x^*) = x^*$. $\square$

There are other ways of proving more general versions of Brouwer's fixed point, but for us to do that we would have to cover Sperner's Lemma (for example), which is not in the scope of this course.

Banach Fixed Point This may be one of the most general and useful theorems to be used during your first year and is used to prove the convergence of the value function in dynamic programming problems, largely used in modern macro and IO models.

Definition 86. $V : X \rightarrow X$ is said to be a **contraction** if $\exists \beta \in (0, 1)$ such that:

$$\forall x, y \in X, \quad d(V(x), V(y)) \leq \beta d(x, y)$$



To gain some intuition over this definition consider the case of real functions $f \in \mathbb{R}^{\mathbb{R}}$ and assume f to be a contraction mapping. We want to take some element $x \in \mathbb{R}$ and observe the behavior of the sequence $(x, f(x), f(f(x)), \dots)$. The contraction property tells us that this sequence happens to be a Cauchy sequence, and as we already proved, Cauchy sequences converge in complete metric spaces (which is the case of the real numbers).

Theorem 10 (Banach Fixed Point). *If $V : X \rightarrow X$ is a contraction on a complete metric space, there exists a unique $x^* \in X$ such that $V(x^*) = x^*$.*

Proof. Fix $x \in X$ and let (y_n) be defined by:

$$y_1 := V(x) \quad \text{and} \quad y_{n+1} := V(y_n)$$

Observe that $\forall n \in \mathbb{N}$:

$$y_n = V^n(x) = \underbrace{[V \circ V \circ \dots \circ V]}_{n\text{-times}}(x) :$$

We claim that (y_n) is a Cauchy sequence. Indeed, let $p > n$:

$$\begin{aligned} d(V^p(x), V^n(x)) &\leq d(V^p(x), V^{p-1}(x)) + \dots + d(V^{n+1}(x), V^n(x)) \\ &\leq \beta^{p-1} d(V(x), x) + \dots + \beta^n d(V(x), x) \\ &\leq (\beta^{p-1} + \dots + \beta^n) d(V(x), x) \\ &\leq \left[\sum_{t=n}^{\infty} \beta^t \right] \times d(V(x), x) \end{aligned}$$

Note that since $\beta \in (0, 1)$ the term in the sum converges to zero and (y_n) is Cauchy. Since X is a complete metric space we know that $y_n \rightarrow x^* \in X$ (this will be our fixed point). Then

for any ϵ there must exist $N \in \mathbb{N}$ such that $d(x^*, y_n) < \frac{\epsilon}{2}$ for any $n \geq N$ and hence

$$\begin{aligned} d(V(x^*), x^*) &\leq d(V(x^*), x_{n+1}) + d(y_{n+1}, x^*) \\ &= d(V(x^*), V(y_n)) \\ &\leq \beta d(x^*, y_n) + d(y_{n+1}, x^*) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \end{aligned}$$

We just proved the existence of the fixed point of a contraction and now proceed to prove uniqueness. Assume that another $\exists \tilde{x} \in X$ such that $V(\tilde{x}) = \tilde{x}$. Since V is a contraction mapping:

$$d(x^*, \tilde{x}) = d(V(x^*), V(\tilde{x})) \leq \beta d(x^*, \tilde{x})$$

This is true if, and only if, $d(x, \tilde{x}) = 0$. □

This is a significant result since **any** contraction in a complete metric space has a unique fixed point. According to Proposition (11) bounded functions are a complete metric space. Hence, a contraction on the space of bounded real functions has a unique fixed point.

Class 9

Problem Set 2

Real Functions

Exercise 1. Prove that if f is continuous at x , then $|f|$ is also continuous at x .

Proof. We need to show that

$$\forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies \|f(t) - f(x)\| < \varepsilon$$

which is exactly the definition of $|f|$ is continuous at x . We know that since f is continuous $\exists \delta > 0$ st,

$$\forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies |f(t) - f(x)| < \varepsilon$$

By the triangle inequality

$$\forall t \in T \setminus \{x\}, \quad |t - x| < \delta \implies \|f(t) - f(x)\| < |f(t) - f(x)|.$$

□

Exercise 2. Take any $-\infty < a < b < \infty$ and let $f \in \mathbb{R}^{[a,b]}$. Prove f is a continuous bijection, then f^{-1} is a continuous bijection defined on $f([a, b])$.

Proof. Proof. Recall that $f^{-1} : f([a, b]) \rightarrow [a, b]$ is the function satisfying

$$f^{-1} \circ f = \text{Id}_{[a,b]} \quad \text{and} \quad f \circ f^{-1} = \text{Id}_{f([a,b])}.$$

In particular, f^{-1} is also a bijection. Fix an arbitrary $y \in f([a, b])$. We want to show that f^{-1} is continuous at y . Since f is bijective, there exists a unique $x \in [a, b]$ such that $y = f(x)$. Let (y_n) be a sequence in $f([a, b]) \setminus \{f(x)\}$ that converges to y . Again, since f is bijective, for each n , there exists a unique $x_n \in [a, b]$ such that $y_n = f(x_n)$. Observe that $x_n \neq x$ since $y_n \neq f(x)$. We want to show that $(f^{-1}(y_n))$ converges to $f^{-1}(y)$, or, equivalently, that (x_n) converges to x . We propose to use Lemma which states that a sequence converges if and only if any subsequence has itself a subsequence that converges. Fix an arbitrary strictly increasing function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$. The sequence $(x_{\varphi(n)})$ admits a converging subsequence since it belongs to the compact interval $[a, b]$. Therefore there exists $\psi : \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing and real number $x(\varphi)$ such that

$$\lim_{n \rightarrow \infty} x_{\varphi \circ \psi(n)} = x(\varphi).$$

Since f is continuous, we have that

$$\lim_{n \rightarrow \infty} f(x_{\varphi \circ \psi(n)}) = f(x(\varphi)).$$

Since $f(x_{\varphi \circ \psi(n)}) = y_{\varphi \circ \psi(n)}$ and (y_n) converges to y , we deduce that $f(x(\varphi)) = y = f(x)$. Since f is injective, this implies that $x(\varphi) = x$ is independent of the choice of φ . We can deduce from Lemma (6) that (x_n) converges to x , or, equivalently, that $(f^{-1}(y_n))$ converges to $f^{-1}(y)$. □

Exercise 3. Is $f(x) = 1/x$ uniformly continuous? Prove or Disprove.

Proof. Is is not uniformly continuous if we take consider the interval $(0, \infty)$. It is enough to show that,

$$\exists \epsilon > 0 \text{ st } \forall \delta > 0, \quad \exists x, y \in (0, \infty) \text{ s.t. } |x - y| < \delta \text{ but } |f(x) - f(y)| \geq \epsilon.$$

Chose $\epsilon = 1$ and let $y = \frac{x}{2}$. Then, $\exists x$ st,

$$|x - y| = \frac{x}{2} < \delta, \quad |f(x) - f(y)| = \frac{1}{x} \geq 1.$$

□

Exercise 4. Prove Lemma (8)

Proof. Fix some $x \in [a, b]$. Since $[a, b]$ is a convex set, there exists some $\lambda \in [0, 1]$ such that $x = \lambda a + (1 - \lambda)b$. Since f is concave:

$$f(x) = f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b) \geq \min\{f(a), f(b)\}$$

We just proved that $m := \min\{f(a), f(b)\}$ is a lower bound of the set $f([a, b])$ which means that $\inf f([a, b]) > -\infty$. □

Exercise 5. Let $[a, b] \in \mathbb{R}$ and $f \in \mathbb{R}^{[a, b]}$ be a concave function. Prove that

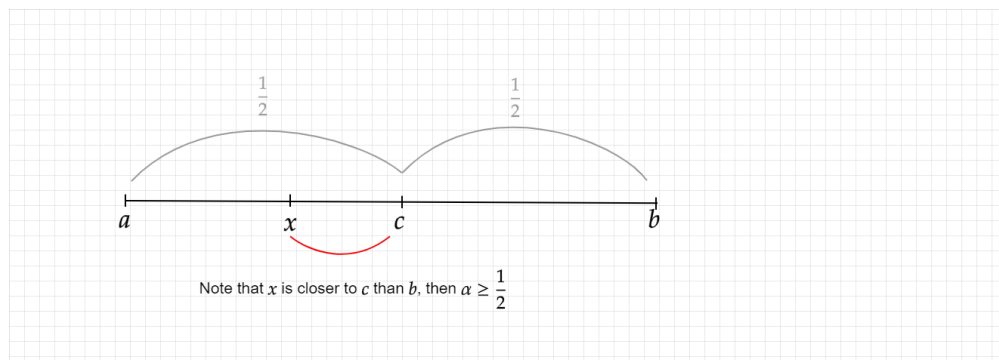
$$\sup f([a, b]) < \infty$$

Proof. Recall that we have already proved that $m := \min\{f(a), f(b)\}$ is a lower bound of the set $f([a, b])$. Denote $c := \frac{a+b}{2}$.

Fix $x \in [a, c]$ and note that $c \in [x, b]$. Therefore, there must exist $\alpha \in [0, 1]$ such that c can be written as:

$$c = \alpha x + (1 - \alpha)b$$

Note that since c is the mid-point of $[a, b]$ and $x \in [a, c]$ we must have that $\alpha \geq 1/2$:



By concavity of f :

$$f(c) - m \geq \alpha[f(x) - m] + (1 - \alpha)[f(b) - m] \geq \frac{1}{2}[f(x) - m] \implies f(x) \leq 2f(c) - m$$

Now fix $y \in [c, b]$ and note that $c \in [a, y]$. Again there must be the case that $\exists \alpha \in [0, 1]$ such that:

$$c = (1 - \alpha)a + \alpha y$$

By the same argument as before:

$$f(c) - m \geq (1 - \alpha)[f(a) - m] + \alpha[f(y) - m] \geq \frac{1}{2}[f(x) - m] \implies f(y) \leq 2f(c) - m$$

We have thus proved that $M := 2f(c) - m$ is an upper bound of $f([a, b])$ which implies that $\sup f([a, b]) < \infty$. $\square$

Exercise 6. Prove the Lemma (9).

Proof. Note that we can write

$$b = \alpha a + (1 - \alpha)c$$

For some $\alpha \in [0, 1]$, or analogously:

$$\alpha = \frac{c - b}{c - a}$$

By concavity of f :

$$f(b) \geq \alpha f(a) + (1 - \alpha)f(c)$$

$$f(b) - f(a) \geq (1 - \alpha)[f(c) - f(a)]$$

$$f(b) - f(c) \geq -\alpha[f(c) - f(a)]$$

But since $1 - \alpha = (b - a)/(c - a)$:

$$\frac{f(b) - f(a)}{b - a} \geq \frac{f(c) - f(a)}{c - a} \quad \text{and} \quad \frac{f(c) - f(b)}{c - b} \leq \frac{f(c) - f(a)}{c - a}.$$

$\square$

Exercise 7. Let I be an open interval and $f \in \mathbb{R}^I$ be a concave function. Prove that for every $[a, b] \subseteq I$, there exists $K > 0$ such that:

$$\forall x, y \in [a, b], \quad |f(x) - f(y)| \leq K|x - y|$$

A function that satisfies this property is said to be **locally Lipschitz**.

Hint. Use Lemma (9)

Proof. Since I is an open interval there exists $\epsilon > 0$ such that $[a - \epsilon, b + \epsilon] \subseteq I$. Let $x, y \in [a, b]$ and assume without loss of generality that $x \leq y$: $a - \epsilon \leq a \leq x \leq y$, we can thus apply Lemma (9) twice to get:

$$\begin{aligned} \frac{f(y) - f(x)}{y - x} &\leq \frac{f(x) - f(a)}{x - a} \leq \frac{f(a) - f(a - \epsilon)}{\epsilon} \\ \frac{f(b + \epsilon) - f(b)}{\epsilon} &\leq \frac{f(b) - f(y)}{b - y} \leq \frac{f(y) - f(x)}{y - x} \end{aligned}$$

We thus conclude that:

$$\begin{aligned} \left| \frac{f(y) - f(x)}{y - x} \right| &\leq \max \left\{ \left| \frac{f(a) - f(a - \epsilon)}{\epsilon} \right|, \left| \frac{f(b + \epsilon) - f(b)}{\epsilon} \right| \right\} \\ &\quad \underbrace{\hspace{10em}}_{=:K} \\ |f(y) - f(x)| &\leq K|y - x| \end{aligned}$$

□

Exercise 8. Let I be an open interval and $f \in \mathbb{R}^I$. Show that if f is concave, then it is continuous.

Hint. Use **Exercise 7**

Proof. Since I is an open interval, there exist a and b such that $x \in (a, b)$. By **Exercise 6**, there exists $K > 0$ such that:

$$\forall x, y \in [a, b], \quad |f(x) - f(y)| \leq K|x - y|$$

Fix $\epsilon > 0$. Choose $\delta := \epsilon/K$, then $\epsilon = K\delta$. Therefore:

$$|x - y| < \delta \implies |f(x) - f(y)| < K\delta = \epsilon$$

□

Metric Spaces

Exercise 9. Define the closed ball as follows:

$$\overline{B}_X(x, \epsilon) := \{y \in X : d(x, y) \leq \epsilon\}$$

Prove that the closed ball $\overline{B}_X(x, \epsilon)$ is closed.

Proof. Let $Y := X \setminus \bar{B}_X(x, \epsilon)$. Fix $y \in Y$. Since $y \notin \bar{B}_X(x, \epsilon)$ we have that $d(x, y) > \epsilon$. Let $\alpha := d(x, y) - \epsilon$. We shall prove that $B_X(y, \alpha) \subseteq Y$. Let $z \in B_X(y, \alpha)$:

$$d(x, y) \leq d(x, z) + d(z, y)$$

This means that:

$$d(x, z) \geq d(x, y) - d(z, y) > d(x, y) - \alpha = \epsilon \implies z \notin \bar{B}_X(x, \epsilon)$$

□

Exercise 10. Prove that in the discrete metric any subset is closed **and** open.

Proof. Let $S \subseteq X$. We claim that S is open. Fix $x \in S$, pick $\epsilon = 1$, then $B(x, 1) := \{x\} \subseteq S$, then S is open. Now we claim that $X \setminus S$ is open. Fix $y \in X \setminus S$ and note that $B(y, 1) := \{y\} \subseteq X \setminus S$, which implies that $X \setminus S$ is open and then S is closed. □

Exercise 11. Provide a counter-example in which the intersection of open sets is **not** open.

Proof. Take $X = \mathbb{R}$ and let:

$$\mathcal{C} := \{(-1/n, 1 + 1/n) : n \in \mathbb{N}\} \implies \bigcap \mathcal{C} = [0, 1].$$

□

Exercise 12. Prove that S is closed in $X \iff cl(S) = S$.

Proof. This is very similar to Proposition (14) but adapted to the definition of the closure (instead of the interior). □

Exercise 13. Prove that $x \in cl(S) \iff \forall \epsilon > 0$ we have $B(x, \epsilon) \cap S \neq \emptyset$

Proof. ($\implies$). Let $x \in cl(S)$. Fix $\epsilon > 0$. If $B(x, \epsilon) \cap S = \emptyset$ then $S \subseteq X \setminus B(x, \epsilon)$. Since $X \setminus B(x, \epsilon)$ is a closed set, it must be the case that $cl(S) \subseteq X \setminus B(x, \epsilon)$. But since $x \in cl(S)$ we have that $x \in X \setminus B(x, \epsilon)$ which means that $x \notin B(x, \epsilon)$ which is a contradiction. We thus conclude that $B(x, \epsilon) \cap S \neq \emptyset$.

($\impliedby$). Assume that for any $\epsilon > 0$ we have $B(x, \epsilon) \cap S \neq \emptyset$. If $x \notin cl(S)$ then $x \in X \setminus cl(S)$ (which is an open set). Therefore, $\exists \epsilon > 0$ such that $B(x, \epsilon) \subseteq X \setminus cl(S)$. In particular, we have that $B(x, \epsilon) \cap cl(S) = \emptyset$. This is a contradiction since $S \subseteq cl(S)$. □

Exercise 14. Prove that

$$x \in bd(S) \iff \forall \epsilon > 0, B(x, \epsilon) \cap S \neq \emptyset \quad \text{and} \quad B(x, \epsilon) \cap X \setminus S \neq \emptyset$$

Proof. Use **Exercise (5)** with Proposition (15) □

Exercise 15. For any subset S of a metric space X . Prove that the following statements are equivalent.

1. $x \in cl(S)$
2. Every open neighborhood of x intersects S
3. There exists a sequence in S that converges to x

Proof. (1) $\implies$ (2). Let V be an open neighborhood of $x \in X$. This means that V is open and $x \in V$. Suppose by contradiction that $V \cap S = \emptyset$. This means that $S \subseteq X \setminus V$. Note that the set $X \setminus V$ is closed and by definition $cl(S) \subseteq S$, therefore $cl(S) \subseteq X \setminus V$. But since $x \in cl(S)$ it must be the case that $x \in X \setminus V$, which is a contradiction since we assumed $x \in V$.

(2) $\implies$ (3). For any $n \in \mathbb{N}$, the open ball $B_X(1, 1/n)$ is an open neighborhood of $x \in X$. It follows from (2) that $\exists s_n \in B_X(1, 1/n) \cap S$. The sequence $(s_n) \in S^{\mathbb{N}}$ converges to x .

(3) $\implies$ (1). Assume that $\exists (s_n) \in S^{\mathbb{N}}$ that converges to x . Assume by contradiction that $x \in V := X \setminus cl(S)$. Since $cl(S)$ is closed V is open, then $\exists \epsilon > 0$ such that $B(x, \epsilon) \subseteq V$. But since $s_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that $\forall n \geq N, s_n \in B(x, \epsilon)$. In particular, we have that $s_N \in S \cap V$. We thus conclude that $s_N \in S$ and $s_N \notin cl(S)$, which is a contradiction. □

Exercise 16. Prove **using sequences** that if $f : X \rightarrow Y$ is continuous and $S \subseteq X$ is a compact set, then $f(S) \subseteq Y$ is a compact set.

Proof. To prove that $f(S)$ is compact we show that any sequence in $f(S)$ admits a subsequence that converges in $f(S)$. Let $(y_n) \in f(S)^{\mathbb{N}}$. For any $n \in \mathbb{N}$ there exists $s_n \in S$ such that $y_n = f(s_n)$. Since S is compact, $\exists \varphi \in \mathbb{N}^{\mathbb{N}}$ strictly increasing, and $s \in S$ such that $\lim s_{\varphi(n)} = s$. Since f is continuous we deduce that $\lim f(s_{\varphi(n)}) = f(s)$. □

Exercise 17. Let $S \neq \emptyset$ be a closed subset of $\mathbb{R}^n$. Fix $x \notin S$. Prove that there exists $y_x \in S$ such that:

$$d(x, S) := \inf\{d(x, y) : y \in S\} = d(x, y_x)$$

*Hint: the set S is **not** necessarily compact! You have to “create” bounds for it...*

Proof. Fix $\bar{y} \in S$ and pose $T := \{y \in S : d(x, y) \leq d(x, \bar{y})\}$. We have that $\inf\{d(x, y) : y \in S\} = \inf\{d(x, y) : y \in S \cap T\}$. We now apply the Weierstrass Theorem for metric spaces and get the existence of y_x . $\square$

Class10

Matrices: Basic definitions

Definition 87. An $(n \times k)$ matrix A is a rectangular array of objects,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & \cdots & a_{nk} \end{pmatrix}_{n \times k}$$

where n is the number of rows and k is the number of columns.

The entry in the i^{th} row and j^{th} column of a matrix A is sometimes referred to as the (i, j) entry of the matrix, and commonly denoted by a_{ij} . Alternative notations for that entry are $A[i, j]$ and $A_{i,j}$.

Definition 88. Two matrices are equal, if and only if,

$$A = B \Leftrightarrow a_{i,j} = b_{i,j} \quad \forall i, j.$$

Definition 89. Let A be an $n \times m$ matrix and A^T be an $m \times n$ matrix. We say that A^T is the transpose of A , if and only if,

$$a_{i,j} = a_{j,i}^T \quad \forall i, j.$$

Definition 90. The sum $(A + B)$ of two $m \times n$ matrices A and B is calculated entry-wise:

$$(A + B)_{i,j} = A_{i,j} + B_{i,j}, \text{ where } 1 \leq i \leq m \text{ and } 1 \leq j \leq n.$$

Definition 91. The product (cA) of a number $c \in \mathbb{R}$, and a matrix A is computed by multiplying every entry of A by c :

$$(cA)_{i,j} = c \cdot A_{i,j}$$

This operation is called scalar multiplication.

Definition 92. If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then their matrix product (AB) is the $m \times p$ matrix whose entries are given by,

$$(AB)_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j} = \sum_{r=1}^n a_{i,r}b_{r,j}$$

Proposition 21. Let A and B be matrices and c be scalar. Then the following properties hold:

1. The operation of taking the transpose is an involution

$$(A^T)^T = A.$$

2. The transpose respects addition

$$(A + B)^T = A^T + B^T$$

3. The order of the factors reverses

$$(AB)^T = B^T A^T$$

Proposition 22. Whenever the size of the matrices is such that the various products are defined, matrix multiplication satisfies:

$$(AB)C = A(BC)$$

$$(A + B)C = AC + BC$$

$$C(A + B) = CA + CB$$

Definition 93. A square matrix is a matrix that has an equal number of rows and columns.

Definition 94. The identity matrix, denoted by I , is a square matrix with ones on the main diagonal and zeros elsewhere.

The $n \times n$ identity matrix I is denoted as:

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

Definition 95. A square matrix A is symmetric, if and only if, $A = A^T$.

Definition 96. A square matrix B is idempotent, if and only, if

$$B = B^2 = B^3 \dots$$

Linear systems

Linear algebra is the branch of mathematics concerning linear equations such as:

$$a_1 x_1 + \cdots + a_n x_n = b.$$

A general linear system of m equations in n unknowns can be written as:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned} \tag{2}$$

In (2) the a_{ij} and b_i are given real numbers. We say that a_{ij} is the coefficient of the unknown x_j in the i^{th} equation.

Definition 97. A solution to (2) is an n -tuple of real numbers $(x_1, x_2, \dots, x_n)$, which satisfies each of the m equations.

For a linear system such as the one above, we are interested in answering the following three questions:

1. Does a solution exist?
2. How many solutions are there?
3. Is there an efficient algorithm that computes actual solutions?

It makes sense to simplify the representation of a linear system by writing two matrices,

Definition 98. The coefficient matrix associated with the linear system of equations (2) is given by,

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Definition 99. The augmented matrix associated with linear system of equations (2) is given by,

$$\hat{A} = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right)$$

Proposition 23. There are three types of elementary row operations that may be performed on the rows of the augmented matrix that do not change the solution set to (2):

1. Swap the positions of two rows.
2. Multiply a row by a non-zero scalar.
3. Add to one row a scalar multiple of another.

Therefore, if one's goal is to solve a system of linear equations, then using these row operations could make the problem easier.

Definition 100. A row of a matrix is said to have k leading zeros if the first k elements of the row are all zeros and the $(k + 1)$ element of the row is not zero.

Definition 101. A matrix is in row echelon form if each row has more leading zeros than the row preceding it.

Definition 102. The first nonzero entry in each row of a matrix in row echelon form is called a pivot.

Definition 103. A row echelon matrix in which each pivot is a 1 and in which each column containing a pivot contains no other nonzero entries are said to be in reduced row echelon form.

Definition 104. We call the Gauss-Jordan Elimination the process of using the three elementary row operations to convert a matrix into reduced-row echelon form.

Definition 105. We say a row of a matrix is nonzero if and only if it contains at least one nonzero entry.

Definition 106. The rank of a matrix is the number of nonzero rows in its row echelon form.

Let A be the coefficient matrix and let $\hat{A}$ be the corresponding augmented matrix. Note that $\text{rank}(A) \leq \text{rank } \hat{A}$.

Proposition 24. A linear system of equations must have either no solution, one solution, or infinitely many solutions.

Proposition 25. A system of linear equations with coefficient matrix A and augmented matrix has a solution if and only if

$$\text{rank } \hat{A} = \text{rank } A.$$

Proposition 26. If a linear system has more than one solution, it has infinitely many.

Proposition 27. If a system of linear equations has more unknowns than equations, it must have either no solution or infinitely many solutions.

Definition 107. A homogeneous system of equations is a linear system of equations (2) such that $b_j = 0, \quad \forall j$.

Proposition 28. A homogeneous system of linear equations that has more unknowns than equations must have infinitely many distinct solutions.

Definition 108. For a square matrix A , if there exists B s.t. $BA = AB = I$, then we say that A is non-singular, and B is the inverse of A . We use A^{-1} to denote the inverse of A .

Proposition 29. If A and B are square matrices such that $AB = I$, then $BA = I$.

Proposition 30. Let A be a square matrix. If A is non-singular, then it has a unique inverse.

Proposition 31. Let the coefficient matrix A be non-singular; then the corresponding linear system has one and only one solution for every choice of right-hand side $(b_1, \dots, b_m)$ in (2).

Proposition 32. Let A be a non-singular matrix. Then $(A^{-1})' = (A')^{-1}$

Definition 109. Let A be a 2×2 matrix. The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is denoted by $\det(A)$, and is defined as

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

and the determinant of a 3×3 matrix is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Vectors

Definition 110. Let $\mathbb{R}^n$ is the set of all lists of length n of elements of $\mathbb{R}$:

$$\mathbb{R}^n = \{(x_1, \dots, x_n) : x_j \in \mathbb{R} \text{ for } j = 1, \dots, n\}.$$

For $(x_1, \dots, x_n) \in \mathbb{R}^n$ and $j \in \{1, \dots, n\}$, we say that x_j is the j^{th} coordinate of $(x_1, \dots, x_n)$.

Definition 111. Addition of $x, y \in \mathbb{R}^n$ is defined by,:

$$x + y = (x_1 + y_1, \dots, x_n + y_n).$$

Definition 112. Subtraction of $x, y \in \mathbb{R}^n$ is defined by,:

$$x - y = (x_1 - y_1, \dots, x_n - y_n).$$

Definition 113. Scalar multiplication of $x \in \mathbb{R}^n$ by $\lambda \in \mathbb{R}$ is defined by,

$$\lambda x = (\lambda x_1, \dots, \lambda x_n).$$

Definition 114. Let $\mathbf{0} = (0, \dots, 0)$ denote the list of length n whose coordinates are all 0.

Definition 115. V is said to be closed under addition if $(v_1 + v_2) \in V$, $\forall v_1, v_2 \in V$.

Definition 116. V is said to be closed under scalar multiplication if $\lambda v \in V$, $\forall v, \lambda \in V \times \mathbb{R}$.

Definition 117. Let V be a set closed both under addition and scalar multiplication. In order for V to be a vector space, the following conditions must hold for all elements $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and any scalars $r, s \in \mathbb{R}$:

1. Commutativity:

$$\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$$

2. Associativity of vector addition:

$$(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z}).$$

3. Additive identity:

$$\mathbf{0} + \mathbf{x} = \mathbf{x} + \mathbf{0} = \mathbf{x}.$$

4. Existence of additive inverse:

$$\mathbf{x} + (-\mathbf{x}) = \mathbf{0}.$$

5. Associativity of scalar multiplication:

$$r(s\mathbf{x}) = (rs)\mathbf{x}.$$

6. Distributivity of scalar sums:

$$(r + s)\mathbf{x} = r\mathbf{x} + s\mathbf{x}.$$

7. Distributivity of vector sums:

$$r(\mathbf{x} + \mathbf{y}) = r\mathbf{x} + r\mathbf{y}.$$

8. Scalar multiplication identity:

$$1\mathbf{x} = \mathbf{x}$$

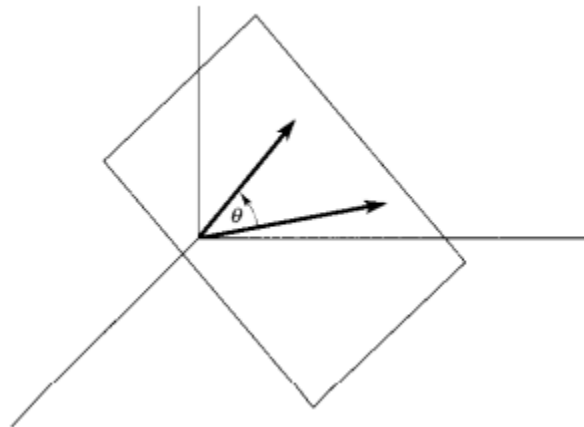
Definition 118. Elements of a vector space are called vectors.

Definition 119. A subspace is a vector space included in another larger vector space.

Definition 120. The dot product of vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is given by,

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

Any two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ determine a plane, as illustrated in the next figure,



In that we can measure the angle θ between $\mathbf{u}$ and $\mathbf{v}$.

Theorem 11. Let $\mathbf{u}, \mathbf{v}$ be two vectors in $\mathbb{R}^n$. Let θ be the angle between them. Then,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos(\theta).$$

Proof. Refer to Simon and Blume, "Mathematics for Economists", page 216. □

Class 11

Linear Independence

Sets are often described as solutions to a system of equations. If there are finitely many solutions, the exact number of solutions gives a satisfactory answer. If there are infinitely many solutions, the size of the solution set is best captured by its dimension. In this chapter, we will give a precise definition of “dimension” for linear spaces. The key underlying concept is that of linear independence:

Definition 121. The line generated or spanned by the vector $\mathbf{v} \in \mathbb{R}^n$ is given by

$$\mathcal{L}[\mathbf{v}] \equiv \{r\mathbf{v} : r \in \mathbb{R}\}.$$

Definition 122. Let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a fixed set of k vectors in $\mathbb{R}^n$. The set spanned by $\mathbf{v}_1, \dots, \mathbf{v}_k$ is given by

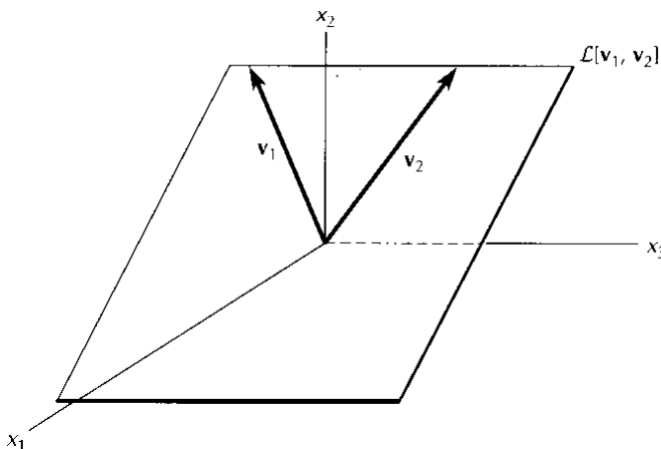
$$\mathcal{L}[\mathbf{v}_1, \dots, \mathbf{v}_k] \equiv \{c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k : c_1, \dots, c_k \in \mathbb{R}\}.$$

Definition 123. For any set V in $\mathbb{R}^n$ we say that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ spans V , if and only if,

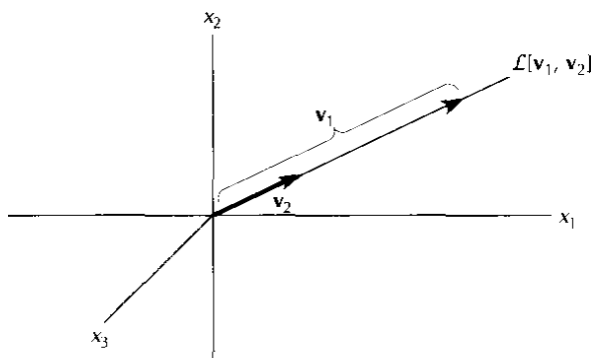
$$V \subseteq \mathcal{L}[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k]$$

Example: Take 2 vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\mathbb{R}^3$. Let's analyze the 2 possible cases:

1. $\mathbf{v}_1$ is not a multiple of $\mathbf{v}_2$. In this case $\mathcal{L}[\mathbf{v}_1, \mathbf{v}_2]$ will be a plane in $\mathbb{R}^3$.



2. $\mathbf{v}_1$ is a multiple of $\mathbf{v}_2$. In this case $\mathcal{L}[\mathbf{v}_1, \mathbf{v}_2]$ will be a line in $\mathbb{R}^3$.



Definition 124. Vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are said to be linearly dependent, if there exist scalars $a_1, a_2, \dots, a_k$, not all zero, such that

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}.$$

Equivalently we can state the definitions in terms of linearly independent vectors:

Definition 125. $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are said to be linearly independent, if given $b_1, b_2, \dots, b_k \in \mathbb{R}$,

$$b_1 \mathbf{v}_1 + b_2 \mathbf{v}_2 + \dots + b_k \mathbf{v}_k = \mathbf{0} \Rightarrow b_1, b_2, \dots, b_k = 0.$$

Lemma 12. If each of the vectors $\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_m \in \mathbb{R}^n$ is a linear combination of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$ then the $\mathbf{y}_j$'s are linearly dependent.

Proof. Proceed by induction on m . Starting with $m = 1$, let $\mathbf{y}_0 = \alpha_0 \mathbf{x}_1$ and $\mathbf{y}_1 = \alpha_1 \mathbf{x}_1$. If either α_0 or α_1 are equal to zero then the $\mathbf{y}_j$'s are linearly dependent (we make the coefficient associated with the zero vector positive). Otherwise, if both α_0 and α_1 are different from zero, note that $\alpha_0 \mathbf{y}_1 - \alpha_1 \mathbf{y}_0 = \alpha_0 \alpha_1 \mathbf{x}_1 - \alpha_1 \alpha_0 \mathbf{x}_1 = \mathbf{0}$, implying linear dependence.

Assume that the claim holds for $m = k - 1$, that is, if each of the vectors $\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_{k-1} \in \mathbb{R}^n$ is a linear combination of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_{k-1} \in \mathbb{R}^n$ then the $\mathbf{y}_j$'s are linearly dependent. We want to show that if each of the vectors $\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_k \in \mathbb{R}^n$ is a linear combination of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$ then the $\mathbf{y}_j$'s are linearly dependent.

In other words, we want to define $\mathbf{y}_j$'s as:

$$\mathbf{y}_j = \sum_{i=1}^k \alpha_{ij} \mathbf{x}_i, \quad j = 0, 1, \dots, k$$

and show that the $\mathbf{y}_j$'s are linearly dependent.

If there exist one j such that all α_{ij} are equal zero, then we know that is enough to show linear dependence. Now assume that not all the α_{ij} 's are zero for every j . Without loss of generality, say $\alpha_{10} \neq 0$ and define, for $j = 1, \dots, k$,

$$\mathbf{z}_j = \mathbf{y}_j - \frac{\alpha_{1j}}{\alpha_{10}} \mathbf{y}_0 = \sum_{i=1}^k \alpha_{ij} \mathbf{x}_i - \frac{\alpha_{1j}}{\alpha_{10}} \sum_{i=1}^k \alpha_{i0} \mathbf{x}_i = \sum_{i=2}^k \left(\alpha_{ij} - \frac{\alpha_{1j}}{\alpha_{10}} \alpha_{i0} \right) \mathbf{x}_i.$$

But now each of the k vectors $\mathbf{z}_1, \dots, \mathbf{z}_k$ is a linear combination of the $k-1$ vectors $\mathbf{x}_2, \dots, \mathbf{x}_k$.

By the induction hypothesis, the $\mathbf{z}_j$'s are linearly dependent, that is, there exist numbers $\beta_1, \dots, \beta_k$, not all zero, such that $\beta_1 \mathbf{z}_1 + \dots + \beta_k \mathbf{z}_k = \mathbf{0}$. But since

$$\mathbf{0} = \sum_{j=1}^k \beta_j \mathbf{z}_j = \sum_{j=1}^k \beta_j \mathbf{y}_j - \frac{\mathbf{y}_0}{\alpha_{10}} \sum_{j=1}^k \beta_j \alpha_{1j}$$

It follows that the $\mathbf{y}_j$'s are linearly dependent. □

Example: Every vector in $\mathbb{R}^n$ is a linear combination of $\mathbf{e}_1, \dots, \mathbf{e}_n$ - where

$$\mathbf{e}_i = (0, \dots, 0, 1, 0, \dots, 0)$$

has a 1 in the i th place and zeros elsewhere for every i . Putting differently

$$\mathcal{L}[\mathbf{e}_1, \dots, \mathbf{e}_n] = \mathbb{R}^n.$$

We can conclude that every collection of $n+1$ vectors is linearly dependent by Lemma (12).

Corollary 2. *Every system of n homogeneous linear equations in $n+1$ unknowns has a nonzero solution.*

Proof. Consider the system of homogeneous equations given by

$$\begin{aligned} \alpha_{1,0} w_0 + \dots + \alpha_{1,n} w_n &= 0 \\ \alpha_{2,0} w_0 + \dots + \alpha_{2,n} w_n &= 0 \\ &\vdots \\ \alpha_{n,0} w_0 + \dots + \alpha_{n,n} w_n &= 0 \end{aligned}$$

we can also rewrite this as,

$$\sum_{j=0}^n \alpha_{ij} w_j = 0, \quad \forall i = 1, \dots, n.$$

Let $\alpha_j = (\alpha_{1j}, \dots, \alpha_{nj})$ for $j = 0, \dots, n$ (Note that these are the coefficients associated with the columns of the linear system of equations). These are $n+1$ vectors in $\mathbb{R}^n$, so by the previous paragraph, they are linearly dependent. Therefore, there are numbers $\hat{w}_0, \hat{w}_1, \dots, \hat{w}_n$, not all zero, such that

$$\sum_{j=0}^n \alpha_j \hat{w}_j = \mathbf{0},$$

but notice that this can be rewritten as,

$$\sum_{j=0}^n (\alpha_{1j}, \dots, \alpha_{nj}) \hat{w}_j = \mathbf{0},$$

$$\sum_{j=0}^n (\alpha_{1j} \hat{w}_j, \dots, \alpha_{nj} \hat{w}_j) = \mathbf{0},$$

$$\left(\sum_{j=0}^n \alpha_{1j} \hat{w}_j, \dots, \sum_{j=0}^n \alpha_{nj} \hat{w}_j \right) = (0, \dots, 0),$$

i.e., the numbers $\hat{w}_0, \hat{w}_1, \dots, \hat{w}_n$ are a nonzero solution of our n homogeneous equations. $\square$

Remark: (2) can be extended to n equations in any number of unknowns greater than n .

Note that if we have a spanning set of vectors, we can always include any linear combination of the vectors to the set and still span the same original set. For the sake of efficiency, if $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, we would like the smallest possible subset of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ that spans V .

Definition 126. Let $S \subseteq \mathbb{R}^n$, the rank of S is the maximum number of linearly independent vectors that can be simultaneously selected from S .

Definition 127. If the rank of S is r then any set of r linearly independent vectors from S is called a basis of S .

Lemma 13. Let $S \subset \mathbb{R}^n$ and let $x_1, \dots, x_r$ be r linearly independent vectors. The vectors $x_1, \dots, x_r \in S$ are a basis of S , if and only if, every vector in S is a linear combination of the x_i 's.

Proof. [$\Leftarrow$] Suppose that every vector $y \in S$ is a linear combination of the x_i 's. By Lemma (2), S contains no larger set of linearly independent vectors, since any set of more than r vectors must be linearly dependent by virtue of being linear combinations of the x_i 's. Therefore, S has rank r and the x_i 's form a basis of S .

[=>] Suppose that the x_i 's form a basis of S . By definition of basis, for any other vector $y \in S$, the collection $\{y, x_1, \dots, x_r\}$ is linearly dependent, implying

$$\alpha_0 y + \sum_{i=1}^r \alpha_i x_i = \mathbf{0}$$

for some scalars $\alpha_0, \alpha_1, \dots, \alpha_r$, not all zero. Moreover, $\alpha_0 \neq 0$, since otherwise the x_i 's would be linearly dependent. Therefore,

$$y = -\frac{1}{\alpha_0} \sum_{i=1}^r \alpha_i x_i$$

and y is a linear combination of the x_i 's, as claimed. □

By Lemma (13), the vectors $e_1, \dots, e_n$ form a basis of $\mathbb{R}^n$, as does any collection of n linearly independent vectors.

Definition 128. The number k of vectors in every basis of V is called the dimension of V .

Definition 129. The column space of matrix A is the span of its column vectors.

Definition 130. The row space of matrix A is the span of its row vectors.

Definition 131. Let A be a matrix. The column rank of a matrix, $\text{colrank}(A)$, corresponds to the maximal number of linearly independent columns of A

Definition 132. Let A be a matrix. The row rank of a matrix, $\text{rowrank}(A)$, corresponds to the maximal number of linearly independent rows of A

Lemma 14. Let A be a $m \times n$ matrix, then $\text{rowrank}(A) = \text{colrank}(A)$.

Proof. Let r be the row rank and s the column rank of a given matrix

$$A = \begin{bmatrix} \alpha_{11} & \cdots & \alpha_{1n} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mn} \end{bmatrix}.$$

For a contradiction, suppose that $r < s$. Choose a row basis for A . Reordering rows if necessary, assume that the row basis consists of the first r rows of A , labeled $a^1, \dots, a^r$. Likewise, choose a column basis for A , which, by reordering columns if necessary, we may assume consists of the first s columns of A , labeled $a_1, \dots, a_s$.

Let $\hat{a}^i = (\alpha_{i1}, \dots, \alpha_{is})$ be the vector consisting of the first s entries of the i th row of A . Since $r < s$, by Corollary (2), constructing a system with more unknowns than equations,

$$\begin{aligned}\alpha_{11}y_1 + \dots + \alpha_{1s}y_s &= 0 \\ \alpha_{21}y_1 + \dots + \alpha_{2s}y_s &= 0 \\ &\vdots \\ \alpha_{r1}y_1 + \dots + \alpha_{rs}y_s &= 0\end{aligned}$$

we know that a nonzero solution vector $\bar{y}$ exists.

Furthermore, since $\{a^1, \dots, a^r\}$ form a row basis for A , by Lemma (13), for every row a^k there exist numbers $\beta_{1k}, \dots, \beta_{rk}$ such that

$$a^k = \sum_{i=1}^r \beta_{ik} a^i, \quad k = 1, \dots, m.$$

Truncating the last $n - s - 1$ entries from each vector in the equations above implies

$$\hat{a}^k = \sum_{i=1}^r \beta_{ik} \hat{a}^i, \quad k = 1, \dots, m,$$

therefore

$$\hat{a}^k \bar{y} = \left(\sum_{i=1}^r \beta_{ik} \hat{a}^i \right) \bar{y} = \sum_{i=1}^r \beta_{ik} (\hat{a}^i \bar{y}) = 0, \quad k = 1, \dots, m.$$

Letting $\bar{y} = (\bar{y}_1, \dots, \bar{y}_s)$, we rewrite this result for a clear exposition as,

$$\begin{aligned}(a_{11}, \dots, a_{1s}) \cdot (\bar{y}_1, \dots, \bar{y}_s) &= 0 \\ (a_{21}, \dots, a_{2s}) \cdot (\bar{y}_1, \dots, \bar{y}_s) &= 0 \\ &\vdots \\ (a_{m1}, \dots, a_{ms}) \cdot (\bar{y}_1, \dots, \bar{y}_s) &= 0\end{aligned}$$

then

$$\begin{aligned}(a_{11}\bar{y}_1 + \dots + a_{1s}\bar{y}_s) &= 0 \\ (a_{21}\bar{y}_1 + \dots + a_{2s}\bar{y}_s) &= 0 \\ &\vdots \\ (a_{m1}\bar{y}_1 + \dots + a_{ms}\bar{y}_s) &= 0\end{aligned}$$

which in terms implies that,

$$\sum_{j=1}^s \bar{y}_j a_j = \mathbf{0}$$

But this implies that the column vectors $a_1, \dots, a_n$ are linearly dependent, which contradicts the assumption that they are a basis.

A symmetric argument exchanging the roles of rows and columns yields the inequality $s \geq r$, whence $r = s$, as was claimed. $\square$

Corollary 3. *If the vectors $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^m$ are linearly independent then for any numbers $\gamma_1, \dots, \gamma_n$ there is a vector $\mathbf{y} \in \mathbb{R}^m$ such that*

$$\mathbf{y}\mathbf{a}_j = \gamma_j, \quad j = 1, \dots, n.$$

Proof. Let A be the matrix whose columns are $\mathbf{a}_1, \dots, \mathbf{a}_n$. Note that each row in this matrix has n elements. By Lemma (14), the rows of A have rank n , so if the vectors $\mathbf{a}^1, \dots, \mathbf{a}^n$ form a row basis of A (after reshuffling of rows, if necessary) then they constitute n linearly independent vectors.

By Lemma (13), since the n rows span $\mathbb{R}^n$, every vector $c = (\gamma_1, \dots, \gamma_n)$ is a linear combination of $\mathbf{a}^1, \dots, \mathbf{a}^n$, that is,

$$c = \sum_{i=1}^n \eta_i \mathbf{a}^i$$

for some numbers $\eta_1, \dots, \eta_n$.

Reorganizing this system of equations yields precisely $\gamma_j = y a_j$ for $j = 1, \dots, n$, as required, with the vector $y = (\eta_1, \dots, \eta_n, 0, \dots, 0)$. $\square$

Now, we will move on to state and prove Farkas' lemma. This key result underpins the linear programming duality and has played a central role in the development of mathematical optimization.

Theorem 12. (Farkas) *Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, either (i) there exists $x \in \mathbb{R}^n$ such that $Ax = b$, or (ii) there exists $y \in \mathbb{R}^m$ such that $yA = \mathbf{0}$ and $yb = 1$, but not both.*

Proof. First, let us see that (i) and (ii) cannot both hold simultaneously. Otherwise, there would exist vectors x and y such that $Ax = b$, $yA = 0$ and $yb = 1$. Multiplying by y gives, though, $0 = 0x = yAx = yb = 1$.

Next, let us show that if (i) fails, then (ii) must hold. To this end, let $a_1, \dots, a_s$ be a column basis for A . Since, by hypothesis, there is no x such that $Ax = b$, it must be the case that the vectors $a_1, \dots, a_s, b$ are together linearly independent, otherwise there would be a solution x to $Ax = b$. By Corollary (2), there exists a vector y such that $ya_j = 0$ for $j = 1, \dots, s$ and $yb = 1$. But since the vectors $a_1, \dots, a_s$ form a column basis for A , every column vector a_k can be written as a linear combination $a_k = \lambda_1 a_1 + \dots + \lambda_s a_s$, therefore $ya_k = y(\lambda_1 a_1 + \dots + \lambda_s a_s) = \lambda_1 ya_1 + \dots + \lambda_s ya_s = 0$ for $k = 1, \dots, n$. In other words, $yA = \mathbf{0}$ and $yb = 1$, i.e., (ii) holds, as required. $\square$

Class 12

Linear Programming

A linear programming problem is a mathematical optimization technique used to find the best possible outcome in a situation where linear relationships exist among different variables. It involves maximizing or minimizing a linear objective function subject to a set of linear constraints.

Definition 133. A linear programming problem can be represented in the following form:

$$\begin{aligned} \text{Sup}_x : \quad & c_1x_1 + c_2x_2 + \dots + c_nx_n \\ & \left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2 \\ \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m \\ x_1 \geq 0, \quad x_2 \geq 0, \quad \dots, \quad x_n \geq 0. \end{array} \right. \end{aligned}$$

Here:

- $x = (x_1, x_2, \dots, x_n)$ represent decision variables.
- $c = (c_1, c_2, \dots, c_n)$ are coefficients of the objective function.
- a_{ij} are coefficients of the constraint inequalities.
- $b = (b_1, b_2, \dots, b_m)$ are constants on the right-hand side of the constraints.

Example 2: A workshop produces widgets and gadgets. It costs \$2 and takes 3 hours to manufacture a widget. It costs \$4 and takes 2 hours to produce a gadget. The workshop has \$220 and 150 hours this week to allocate for manufacturing these products. If each widget sells for \$6 and each gadget sells for \$7, then what is the optimal production strategy to maximize profit?

Let w represent the number of widgets produced, and let g represent the number of gadgets produced. Collectively, these constraints form a system of inequalities:

$$\left\{ \begin{array}{l} 2x_1 + 4x_2 \leq 220 \\ 3x_1 + 2x_2 \leq 150 \\ \qquad \qquad \qquad x_1 \geq 0 \\ \qquad \qquad \qquad x_2 \geq 0. \end{array} \right.$$

Where we want to find the supremum of the objective function given by:

$$p(x_1, y) = 6x_1 + 7x_2.$$

Note that we can write any linear programming problem using matrix notation.

Example: Define (A, b, c) st,

$$V = \sup_{x \geq 0} \{cx : Ax \leq b\}$$

where:

$$c = \begin{bmatrix} 6 & 7 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 220 \\ 150 \end{bmatrix}$$

As we shall see, every linear programming program has another intimately related one—they come in pairs.

Definition 134. Fix (A, b, c) such that $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^{m \times 1}$ and $c \in \mathbb{R}^{1 \times n}$. The primal is the linear program,

$$V = \sup_{x \geq 0} \{cx : Ax \leq b\}$$

And the dual is the linear program,

$$W = \inf_{y \geq 0} \{yb : yA \geq c\}.$$

where $x \in \mathbb{R}_+^{n \times 1}$ and $y \in \mathbb{R}_+^{1 \times m}$.

Definition 135. A linear program is in "standard form" if it looks like a primal or dual for some (A, b, c) .

It is easy to see that every linear program is expressible in standard form.

Example: Associated with the primal of the example previously discussed, the dual is,

$$W = \inf_{y \geq 0} \{yb : yA \geq c\}.$$

where $y = [y_1 \ y_2]$ and (A, b, c) are the same as before.

Definition 136. A linear program is called infeasible if its constraint set is empty.

By convention, we say that $V = -\infty$ if the primal is infeasible and $W = +\infty$ if the dual is infeasible.

Definition 137. Any feasible vector is called a feasible solution.

Before we move on, we state a slightly different version of Farkas Lemma that the one we proved in class.

Theorem 13. Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, either (i) there exists $x \in \mathbb{R}^n$ such that $x \geq 0$ and $Ax \leq b$, or (ii) there exists $y \in \mathbb{R}_+^m$ such that $yA \geq 0$ and $yb < 0$, but not both.

We now state a theorem showing an important relationship between the primal and the dual of a linear programming problem.

Theorem 14. (Weak Duality) $V \leq W$ for any triple (A, b, c) .

Proof. If the primal is infeasible, then $V = -\infty$, so necessarily $V \leq W$ regardless of W , and if the dual is infeasible then $W = +\infty$; again $V \leq W$.

Now assume that neither the primal nor the dual is infeasible. That is, there exist vectors $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ such that $x, y \geq 0$, $Ax \leq b$ and $yA \geq c$.

This implies that $(b - Ax) \geq 0$. Multiplying by $y \geq 0$ gives $y(b - Ax) \geq 0$, or $yb \geq yAx$. Similarly, multiplying $(yA - c) \geq 0$ by $x \geq 0$ gives $(yA - c)x \geq 0$, or $yAx \geq cx$. Collecting these inequalities yields,

$$-\infty < cx \leq yb < \infty, \quad \text{for every feasible pair } (x, y).$$

By taking the supremum of the LHS of $cx \leq yb$ with respect to feasible x given yb , we arrive to,

$$V \leq yb < \infty$$

Taking the infimum of the RHS of $cx \leq yb$ with respect to feasible y given x we arrive to,

$$-\infty < cx \leq W.$$

Taking the infimum of the RHS of $V \leq yb$ with respect to feasible y yields,

$$V \leq W.$$

□

The proof of Theorem (14) above suggests a classification of linear programs into three kinds:

- Infeasible (I).
- Feasible and Bounded (B).
- Feasible and Unbounded (U).

The only possibilities for a primal and its dual are those described in Table 1 below.

		Dual		
		I	B	U
Primal	I	✓		✓
	B		✓	
	U	✓		

We now provide some of the proofs that allow us to complete the table.

Proposition 33. *If the primal is infeasible, then the dual is either infeasible or unbounded.*

Proof. If the primal is infeasible, then there does not exist a vector $x \geq 0$ such that $Ax \leq b$. By Farkas' Lemma, there exists a vector $y \geq 0$ such that $yA \geq 0$ and $yb < 0$. If the dual is infeasible, the proposition is proved, so suppose that the dual is feasible. For any feasible dual solution z and any $\lambda \geq 0$, the vector $z + \lambda y$ is also feasible, since $z + \lambda y \geq 0$ and $(z + \lambda y)A = zA + \lambda yA \geq c + \lambda yA \geq c$. Moreover, $(z + \lambda y)b = zb + \lambda yb$ is unbounded: since $yb < 0$, for any bound $K \in \mathbb{R}$ there exists λ large enough that $zb + \lambda yb < K$. $\square$

Proposition 34. *If the dual is bounded, then its primal is bounded.*

Proof. The contrapositive of Proposition (33) is this: if the dual is bounded then its primal is feasible (Could be bounded or unbounded). We know that the primal cannot be unbounded by Theorem (14), hence both primal and dual are bounded. $\square$

Proposition 35. *If primal is feasible but unbounded, then the dual is infeasible.*

Proof. If the primal is feasible but unbounded, then $V = +\infty$. If its dual were feasible, this would contradict Theorem (14). Moreover, since the dual is infeasible, $W = +\infty$. $\square$

Proposition 36. *If dual is feasible but unbounded, then the primal is infeasible.*

Proof. If the dual is feasible but unbounded, then $W = -\infty$. If its primal were feasible, this would contradict Theorem (14). Moreover, since the primal is infeasible, $V = -\infty$. $\square$

Theorem 15. (Strong Duality). *Unless both the primal and the dual are infeasible, $V = W$. Moreover, both the primal and the dual have optimal solutions whenever either is bounded.*

Proof. Given our previous results, we just need to analyze the case where the primal and the dual are bounded.

By Farkas' Lemma, primal feasibility implies that there is no vector $u \geq 0$ such that $uA \geq 0$ and $ub < 0$, whereas dual feasibility implies that there is no vector $v \geq 0$ such that $Av \leq 0$ and $cv > 0$. Call these the primal and dual feasibility conditions.

Given that both primal and dual are feasible, we must show that their values coincide; that is, there exist vectors $x, y \geq 0$ such that $Ax \leq b, yA \geq c$ and $yb \leq cx$. By Theorem 1, the inequality $cx \leq yb$ already holds.

Let us now collect these inequalities into a single system. Define $z = (x, y') \in \mathbb{R}^{(n+m) \times 1}$, $d = (b, -c', 0) \in \mathbb{R}^{(m+n+1) \times 1}$ and $B \in \mathbb{R}^{(m+n+1) \times (n+m)}$ by

$$B = \begin{bmatrix} A & 0^{m \times m} \\ 0^{n \times n} & -A' \\ -c & b' \end{bmatrix}.$$

By construction, there exists a vector $z \geq 0$ such that $Bz \leq d$, if and only if, the primal and dual values coincide at x and y , respectively.

By Farkas Lemma, either this system of inequalities holds or there is a vector $w \geq 0$ such that $wB \geq 0$ and $wd < 0$. Writing $w = (u, v', \lambda) \in \mathbb{R}^{1 \times (m+n+1)}$, where $\lambda \in \mathbb{R}$, the inequalities $wB \geq 0$ disaggregates into,

$$uA - \lambda c \geq 0, \quad -v'A' + \lambda b' \geq 0,$$

and $wd < 0$ disaggregate into,

$$ub - v'c' < 0,$$

or equivalently

$$uA \geq \lambda c, \quad Av \leq \lambda b, \quad ub < cv.$$

We will now show that there does not exist a vector $w \geq 0$ that satisfies these inequalities.

To see why, suppose that $w = (u, v', \lambda)$ satisfies the system. If $\lambda > 0$ then, dividing by λ and substituting $\hat{u} = u/\lambda \geq 0, \hat{v} = v'/\lambda \geq 0$ into the system of inequalities above gives

$$\hat{u}A \geq c, \quad A\hat{v} \geq b, \quad \hat{u}b < c\hat{v}.$$

This contradicts Theorem (14), as it states that there is a primal feasible solution $\hat{u}$ and a dual feasible solution $\hat{v}$ such that weak duality fails. Therefore, $\lambda > 0$ leads to a contradiction. If instead $\lambda = 0$, the inequalities above become

$$uA \geq 0, \quad Av \geq 0, \quad ub < cv.$$

But this contradicts the primal and dual feasibility conditions since they stipulate that $ub \geq 0 \geq cv$.

Therefore, the alternative to these inequalities must hold; in other words, there exist primal and dual feasible solutions whose values coincide.

By weak duality, these variables are optimal for the respective linear programs in which they are feasible. $\square$

Theorem (15) and its proof facilitate the analysis of linear programs substantially. To find an optimal solution, it is enough to find feasible primal and dual solutions with the same value. Moreover, to guarantee strong duality, that is, $V = W$, it is enough to establish that either the primal or the dual is feasible.

Introduction to Convex Analysis

Linear Operators

Definition 138. Let X and Y be two linear spaces. A function $L : X \rightarrow Y$ is called a linear operator if

$$L(\alpha x + x') = \alpha L(x) + L(x')$$

for all $x, x' \in X$ and $\alpha \in \mathbb{R}$.

We can expand the above definition and write it for any finite set $T \subseteq X$ and any $\lambda \in \mathbb{R}^T$

$$L\left(\sum_{x \in T} \lambda(x)x\right) = \sum_{x \in T} \lambda(x)L(x)$$

Definition 139. A real-valued linear operator $L : X \rightarrow \mathbb{R}$ is called a linear functional on X .

Example. For any $m, n \in \mathbb{N}$, let $A = [a_{ij}]_{m \times n} \in \mathbb{R}^{m \times n}$ be a matrix. Consider $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by:

$$L(x) := Ax$$

Where

$$Ax = \left(\sum_{j=1}^n a_{1j}x_j, \dots, \sum_{j=1}^n a_{mj}x_j \right)$$

Example. Let $L : \mathbb{R}^n \rightarrow \mathbb{R}$. L is said to be a linear functional if, and only if, there exist real numbers $\alpha_1, \dots, \alpha_n$ such that:

$$\forall x \in \mathbb{R}^n, \quad L(x) = \alpha_1 x_1 + \dots + \alpha_n x_n = \alpha \cdot x$$

Definition 140. A subset H of a linear space X is a hyperplane in X if, and only if,

$$H = \{x \in X : L(x) = \alpha\}$$

for some $\alpha \in \mathbb{R}$ and a non-zero linear functional L on X .

The hyperplane H induces two closed half-spaces

$$\{x \in X : L(x) \geq \alpha\} \quad \text{and} \quad \{x \in X : L(x) \leq \alpha\}$$

And two open half-spaces

$$\{x \in X : L(x) > \alpha\} \quad \text{and} \quad \{x \in X : L(x) < \alpha\}$$

Convex Sets

We now recall the definition of a convex set.

Definition 141. A subset S of a linear space X is said to be convex if

$$\forall x, y \in S, \quad \forall \lambda \in (0, 1), \quad \lambda x + (1 - \lambda)y \in S$$

Remark. The set of elements **between** x and y is said to be the *line segment between x and y* .

$$\{z \in X : z = \lambda x + (1 - \lambda)y, \quad \lambda \in [0, 1]\}$$

The line through x and y is the set

$$\{z \in X : z = \lambda x + (1 - \lambda)y, \quad \lambda \in \mathbb{R}\}$$

Lemma 15. The intersection of convex sets is a convex set.

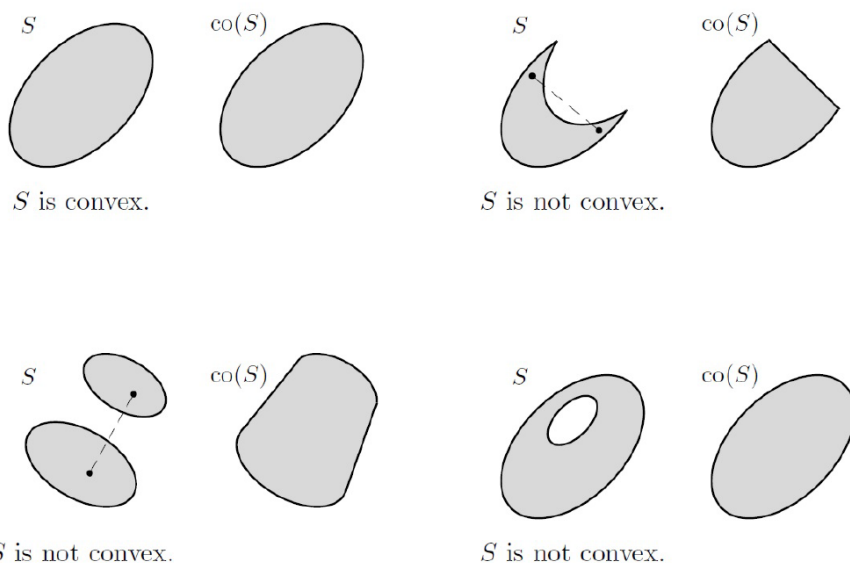
Proof. Let $\mathcal{C}$ be the class of convex subsets of X . Denote by $C^* := \bigcap \mathcal{C}$. Let $x, y \in C^*$ arbitrary and $\lambda \in (0, 1)$. Fix an arbitrary $C \in \mathcal{C}$. Since C is convex, we have $\lambda x + (1 - \lambda)y \in C$. Since this is true for any $C \in \mathcal{C}$, we get that $\lambda x + (1 - \lambda)y \in C^*$. $\square$

Definition 142. Fix S any set in a linear space X . The convex hull of S is the smallest convex set containing S .

Remark. Putting it in mathematical terms, let $\mathcal{S}$ denote the class of all convex subsets of X that contain S :

$$\mathcal{S} := \{C \subseteq X : S \subseteq C, \quad C \text{ is convex}\}$$

The class $\mathcal{S}$ is non-empty since it contains X . Observe that the set $\bigcap \mathcal{S}$ is the smallest convex set containing S . In other words, we have $\text{co}(S) = \bigcap \mathcal{S}$.



Proposition 37. If $T = \{x_1, \dots, x_k\}$ is a finite set, then

$$co(T) = \left\{ \sum_{i=1}^k \lambda_i x_i : (\lambda_1, \dots, \lambda_k) \in \mathbb{R}_+^k \text{ and } \sum_{i=1}^k \lambda_i = 1 \right\}$$

Proof. Denote by $\Delta(T) := \{\lambda \in \mathbb{R}_+^T : \sum_{x \in T} \lambda(x) = 1\}$ the simplex of T . Let

$$C := \left\{ \sum_{x \in T} \lambda(x)x : \lambda \in \Delta(T) \right\}$$

Fix an arbitrary $t \in T$. Choosing $\lambda := \mathbb{1}_{\{t\}}$, we get that $t \in C$. It then follows that C contains T . Let's prove that C is convex. Fix $y, z \in C$ and $\lambda \in (0, 1)$. There exist $\lambda_y \in \Delta(T)$ and $\lambda_z \in \Delta(T)$ such that

$$y = \sum_{x \in T} \lambda_y(x)x \quad \text{and} \quad z = \sum_{x \in T} \lambda_z(x)x$$

It then follows that

$$\lambda y + (1 - \lambda)z = \sum_{x \in T} \underbrace{[\lambda \lambda_y(x) + (1 - \lambda)\lambda_z(x)]}_{=:\mu(x)} x$$

and we get that $\lambda y + (1 - \lambda)z \in C$ since $\mu \in \Delta(T)$. By definition of the convex hull (the smallest convex set that contains T), we have $co(T) \subseteq C$ (since C is convex and $T \subseteq C$). Let's prove the converse inclusion. Fix $y \in C$. There exists $\lambda \in \Delta(T)$ such that

$$y = \sum_{x \in T} \lambda(x)x$$

Let D be an arbitrary convex set containing T . Since D is convex, it must contain y ⁷. We have thus proved that $C \subseteq D$. Since this is true for any convex set D containing T , we get that $C \subseteq co(T)$. $\square$

Definition 143. For any convex subset S of a linear space, we have

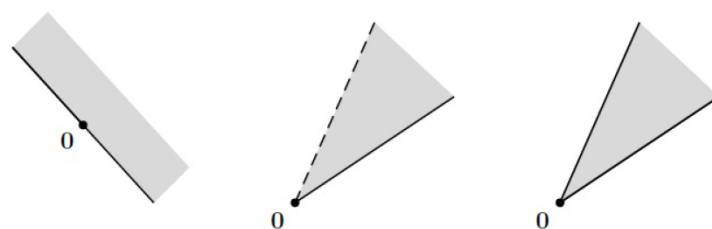
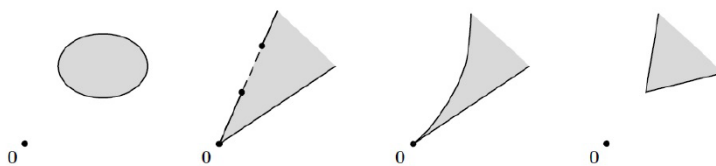
$$\text{aff}(S) = \{\lambda x + (1 - \lambda)y : \lambda \in \mathbb{R} \text{ and } x, y \in S\}$$

Definition 144. A non-empty subset C of a linear space X is said to be a cone if

$$\forall \lambda \geq 0, \quad \lambda C \subseteq C$$

Definition 145. If $C + C \subseteq C$ it is said to be a convex cone.

⁷ Note that $x_1, x_2 \in T \Rightarrow x_1, x_2 \in D$, take $\lambda \in (0, 1)$, then $\lambda x_1 + (1 - \lambda)x_2 \in D$.

Convex cones in $\mathbb{R}^2$ Sets in $\mathbb{R}^2$ which are not cones

Give an example of a non-convex cone!

Lemma 16. If X is a linear space and $L : X \rightarrow \mathbb{R}$ is a linear functional, then

$$\{x \in X : L(x) \geq 0\}$$

is a convex cone.

Proof. Take $y, z \in X$ such that $L(y) \geq 0$ and $L(z) \geq 0$. Since X is a linear space we must have that $y + z \in X$. Then $L(y + z) \in \mathbb{R}$. Finally, using linearity $L(y + z) = L(y) + L(z) \geq 0$. $\square$

Proposition 38. A non-empty subset S of a linear space is a convex cone if, and only if,

$$\sum_{x \in T} \lambda(x)x \in S$$

for any non-empty finite subset T of S and any $\lambda \in \mathbb{R}_+^T$

Proof. ($\implies$). Fix $\emptyset \neq T \subseteq S$ and $\lambda \in \mathbb{R}_+^T$.

$$\sum_{x \in T} \lambda(x)x = \underbrace{\sum_{z \in T} \lambda(z)}_{=: \mu} \times \underbrace{\sum_{x \in T} \frac{\lambda(x)}{\sum_{z \in T} \lambda(z)} x}_{=: y}$$

Indeed, $y \in S$ since S is convex. Since S is a cone we must have $\mu y \in S$.

($\Leftarrow$). Take $x, y \in S$ and $\mu \in (0, 1)$. Since for any non-empty subset $T \subseteq S$ we must have that $\sum_{x \in T} \lambda(x)x \in S$ this must hold for $T := \{x, y\}$ and $\lambda := \mathbb{1}_x \mu + \mathbb{1}_y(1 - \mu)$ and:

$$\sum_{z \in T} \lambda(z)z = \mu x + (1 - \mu)y \in S$$

We thus conclude that S is convex. Now let $\xi \in \mathbb{R}_+$. Define $\lambda := \xi \mathbb{1}_x$ and note that by assumption $\lambda x \in S$, then S is a cone. $\square$

Separating Hyperplanes

Consider A and B two non-empty subsets in a linear space X , and a linear functional $L : X \rightarrow \mathbb{R}$. We write $L(A) \geq L(B)$ to denote

$$L(a) \geq L(b), \quad \forall (a, b) \in A \times B$$

Observe that $L(A) \geq L(B)$ if, and only if, there exists $\alpha \in \mathbb{R}$ such that

$$L(a) \geq \alpha \geq L(b), \quad \forall (a, b) \in A \times B$$

The above property is denoted by

$$L(A) \geq \alpha \geq L(B).$$

We write $L(A) > L(B)$ to denote

$$L(A) \geq L(B) \quad \text{and} \quad L(x) > L(y) \quad \text{for some} \quad (x, y) \in A \times B$$

Theorem 16 (Strict Separating Hyperplane). *Suppose that $B \subseteq \mathbb{R}^n$ is convex and closed and that $x \notin B$. Then there is $p \in \mathbb{R}^n$ with $p \neq 0$, and a value $c \in \mathbb{R}$, such that $p \cdot x > c$ and $p \cdot y < c$ for every $y \in B$.*

Before proceeding with the proof of the theorem note that “there exists $p \neq 0$ ” is basically the same as saying “there exists a non-zero linear operator $L : \mathbb{R}^n \rightarrow \mathbb{R}$ ” since taking a dot product with p is indeed a linear operator. Therefore, we could have said that $L(x) > c$ and $L(y) < c$ for any $y \in B$.

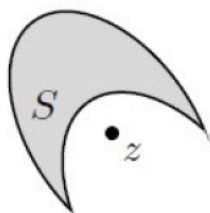


Figure 5.3: Convexity is important

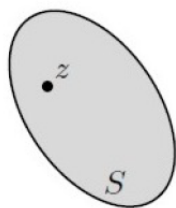


Figure 5.4: Empty intersection is important

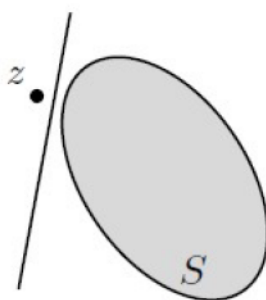
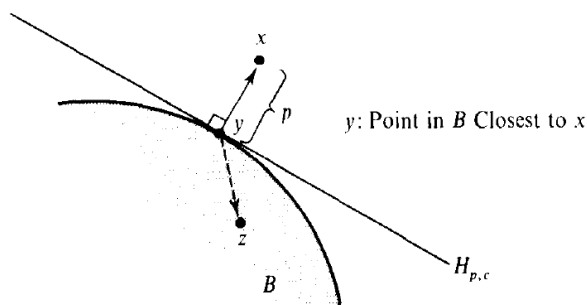


Figure 5.5: Strict separation

Proof. Let $p = x - y$, where we refer to $y \in B$ as the point in B closest to x . Also, let $c' = p \cdot y$. Note that

$$p \cdot x - c' = p \cdot x - p \cdot y = p \cdot (x - y) = (x - y) \cdot (x - y) = \|x - y\|^2 > 0$$

Also, note that for any $z \in B$ the vectors p and $z - y$ cannot make an acute angle, that is, $p \cdot (z - y) = p \cdot z - c' \leq 0$. Finally, let $c = c' + \epsilon$ where $\epsilon > 0$ is small enough for $p \cdot x > c' + \epsilon = c$ to hold.



□

Theorem 17 (Supporting Hyperplane Theorem). Suppose that $B \subseteq \mathbb{R}^n$ is convex and that $x \notin \text{int}(B)$. Then, there is $p \in \mathbb{R}^n$ with $p \neq 0$ such that $p \cdot x \geq p \cdot y$ for every $y \in B$.

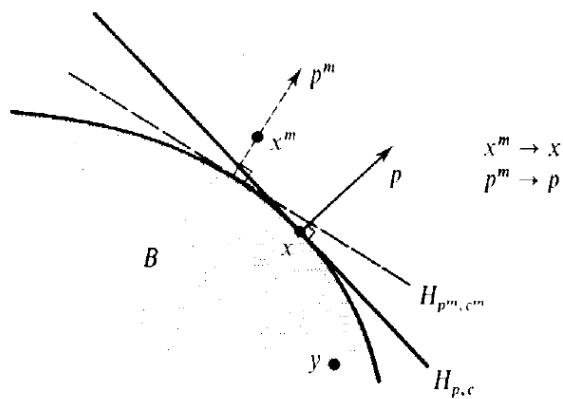
Proof. Suppose $x \notin \text{int}(B)$. It is intuitive that we can find $x_m \rightarrow x$ such that, for all m , x_m is not an element of the closure of the set B . By the Separating Hyperplane Theorem, for each m there exist a $p_m \neq 0$ and $c_m \in \mathbb{R}$ such that

$$p_m \cdot x > c_m \geq p_m \cdot y$$

for every $y \in B$. WLOG we can suppose that $\|p_m\| = 1$ for each m . Thus, extracting a subsequence if necessary we can assume that there is $p \neq 0$ and $c \in \mathbb{R}$ such that $p_m \rightarrow p$ and $c_m \rightarrow c$. Taking the limits in the above inequality we get that

$$p \cdot x \geq c \geq p \cdot y$$

for every $y \in B$



□

Class 14

Problem Set 3

Exercise 1.

Definition 146. The set Y exhibits constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for any scalar $\alpha \geq 0$.

Definition 147. The set Y exhibits nonincreasing returns to scale if for any $y \in Y$, we have $\alpha y \in Y$ for all scalars $\alpha \in [0, 1]$.

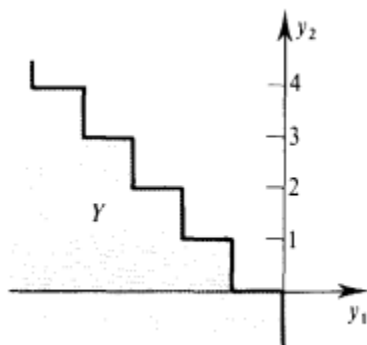
Definition 148. The set exhibits non-decreasing returns to scale if for any $y \in Y$, we have $\alpha y \in Y$ for any scale $\alpha \geq 1$.

Definition 149. Suppose that $y \in Y$ and $y' \in Y$, if the set Y satisfies additivity $y + y' \in Y$.

Definition 150. If the set Y is convex and satisfies constant returns to scale we say that the set is a convex cone.

1. Provide an example of a set that satisfies (148) but not (147)
2. Provide an example of a set that satisfies (147) but not (148).
3. Provide an example of a set that is not convex but satisfies the additivity property.

To illustrate the third case,



How do we formalize this set? Hint: use a floor function and show that the desired properties hold.

Exercise 2 Prove or Disprove the following proposition:

Proposition 39. *The set Y satisfies additivity and the nonincreasing returns condition, if and only if, Y is a convex cone.*

Proof. The definition of a convex cone directly implies the nonincreasing returns and additivity properties. Conversely, we want to show that if nonincreasing returns and additivity hold, then for any $y, y' \in Y$ and any $\alpha > 0$, and $\beta > 0$, we have $\alpha y + \beta y' \in Y$. To this effect, let k be any integer such that $k > \text{Max}\{\alpha, \beta\}$. By additivity, $ky \in Y$ and $ky' \in Y$. Since $(\alpha/k) < 1$ and $\alpha y = (\alpha/k)ky$, the nonincreasing returns condition implies that $\alpha y \in Y$. Similarly, $\beta y' \in Y$. Finally, again by additivity, $\alpha y + \beta y' \in Y$. $\square$

Exercise 3 Consider the production set given by:

$$Y = \{(y_1, y_2, y_3) : y_1 \leq 0, y_2 \leq 0, y_3 \leq \min\{\sqrt{-y_1}, \sqrt{-2y_2}\}\}$$

Verify whether Y exhibits nondecreasing returns to scale, nonincreasing returns to scale, or neither.

Proof. To show that the production set does not exhibit NDRS, take the point $(-16, -8, 4)$ and $\lambda = 4$. You will find that $\lambda(-16, -8, 4) = (-64, -32, 16) \notin Y$.

To show that the production set does exhibit NIRS, we want to show that for any $y \in Y$, and for any $0 < \lambda < 1$ we have that $\lambda y \in Y$.

For that we need to show only that:

$$\lambda y_3 \leq \min\{\sqrt{-\lambda y_1}, \sqrt{-2\lambda y_2}\}$$

Since $y \in Y$, and $\lambda \in (0, 1)$:

$$\begin{aligned} y_3 &\leq \min\{\sqrt{-y_1}, \sqrt{-2y_2}\} \\ \lambda y_3 &\leq \lambda \min\{\sqrt{-y_1}, \sqrt{-2y_2}\} \\ &\leq \sqrt{\lambda} \min\{\sqrt{-y_1}, \sqrt{-2y_2}\} \\ &= \min\{\sqrt{\lambda}\sqrt{-y_1}, \sqrt{\lambda}\sqrt{-2y_2}\} \\ &= \min\{\sqrt{-\lambda y_1}, \sqrt{-2\lambda y_2}\} \end{aligned}$$

$\square$

Definition 151. We use $x \vee y$ to denote the supremum and of two vectors $x, y \in R_+^n$. That is,

$$x \vee y = (\max\{x_1, y_1\}, \max\{x_2, y_2\}, \dots, \max\{x_n, y_n\}),$$

Definition 152. We use $x \wedge y$ to denote the infimum of two vectors $x, y \in R_+^n$. That is,

$$x \wedge y = (\min\{x_1, y_1\}, \min\{x_2, y_2\}, \dots, \min\{x_n, y_n\}),$$

Definition 153. Function $f : R_+^n \rightarrow R$ is supermodular if

$$f(x \vee y) - f(x) \geq f(y) - f(x \wedge y),$$

for every $x, y \in R_+^n$

Exercise 4

1. Show that the Leontief function $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$, where $f(x) = \min\{x_1, x_2\}$ is supermodular.

Proof. Take any $x = (x_1, x_2)$ and $y = (y_1, y_2)$.

First, note that,

$$f(x \vee y) = \min\{\max\{x_1, y_1\}, \max\{x_2, y_2\}\} \geq \max\{\min\{x_1, x_2\}, \min\{y_1, y_2\}\} = \max\{f(x), f(y)\}.$$

Then, note that

$$f(x \wedge y) = \min\{\min\{x_1, y_1\}, \min\{x_2, y_2\}\} = \min\{x_1, y_1, x_2, y_2\} = \min\{f(x), f(y)\}$$

Combining these equation yields,

$$f(x \vee y) + f(x \wedge y) \geq \min\{f(x), f(y)\} + \max\{f(x), f(y)\} = f(x) + f(y).$$

□

2. Show that the Leontief function $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$, where $f(x) = \min_i \{\alpha_i x_i\}$ with $\alpha > 0$ is super modular.

Proof. We want to show that $f(x \vee y) \geq \max\{f(x), f(y)\}$. To show this, note that,

$$\begin{aligned} f(x \vee y) &= \min_i \{\alpha_i \max\{x_i, y_i\}\} \\ &\geq \min_i \{\alpha_i x_i\} = f(x) \end{aligned}$$

By following the same steps we are able to show that,

$$f(x \vee y) \geq f(y).$$

Combining both results yields,

$$f(x \vee y) \geq \max\{f(x), f(y)\}.$$

Next, we want to show that $f(x \wedge y) = \min\{f(x), f(y)\}$. To show this, note that,

$$\begin{aligned} \min\{f(x), f(y)\} &= \min\{\min\{a_1 x_1, \dots, a_n x_n\}, \min\{a_1 y_1, \dots, a_n y_n\}\} \\ &= \min\{\min\{a_1 x_1, a_1 y_1\}, \dots, \min\{a_n x_n, a_n y_n\}\} \\ &= \min_i \{\alpha_i \min\{x_i, y_i\}\} \\ &= f(x \wedge y) \end{aligned}$$

□

3. Give an example of a strictly increasing function in $\mathbb{R}_+^2 \rightarrow \mathbb{R}$ that is not supermodular.

$$f(x, y) = \ln(x + y)$$

Exercise 5. Watch the following sequence of videos ([Click here](#)).

Exercise 6. Consider two closed and convex production sets $Y_1, Y_2 \subset \mathbb{R}^L$ such that $\mathbf{0} \in Y_1$ and $\mathbf{0} \in Y_2$,

$$s_i^*(p) := \{y^* \in Y : py^* \geq py, \quad \forall y \in Y_i\}.$$

and

$$\pi_i^*(p) := \sup_{y \in Y} py.$$

Prove that $Y_1 \subset Y_2$ if and only if $\pi_1^*(p) \leq \pi_2^*(p)$ for every $p \in \mathbb{R}^L$.

Hint: A duality property holds for a closed and convex set $K \in \mathbb{R}^n$:

$$K = \{x \in \mathbb{R}^n : px \leq \mu_K(p), \quad \forall p\}$$

Hint: You have two convex sets in this problem. Moreover, if you negate one of the directions of this proof you get that: $\exists p \in \mathbb{R}^L$ such that $\pi_1^(p) > \pi_2^*(p)$*

Exercise 7. Let $Y \subset \mathbb{R}^L$ be a finite set, $Y = \{y^1, y^2, \dots, y^n\}$. Let $\hat{Y}$ be the convex hull of Y , that is the set of all convex combinations of production plans $y^1, \dots, y^n$. Further, let $\pi_Y^*, s_Y^*, \pi_{\hat{Y}}^*$, and $s_{\hat{Y}}^*$ be the associated functions defined in the previous exercise for Y and $\hat{Y}$, respectively.

1. Show that $\pi_Y^* = \pi_{\hat{Y}}^*$ for every $p \in \mathbb{R}^L$.

Proof. Proving this requires two auxiliary claims. Begin by noting that $Y \subset \hat{Y}$, and thus $\forall p \in \mathbb{R}^L$, since py is being maximized over a larger set, $\pi_{\hat{Y}}^* \geq \pi_Y^*(p)$. Now take any $p \in \mathbb{R}^L$ and consider $\pi_Y^*(p)$. Note if $\pi_Y^*(p) = \infty$, then it must be that $\pi_{\hat{Y}}^*(p) \geq \pi_Y^*(p)$. If $\pi_Y^*(p) \neq \infty$, then $s_Y^*(p) \neq \emptyset$ and thus we can write $\pi_Y^*(p) = py^*$ where $y^* \in s_Y^*(p)$. Also note that $\forall \hat{y} \in \hat{Y}$, we can write $\hat{y} = \sum_{i=1}^n \alpha^i y^i$ where $\alpha \in \Delta^n$. Then by definition,

$$py^* = \pi_Y^*(p) = \sup_{y \in Y} py$$

which implies,

$$\begin{aligned} py^* &\geq py, \quad \forall y \in Y \\ py^* &\geq p \left(\sum_{i=1}^n \alpha^i y^i \right), \quad \forall \alpha \in \Delta^n \\ py^* &\geq p\hat{y}, \quad \forall \hat{y} \in \hat{Y} \end{aligned}$$

therefore,

$$\pi_Y^*(p) = py^* \geq \pi_{\hat{Y}}^*(p).$$

Combining these results, in any case and $\forall p \in \mathbb{R}^L$, we have $\pi_Y^* = \pi_{\hat{Y}}^*$. □

2. Show that $s_Y^* \subseteq s_{\hat{Y}}^*$ for every $p \in \mathbb{R}^L$. Show that the inclusion is strict for some price vectors if $n \geq 2$.

Proof. Take any $p \in \mathbb{R}^L$ and consider $s_Y^*(p)$. Note if $\pi_Y^*(p) = \infty$, which by part (a) implies $\pi_{\hat{Y}}^*(p) = \infty$, then $s_Y^*(p) = s_{\hat{Y}}^*(p) = \emptyset$, so $s_Y^*(p) \subseteq s_{\hat{Y}}^*(p)$.

Consider now when $s_Y^*(p) \neq \emptyset$. Take any $y^* \in s_Y^*(p) \subset Y$ and note by definition $\pi_Y^*(p) = py^*$, and so by part (a), $py^* = \pi_{\hat{Y}}^*(p)$. Thus, by definition, $y^* \in s_{\hat{Y}}^*(p) \subset \hat{Y}$. Since $p \in \mathbb{R}^L$ and $y^* \in s_Y^*(p)$ were taken arbitrarily, $s_Y^*(p) \subseteq s_{\hat{Y}}^*(p) \forall p \in \mathbb{R}^L$.

If $n \geq 2$, we want to show that $s_Y^*(p) \subset s_{\hat{Y}}^*(p)$ for some $p \in \mathbb{R}^L$. Let $n \geq 2$ and take any $p \in \mathbb{R}^L$ such that $\#s_Y^* \geq 2$ (e.g. $p_l = 0 \forall l$). Take any $y_a, y_b \in s_Y^*(p) \subset Y$ and any $\lambda \in [0, 1]$ such that $\lambda y_a + (1 - \lambda)y_b = y_\lambda \in \hat{Y}$ but $y_\lambda \notin Y$. We know this value of $\lambda \in (0, 1)$ exists since Y is finite. Further, since $py_a = py_b = \pi_Y^*(p)$ we know that $py_\lambda = \pi_Y^*(p)$, so by part (a), $py_\lambda = \pi_{\hat{Y}}^*(p)$ and thus $y_\lambda \in s_{\hat{Y}}^*(p) \subset \hat{Y}$. $\square$

Exercise 8.

Definition 154. Let the Kernel of a linear operator L be defined as follows

$$\text{Ker}(L) := \{x \in X : L(x) = 0\}$$

Prove that a linear operator $L : X \rightarrow Y$ is injective if, and only if, $\text{Ker}(L) = \{0\}$.

Proof. ($\implies$). First note that we must have that $L(x + 0) = L(x) + L(0) = L(x)$. Let $x \in \text{Ker}(L)$, then $L(x) = L(0) = 0$, but since L is injective we must have that $x = 0$.

($\impliedby$). Let $x, x' \in X$ such that $L(x) = L(x')$. By linearity $L(x - x') = 0$, then $x - x' \in \text{Ker}(L)$, but since the Kernel is a singleton we must have that $x = x'$. $\square$

Exercise 9. Show that a convex set S is contained in one of the open half-spaces induced by a hyperplane in a Euclidean space X if, and only if, $S \cap H = \emptyset$

Proof. Recall that H is a hyperplane of X if, and only if, there exists a non-zero linear function $L : X \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$ such that $H = \{x \in X : L(x) = \alpha\}$. Fix a convex set S of X .

- First, assume that S is contained in one of the open half-spaces induced by a hyperplane H , say $S \subset \{x \in X : L(x) < \alpha\}$. Then we obviously get that $S \cap H = \emptyset$ since we cannot have $L(x) < \alpha$ and $L(x) = \alpha$.
- Assume now that $S \cap H = \emptyset$. Assume by way of contradiction that there exists $x \in S$ and there exists $y \in S$ with $L(y) > \alpha$. Denote by f the real function on $[0, 1]$ defined by

$$\forall t \in [0, 1], \quad f(t) = L(tx + (1 - t)y).$$

Since L is linear, we have $f(t) = L(y) + t[L(x) - L(y)]$ for all t . Let

$$\tau := \frac{L(y) - \alpha}{L(y) - L(x)}.$$

We have $\tau \in (0, 1)$ and $f(\tau) = \alpha$. Let $z := \tau x + (1 - \tau)y$, and we get that $L(z) = \alpha$, that is $z \in H$. Moreover, by convexity of S , we get that $z \in S$. We have thus proved that $S \cap H \neq \emptyset$, a contradiction

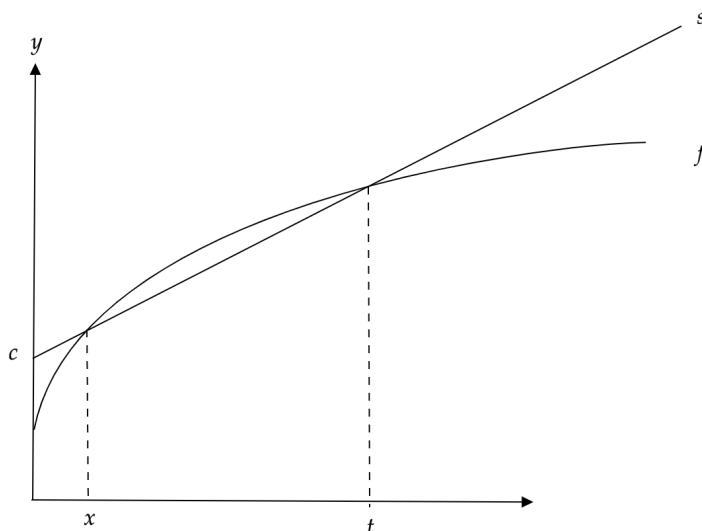
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Class 15

Differential Calculus

First Order Derivatives

Definition 155. A secant line (or simply secant) is a line passing through two points of a curve.



Example: We define the secant between x and t in the above figure as the function:

$$s(t, x) := \left\{ (z, y) \in \mathbb{R} \times \mathbb{R} : z \in \mathbb{R}, y = c + \frac{f(t) - f(x)}{t - x} z \right\}$$

The slope of the secant gives you the *average rate of change* of f from x to t . Let's say that the point t shifts to the left. Naturally, the new secant will be different. We can thus define the growth rate map $G_{f,x} : D \setminus \{x\} \rightarrow \mathbb{R}$ by:

$$G_{f,x}(t) := \frac{f(t) - f(x)}{t - x}$$

Definition 156. Let $f \in \mathbb{R}^D$ where $D \subset \mathbb{R}$. The function f is differentiable at x if, and only if, there exists a linear function $L_x : \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$\lim_{t \rightarrow x} \frac{f(t) - f(x) - L_x(t - x)}{t - x} = 0$$

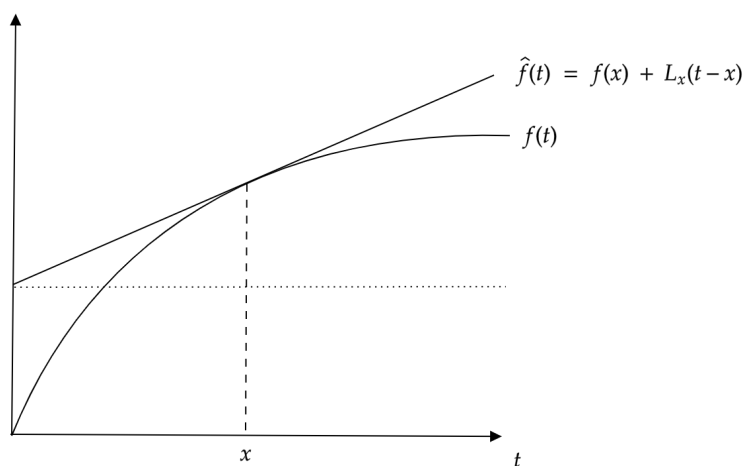
We call L_x a derivative of f at x .

Definition 157. If $L_x \in \mathbf{C}(D)$ we say that f is continuously differentiable.

Definition 158. The tangent of f at x , denoted by $\hat{f}$ is given by:

$$\hat{f}(t) := f(x) + L_x(t - x)$$

As t tends to x the secant tends to the *tangent* of f at x .

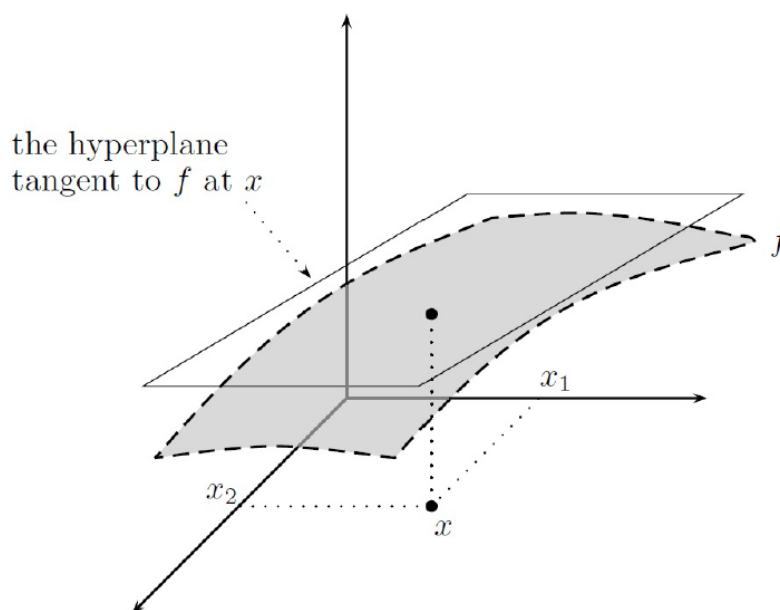


We just found a linear function that maps $t \in \mathbb{R}$ to the elements in the tangent line of f at x . The slope of this line is our geometrical notion of a derivative.

What is the n -dimensional equivalent of a tangent line? For instance, take $\mathbb{R}^2$. We want to know whether there exists a linear function $L_x : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that:

$$\lim_{t \rightarrow x} \frac{f(t) - f(x) - L_x(t - x)}{\|t - x\|_2} = 0$$

In this case the map $t \mapsto f(x) + L_x(t - x)$ is a hyperplane tangent to f at x , and the linear operator L_x is called a derivative.



We will thus state a refinement of our definition of a derivative to $\mathbb{R}^n$.

Definition 159. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. For any x the map is said to be differentiable at x if there is a continuous linear operator $L_x : \mathbb{R}^n \rightarrow \mathbb{R}$ such that:

$$\lim_{t \rightarrow x} \frac{f(t) - f(x) - L_x(t - x)}{\|t - x\|_n} = 0$$

Definition 160. We call the linear operator $L_x : \mathbb{R}^n \rightarrow \mathbb{R}$ the derivative of f at x .

We have defined the derivative as a linear operator that helps to create a hyperplane tangent to a point. To refine our geometrical intuition we will define a directional derivative. Let

$$e^i = (0, \dots, \underbrace{1}_{i\text{-th entry}}, \dots)$$

Definition 161. The i -th partial derivative of f at x is defined as

$$\frac{\partial f(x)}{\partial x_i} := \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon e^i) - f(x)}{\epsilon} \in \mathbb{R}$$

Definition 162. We define the gradient of a function f at x to be

$$\nabla f(x) := \left(\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n} \right)$$

The partial derivative preserves our geometrical intuition of a derivative in $\mathbb{R}$. The way to see it is “cutting a slice” of the function holding all $j \in \{1, \dots, n\} \setminus \{i\}$ entries of x constant and computing the slope of f in the i -th direction.

But what is the relationship between the derivative and the partial derivatives of f ?

Lemma 17. Let S be any open set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. If f is differentiable at x , then all its partial derivatives at x exists, and we have:

$$L_x(t_1, \dots, t_n) = \sum_{i=1}^n \frac{\partial f(x)}{\partial x_i} t_i \quad (= \nabla f(x) \cdot t)$$

for all $t = (t_1, \dots, t_n) \in \mathbb{R}^n$.

In an intuitive way, L_x is giving us the behavior of a function in the neighborhood of a point x by moving in the *direction* of v .

Definition 163. Let $S \subseteq \mathbb{R}^n$ be an open set. Suppose now $f_i : S \rightarrow \mathbb{R}$ for $i = 1, 2, \dots, k$ be differentiable functions and $t \in \mathbb{R}^n$. Define f as follows:

$$f(t_1, \dots, t_n) := (f_1(t_1, \dots, t_n), \dots, f_k(t_1, \dots, t_n))$$

The Jacobian matrix of f at x is defined by

$$\mathbf{J}_{f,x} := \nabla f(x) := \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} & \dots & \frac{\partial f_1(x)}{\partial x_n} \\ \frac{\partial f_2(x)}{\partial x_1} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial f_k(x)}{\partial x_1} & \dots & \dots & \frac{\partial f_k(x)}{\partial x_n} \end{bmatrix}_{k \times n}$$

Proposition 40. Let S be an open and convex subset of $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be continuously differentiable. Define $F : (0, 1) \rightarrow \mathbb{R}$ by

$$\forall t \in (0, 1), \quad F(t) := f(tx + (1-t)y)$$

The function F is differentiable in $(0, 1)$ and

$$\forall t \in (0, 1), \quad F'(t) := \nabla f(tx + (1-t)y) \cdot (x - y)$$

Implicit Function Theorem

The setting for the *implicit function theorem* (IFT) is as follows. Let $X \subseteq \mathbb{R}^n$ $f : X \rightarrow \mathbb{R}$ and $Q \subseteq \mathbb{R}^m$. We have an equation ⁸ depending on n endogenous variables $x = (x_1, \dots, x_n)$ and m parameters $q = (q_1, \dots, q_m)$:

$$f(x_1, \dots, x_n; q_1, \dots, q_m) = 0$$

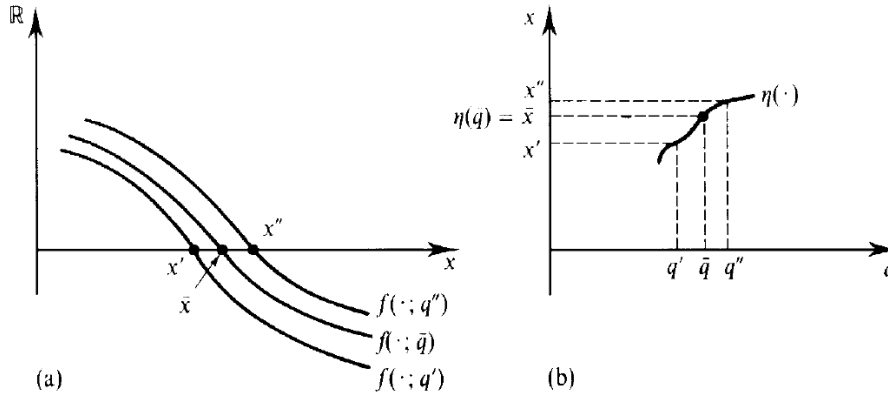
Suppose $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n) \in X$ and $\bar{q} = (\bar{q}_1, \dots, \bar{q}_m) \in Q$ are such that $f(\bar{x}, \bar{q}) = 0$. We are interested in the possibility of solving for x as a function of q *locally* around $\bar{q}$, formally:

⁸ This exposition could also be extended to allow n equations. For pedagogical purposes, we work on the simplest case here.

Definition 164. Suppose $\bar{x} \in X$ and $\bar{q} \in Q$ are such that $f(\bar{x}, \bar{q}) = 0$. We say that we can solve $f(x, q) = 0$ locally at $(\bar{x}, \bar{q})$ if there are $B_X(\bar{x}, \epsilon) \subseteq X$, $B_Q(\bar{q}, \delta) \subseteq Q$, and n uniquely determined functions $\eta_i : B_X(\bar{x}, \epsilon) \rightarrow B_Q(\bar{q}, \delta)$ such that:

$$f(\eta_1(q), \dots, \eta_n(q)) = 0, \quad \forall q \in B_Q(\bar{q}, \delta)$$

$$\eta_i(\bar{q}) = \bar{x}_i, \quad i = 1, \dots, n$$



The implicit function theorem gives a sufficient condition for the existence of the functions η_i and tells us the first-order comparative statics effects of q on x at a solution.

Theorem 18 (Implicit Function Theorem). Suppose that f is continuously differentiable with respect to its $n + m$ arguments and $(\bar{x}, \bar{q})$ is such that $f(\bar{x}, \bar{q}) = 0$. If the

$$\nabla f(\bar{x}, \bar{q}) = (\partial f(\bar{x}, \bar{q})/\partial x_1, \dots, \partial f(\bar{x}, \bar{q})/\partial x_n) \neq 0.$$

Then, the system can be locally solved at $(\bar{x}, \bar{q})$ by the function $\eta : B_Q(\bar{q}, \delta) \rightarrow B_X(\bar{x}, \epsilon)$ that is continuously differentiable. Moreover, the first-order effects of q on x at $(\bar{x}, \bar{q})$ are given by

$$\nabla_q \eta(\bar{q}) = -[\nabla_x f(\bar{x}, \bar{q})]^{-1} \nabla_q f(\bar{x}, \bar{q})$$

We now use the IFT to provide further geometrical intuition for the gradient of a function.

Theorem 19. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuously differentiable function on a neighborhood of (x_0, y_0) . Suppose that $\partial f(x_0, y_0)/\partial x \neq 0$ and $\partial f(x_0, y_0)/\partial y \neq 0$. Then $\nabla f(x_0, y_0)$ is perpendicular to the level curve of f at (x_0, y_0) .

Higher Order Derivatives

Let f be a function from the open set $S \subseteq \mathbb{R}^n$ to $\mathbb{R}$, as usual. Throughout this subsection, we assume f to be differentiable on S .

Suppose now that the derivative L_x is itself differentiable at x , such that for each i , the function $\partial f/\partial x_i : S \rightarrow \mathbb{R}$ is differentiable at x . When this is the case we say that f is twice differentiable at x , and we can apply the definition of the partial derivative with respect to

e^j from the previous subsection to $\partial f / \partial x_i$ to get the second order derivative denoted by:

$$\frac{\partial^2 f(x)}{\partial x_i \partial x_j} \quad \forall j \neq i$$

$$\frac{\partial^2 f}{\partial x_i^2}, \quad j = i$$

Definition 165. Let f be a twice differentiable function from S to $\mathbb{R}$. The Hessian matrix is defined as

$$\mathbf{H}_{f,x} := \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \cdots & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}_{n \times n}$$

Differentiable Concave Maps

Proposition 41. Let $S \subseteq \mathbb{R}^n$ be an open and convex set. Let $f : S \rightarrow \mathbb{R}$ be a continuously differentiable function. Then, f is concave if, and only if,

$$\forall x, y \in S, \quad f(y) - f(x) \leq \nabla f(x) \cdot (y - x)$$

Corollary 4. Let $S \subseteq \mathbb{R}^n$ be an open and convex set and $f : S \rightarrow \mathbb{R}$ be a continuously differentiable function. Then f is concave if, and only if,

$$\forall x, y \in S, \quad (\nabla f(y) - \nabla f(x)) \cdot (y - x) \leq 0$$

Subgradients

Definition 166. We say that y is the right-limit of f at x if $\lim_{n \rightarrow \infty} f(x_n) = y$ for all sequences $D \setminus \{x\}$ such that $x_n \downarrow x$ and denote $y = f(x^+)$

Definition 167. We say that z is the left-limit of f at x if $\lim_{n \rightarrow \infty} f(x_n) = z$ for all sequences $D \setminus \{x\}$ such that $x_n \uparrow x$ and denote $z = f(x^-)$

To simplify our exposition let D be an open interval, therefore it has no endpoints.

Definition 168. The function f is said to be right-differentiable at x if $G_{f,x}(x^+)$ exists in $\mathbb{R}$. We then define the right-derivative at x as:

$$f'_+(x) := G_{f,x}(x^+)$$

Definition 169. The function f is said to be left-differentiable at x if $G_{f,x}(x^-)$ exists in $\mathbb{R}$. We then define the left-derivative at x as:

$$f'_- := G_{f,x}(x^-)$$

We can now adapt the definition of a derivative given in (156):

Definition 170. f is differentiable at x if it is both right and left differentiable at x and $f'_+(x) = f'_-(x)$.

But what if the left and right differentials don't coincide at x ? In this case, the derivative doesn't exist. If you look at the kink of the function f at x you can see that it has not **only** one tangent at that point, but infinitely many. We will later define a special "set of tangent lines".

Definition 171. A vector p is a subgradient of f at x if

$$f(x) - f(y) \leq p \cdot (x - y), \quad \forall y \in \mathbb{R}^n$$

Definition 172. The subdifferential of f at x , denoted by $\partial f(x)$, is the set of all subgradients of f at x

Proposition 42. The subdifferential of a function f , ∂f is closed and convex.

Class 16

Correspondences

Definition 173. Let 2^Y denote the power set of Y , i.e., the collection of all subsets of Y .

Definition 174. A correspondence γ , from X to Y , is a function from X to the family of subsets of Y . We denote this by $\gamma : X \rightarrow\rightarrow Y$.

Example: The Budget Set is a Correspondence. A consumer uses her income to buy any combination of the L goods at given prices. Assume, prices are a vector in $\mathbf{R}_+^L$. Possible consumption choices are described by a vector in $\mathbf{R}_+^L$. Given prices and income, the consumer can only choose combinations of the goods that she can afford:

$$\left\{ \mathbf{x} \in \mathbf{R}_+^L : \mathbf{p} \cdot \mathbf{x} = \sum_{l=1}^L p_l x_l \leq I \right\}$$

Define $\gamma : X \rightarrow\rightarrow Y$ where $X := \mathbf{R}_+^L \times \mathbf{R}_{++}$ and $Y := \mathbf{R}_+^L$ as follows:

$$\gamma(\mathbf{p}, I) = \left\{ \mathbf{x} \in \mathbf{R}_+^L : \mathbf{p} \cdot \mathbf{x} \leq I \right\}$$

Definition 175. For a correspondence $\gamma : X \rightarrow\rightarrow Y$, let $Gr\gamma$ denote the graph of γ , i.e.,

$$Gr\gamma = \{(x, y) \in X \times Y : y \in \gamma(x)\}.$$

Example: $\gamma_1 : \mathbb{R}_+ \rightarrow\rightarrow \mathbb{R}$ is defined by

$$\gamma_1(x) = [-\sqrt{x}, \sqrt{x}]$$

the graph of the correspondence is:

$$Gr\gamma_1 = \{(x, y) \in X \times Y : y \in [-\sqrt{x}, \sqrt{x}]\}.$$

Example: $\gamma_2 : \mathbb{R}_+ \rightarrow\rightarrow \mathbb{R}$ is defined by

$$\gamma_2(x) = \begin{cases} \left[0, \frac{1}{x}\right] & \text{if } x > 0 \\ 0 & \text{if } x = 0. \end{cases}$$

The graph of the correspondence is:

$$Gr\gamma_2 = \left\{ (x, y) \in X \times Y : \begin{array}{ll} y \in \left[0, \frac{1}{x}\right] & \text{if } x > 0, \\ y = 0 & \text{otherwise} \end{array} \right\}.$$

Definition 176. Let $\gamma : X \rightarrow Y$, and $E \subset X$. The image of E under γ is defined by

$$\gamma(E) = \bigcup_{x \in E} \gamma(x)$$

Definition 177. The upper inverse of $F \subset Y$ under γ , denoted $\gamma^+[F]$, is defined by

$$\gamma^+[F] = \{x \in X : \gamma(x) \subset F\}.$$

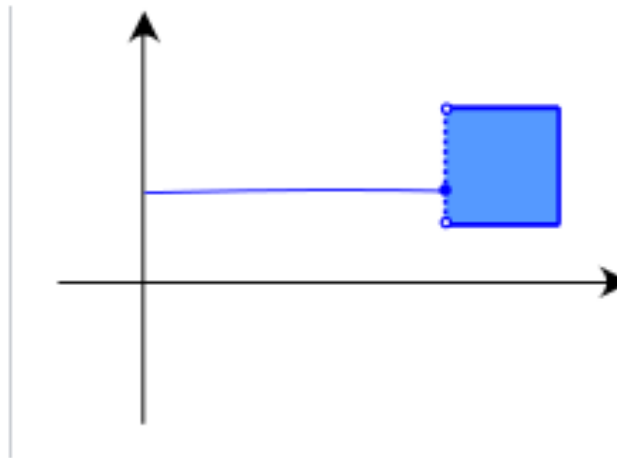
Definition 178. The lower inverse of F under γ , denoted $\gamma^-[F]$, is defined by

$$\gamma^-[F] = \{x \in X : \gamma(x) \cap F \neq \emptyset\}.$$

Definition 179. A correspondence $\gamma : X \rightarrow Y$ is said to be upper-hemicontinuous (u.h.c) at x , if for any open subset V , $V \subset Y$, with $\gamma(x) \subset V$, there exists $\epsilon > 0$ st,

$$\gamma(a) \subset V, \quad \forall a \in B(x, \epsilon).$$

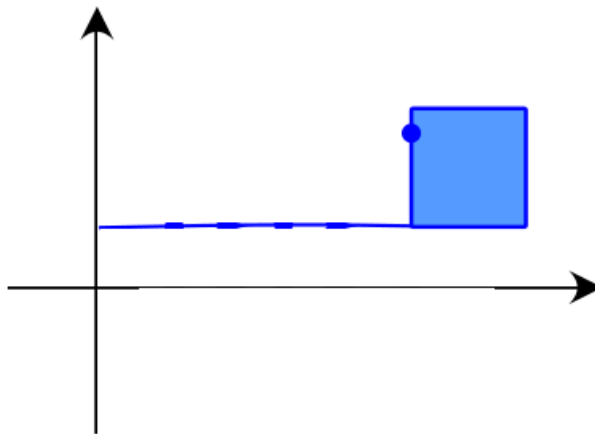
Example:



This correspondence is lower hemicontinuous everywhere, but not upper hemicontinuous at x .

Definition 180. A correspondence $\gamma : X \rightarrow Y$ is said to be lower-hemicontinuous (l.h.c) at x , if for any open subset V , $V \subset Y$, with $\gamma(x) \cap V \neq \emptyset$, there exists $\epsilon > 0$ st,

$$\gamma(a) \cap V \neq \emptyset, \quad \forall a \in B(x, \epsilon).$$



This correspondence is upper-hemicontinuous everywhere, but not lower-hemicontinuous at x .

Definition 181. A correspondence is called continuous if it is both upper and lower hemicontinuous.

Definition 182. The correspondence $\gamma : E \rightarrow\rightarrow F$ is said to be closed at x when, $x_n \rightarrow x$, $y_n \in \gamma(x_n)$ and $y_n \rightarrow y$, implies that,

$$y \in \gamma(x).$$

Definition 183. A correspondence is said to be closed if it is closed at every point of its domain.

Definition 184. The correspondence γ is said to be open if $G\gamma$ is open.

What is the relationship between u.h.c and closed? A correspondence may be closed without being upper-hemicontinuous, and viceversa.

Example: Define,

$$\gamma(x) = \begin{cases} \{1/x\} & \text{if } x \neq 0 \\ \{0\} & \text{if } x = 0. \end{cases}$$

Defined in this way, γ is closed but not u.h.c. Note that if we define γ to be such that,

$$\gamma(x) = \begin{cases} [0, 1/x] & \text{if } x \neq 0 \\ \{0\} & \text{if } x = 0. \end{cases}$$

It would not be closed.

Example: Define $\gamma : \mathbb{R} \rightarrow\rightarrow \mathbb{R}$ via $\gamma(x) = (0, 1)$. γ is u.h.c but not closed.

Proposition 43. Let $E \subset \mathbb{R}^m$, $F \subset \mathbb{R}^k$ and let $\gamma : E \rightarrow\rightarrow F$. If F is compact, and γ is closed, then γ is u.h.c.

Proof. Suppose not. Then there is some x , and some open interval $V \subset F$, with $\gamma(x) \subset V$, st $\forall \epsilon > 0$, there is a $z \in B(x, \epsilon)$ with $\gamma(z) \not\subset V$. Thus, we can find $z_n \rightarrow x$, $y_n \in \gamma(z_n)$ with $y_n \notin V$ for all n . Since F is compact, $\{y_n\}$ has a convergent sub-sequence. Denote the limit point of the subsequence y . Since V is open, V^c is closed, implying that $y \notin V$ and therefore $y \notin \gamma(x)$. Since γ is closed, we arrive to a contradiction. $\square$

Sequential Characterizations

Proposition 44. Let $E \subset \mathbb{R}^m$, $F \subset \mathbb{R}^k$ and let $\gamma : E \rightarrow \rightarrow F$ be a non-empty valued correspondence,

1. If γ is compact-valued, then γ is uhc at x , if and only if, for every $x_n \rightarrow x$ and $y_n \in \gamma(x_n)$ there is a convergent subsequence of y_n with limit in $\gamma(x)$.
2. Then γ is lhc, if and only if, $x_n \rightarrow x$ and $y \in \gamma(x)$ imply that there is a sequence $y_n \in \gamma(x_n)$ with $y_n \rightarrow y$.

Proof. We will only cover the "only if" part of the second statement. Suppose γ is not l.h.c. Then, there exists x and an open set $V \subset F$, such that $V \cap \gamma(x) \neq \emptyset$, but for any open neighborhood T of x , there exists $a \in T$, st $V \cap \gamma(a) = \emptyset$. Take any convergent sequence associated with this open neighborhoods T of x as in Proposition (43). Call this sequence $\{x_n\}$, note that by construction, $x_n \rightarrow x$. In addition, for any sequence $\{z_n\}$ such that $z_n \in \gamma(x_n)$, we know that $z_n \notin V$. Since V^c is closed, the limit point (If it exists) of $\{z_n\}$ is not in V . This also implies that the limit point is not in $\{y : V \cap \gamma(x)\}$. Given $x_n \rightarrow x$ and any $y^* \in \{y : V \cap \gamma(x)\}$, there is not way of constructing $y_n \in \gamma(x_n)$ with $y_n \rightarrow y^*$ $\square$

Try to apply this characterization for the following example:

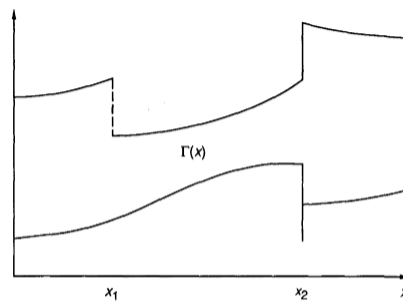


Figure 3.2

In this example we should be able to prove that x_1 is l.h.c and not u.h.c, while x_2 is u.h.c but not l.h.c.

Class 17

Optimization

First Order Conditions

In this subsection, we prove some properties that could have been proved in the Real Functions chapter, but we chose not to for exposure reasons. Let f be a function from $D \subseteq \mathbb{R}^n$. Let $(\mathbb{R}^n, d_2)$ be a metric space.

Definition 185. A point $x \in D$ is said to be a **local maximum** of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ on D if there is $\epsilon > 0$ such that $f(x) \geq f(y)$, for any $y \in D \cap B(x, \epsilon)$. We *do* care about the constraint.

Example. Let $D = [0, 1] \subseteq \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x$. Is $x = 1$ a local maximum on D ? Yes.

Now we state and prove a well-known calculus result, that the derivative of f must be zero at every local maximum or minimum. Note that this is just a **necessary** condition for a local maximum, but it is in no way also sufficient. That is, it doesn't mean that some point is an local maximum just because the derivative of f at that point is zero.

Example. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^3$. Is $x = 0$ a local maximum?

Example. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = c$ for some $c \in \mathbb{R}$. Is there any local maximum?

Theorem 20 (First Order Condition). Let $f \in \mathbb{R}^D$ be a differentiable function. Suppose that $x^* \in \text{int}(D)$ is a local maximum of f on D . Then $\nabla f(x^*) = 0$.

Proof. We start proving it for the simplest case where $D \subseteq \mathbb{R}^n$ with $n = 1$, then we generalize it to a multivariate calculus context. Let (y_n) and (z_n) be two sequences converging to x^* , with $y_n < x^*$ and $z_n > x^*$ for all $n \in \mathbb{N}$. Since f is differentiable:

$$\lim_{n \rightarrow \infty} \frac{f(y_n) - f(x^*)}{y_n - x^*} = f'(x^*)$$

Fix $\epsilon > 0$. Since both (y_n) and (z_n) converge to x^* we must have that for n large enough $y_n, z_n \in B(x^*, \epsilon)$, and since x^* is a local maximum: $f(x^*) \geq f(y_n)$ and $f(x^*) \geq f(z_n)$. We thus have the following inequalities:

$$\frac{f(y_n) - f(x^*)}{y_n - x^*} \geq 0 \geq \frac{f(z_n) - f(x^*)}{z_n - x^*}$$

Passing to the limit when $n \rightarrow \infty$ and using the Sandwich Lemma we get that $f'(x^*) = 0$. We now prove the same result for $n > 1$, that is, $\nabla f(x^*) = 0$. Recall the definition of the gradient: $\nabla f(x) = (\partial f/\partial x_1, \dots, \partial f/\partial x_n)$. Let $e_i \in \mathbb{R}^n$ be the i -th unit vector, that is, the vector with a 1 in the i -th entry and zeros in all elsewhere. Fix $i \in \{1, 2, \dots, n\}$ and define a real function g as follows:

$$g(t) := f(x^* + te_i)$$

Note that $g(0) = f(x^*)$. Moreover, for any $t_n \rightarrow 0$ we have that for each $n \in \mathbb{N}$:

$$\frac{g(t_n) - g(0)}{t_n} = \frac{f(x^* + t_n e_i) - f(x^*)}{t_n}$$

As $n \rightarrow \infty$ the RHS of the above equation converges to $\partial f(x^*)/\partial x_i$. Therefore, g is differentiable at 0 and $g'(0) = \partial f(x^*)/\partial x_i$. Note that $d(x^* + te_i, x^*) = |t|$, so for t small enough $x^* + te_i \in B(x^*, \epsilon)$. By the definition of the local maximum:

$$g(t) = f(x^* + te_i) \leq f(x^*) = g(0)$$

This means that g has a local maximum at 0. Then, by the $n = 1$ case we have that $g'(0) = \partial f(x^*)/\partial x_i = 0$. $\square$

The intuition of the above condition and its proof is simple. If the function is at its local maximum and the derivative is **not** zero moving to the left or to the right (depending on the signal of the derivative) in the domain would increase the function's value. If we are in $\mathbb{R}^n$ and **any** of the $i \in \{1, 2, \dots, n\}$ derivatives are non-zero, going in this forward (or backward) in this direction would increase the function's value.

This is a good hint on what we need to ask for a function in economic models (profit, utility, etc) in order to have uniqueness. In the next subsection, we add more structure to the functions governing the constraints on our optimization problems.

Theorem 21 (Rolle). Let $[a, b] \in \mathbb{R}$ and $f \in \mathbf{C}[a, b]$ that is differentiable on (a, b) .

$$f(a) = f(b) \implies \exists c \in (a, b), \quad f'(c) = 0$$

Proof. Since f is continuous on $[a, b]$ we can apply Weierstrass theorem to conclude that there exists $c, c' \in [a, b]$ such that:

$$f(c') \leq f(x) \leq f(c), \quad x \in [a, b]$$

If $c, c' \in (a, b)$ (interior) then by the FOC proposition above we have that $f'(c) = 0$ and $f'(c') = 0$. If $c, c' \in \{a, b\}$ and $f(a) = f(b)$ by assumption, we get that f is constant on $[a, b]$, then $f'(c) = 0$ for any $c \in (a, b)$. $\square$

Corollary 5 ("Baby" Mean Value Theorem). Let $[a, b] \in \mathbb{R}$ and $f \in \mathbf{C}[a, b]$ that is differentiable on (a, b) . There exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. Let $G : [a, b] \rightarrow \mathbb{R}$ be defined by:

$$G(t) := f(t) - \frac{f(b) - f(a)}{b - a}(t - a)$$

Observe that $G(a) = f(a)$ and $G(b) = f(a)$. By Rolle Theorem (21) we get that there exists $c \in (a, b)$ such that $G'(c) = 0$. $\square$

Theorem 22 (First Mean Value Theorem). *Let $D \subseteq \mathbb{R}^n$ be an open and convex set. Let $f : D \rightarrow \mathbb{R}$ be continuously differentiable. For any $x \neq y \in D$, there exists $z \in \text{co}\{x, y\} \setminus \{x, y\}$ such that*

$$f(x) - f(y) = \nabla f(z) \cdot (x - y)$$

Proof. Let $F : [0, 1] \rightarrow \mathbb{R}$ be defined by $F(t) := f(tx + (1 - t)y)$. The function F is differentiable on $(0, 1)$. Applying the “baby” version of the MVT for functions in $\mathbb{R}$, there exists $c \in (0, 1)$ such that $F(1) - F(0) = F'(c)$. This means that

$$f(x) - f(y) = \nabla f(cx + (1 - c)y) \cdot (x - y).$$

Observe that $z := cx + (1 - c)y$ belongs to $\text{co}\{x, y\} \setminus \{x, y\}$ $\square$

Now we use the MVT to complete the proof of (4).

Corollary 6. *Let $D \subseteq \mathbb{R}^n$ be an open and convex set and $f : D \rightarrow \mathbb{R}$ be a continuously differentiable function. Then f is concave if, and only if,*

$$\forall x, y \in D, \quad (\nabla f(y) - \nabla f(x)) \cdot (y - x) \leq 0$$

Proof. We start with the proof of ($\implies$). Let $x, y \in D$ and f a concave function. By (41) interchanging x and y we get that

$$\begin{aligned} \nabla f(y) \cdot (y - x) &\leq f(y) - f(x) \leq \nabla f(x) \cdot (y - x) \\ (\nabla f(y) - \nabla f(x)) \cdot (y - x) &\leq 0 \end{aligned}$$

We now prove the ($\impliedby$) part. Assume that

$$\forall x, z \in D, \quad \nabla f(z) \cdot (z - x) \leq \nabla f(x) \cdot (z - x)$$

Let $x, y \in D$. By the MVT there exists $\lambda \in (0, 1)$ such that

$$f(y) - f(x) = \nabla f(z) \cdot (y - x)$$

Where $z = (1 - \lambda)x + \lambda y$. Observe that $z - x = \lambda(y - x)$. This implies that

$$\nabla f(z) \cdot (y - x) = \frac{1}{\lambda} \nabla f(z) \cdot (z - x) \leq \frac{1}{\lambda} \nabla f(x) \cdot (z - x) \leq \nabla f(x) \cdot (y - x)$$

We then get that

$$f(y) - f(x) \leq \nabla f(x) \cdot (y - x)$$

Since this is true for any $x, y \in D$, by Proposition (41) we get that f is concave. $\square$

We now add some further structure to the objective function f . If f is strictly quasi-concave the maximum of f is unique.

Proposition 45. *Let $D \subseteq \mathbb{R}^n$ be a convex set, and $f \in \mathbb{R}^D$ be a strictly quasi-concave function. If f attains his maximum at some point in D , then this point is unique.*

Proof. Suppose not, that is, $x, y \in D$ are both maximum of f . Fix $\lambda \in (0, 1)$. Since D is convex it must be the case that $z := \lambda x + (1 - \lambda)y \in D$. Since f is strictly quasi-concave:

$$f(z) > \max\{f(x), f(y)\}$$

Thus, we conclude that f does not attain its maximum on x nor y , a contradiction. $\square$

Second Order Conditions

Corollary 7 (“Baby” Second Mean Value Theorem). *Let $[a, b] \in \mathbb{R}$. If $f \in R^{[a, b]}$ is continuously differentiable and twice differentiable on (a, b) , then there exists $c \in (a, b)$ such that*

$$f(b) = f(a) + f'(a)(b - a) + \frac{1}{2}f''(c)(b - a)^2$$

Proof. Define $g : [a, b] \rightarrow \mathbb{R}$ by

$$g(t) := f(b) - f(a) - f'(t)(b - t) - \frac{1}{2}M(b - t)^2$$

Where M is chosen such that $g(a) = 0$. Since $g(b) = 0$, we can apply Rolle’s Theorem (21) to get the existence of $c \in (a, b)$ such that $g'(c) = 0$:

$$g'(t) = (b - t)(M - f''(t))$$

We conclude that $M = f''(c)$. $\square$

Theorem 23 (Second Mean Value Theorem). *Let $D \subseteq \mathbb{R}^n$ and $f : D \rightarrow \mathbb{R}$ be twice continuously differentiable. For any $x \neq y \in D$, there exists $z \in \text{co}\{x, y\} \setminus \{x, y\}$ such that*

$$f(x) - f(y) = \nabla f(y) \cdot (x - y) + \frac{1}{2}(x - y)^T [\mathbf{H}_{f, z}](x - y)$$

Definition 186. *The $n \times n$ matrix H is negative semidefinite if*

$$z^T H z \leq 0$$

For all $z \in \mathbb{R}^n$.

Theorem 24. Let $D \subseteq \mathbb{R}^n$ and $f : D \rightarrow \mathbb{R}$ be a twice continuously differentiable function. f is concave if and only if $\mathbf{H}_{f,x}$ is negative semidefinite for every $x \in D$.

Proof. We argue only necessity. Suppose that f is concave. Fix $x \neq z \in D$ and with $z \neq 0$. Define the function $g : \mathbb{R} \rightarrow \mathbb{R}$ as $g(t) := f(x + tz)$. By the second MVT:

$$f(x + tz) - f(x) - \nabla f(x) \cdot (tz) = \frac{t^2}{2} z^T \mathbf{H}_{f, x+\beta z} z$$

□

Nonlinear Programming

Let $\emptyset \neq D \subseteq \mathbb{R}^n$ be an open and convex set, $f : D \rightarrow \mathbb{R}$ be a continuous function and $g = (g_1, g_2, \dots, g_l) : D \rightarrow \mathbb{R}^l$ be all differentiable functions. We refer to f as the **objective function**. The functions g_i will describe the l inequality constraints of the optimization problem and will be named **constraint functions**. This is the canonical nonlinear programming problem:

$$\begin{aligned} \max f(x) \\ \text{st} : g(x) \geq 0 \\ x \in D \end{aligned} \tag{NLP}$$

By the Weierstrass theorem, we already know that a function attains its maximum in a compact set. It follows immediately that if the constraint set $C := \{x \in D : g(x) \geq 0\}$ is compact the problem (NLP) has a solution. Moreover, if f is strictly quasi-concave and C is a convex set, this solution is unique. By Proposition (12), we know that C will be a convex set under the assumption of $(g_1, \dots, g_l)$ being quasi-concave functions.

Let $x^* \in C$ denote the solution of (NLP). At this point, some of the constraints may hold with equality, we call those constraints **binding** and some may hold with strict inequality which we call **slack**. Let $K \subset \{1, 2, \dots, l\}$ denote the set of binding constraints at x^* and for ease of notation we denote $|K| = k$. Without any loss of generality, we reorder the constraints to write:

$$\begin{aligned} g_i(x^*) &= 0, \quad i = 1, \dots, k \\ g_i(x^*) &> 0, \quad i = k+1, \dots, l \end{aligned}$$

Analogously we can write the following definition.

Definition 187. We say that $x^* \in C$ satisfies the constraint qualification if x^* achieves the maximum of the following linear programming problem

$$\begin{aligned} \max \nabla f(x^*) \cdot x \\ \text{st} : \quad \nabla g_i(x^*) \cdot (x - x^*) \geq 0, \quad \forall i \in \{1, 2, \dots, k\} \\ x \in \mathbb{R}^n \end{aligned}$$

Some references state the following condition as a definition for a point to satisfy constraint qualification:

Definition 188. Denote $g_K = (g_i)_{i \in K}$. We say that $x^* \in C$ satisfies constraint qualification if $\text{rank}(\nabla g_K(x^*)) = k$.

In the problem set, you will prove that Definition (190) actually implies Definition (189).

Theorem 25 (Kuhn-Tucker). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ be $\mathbf{C}^1$ functions for all $i \in \{1, 2, \dots, l\}$. Suppose x^* is a local maximum of f on

$$C = \{x \in \mathbb{R}^n : g_i(x) \geq 0, i = 1, \dots, l\}$$

And that x^* satisfies constraint qualification. Then, there exists a vector $\lambda \in \mathbb{R}^l$ such that the following conditions hold

$$\lambda_i \geq 0 \quad \text{and} \quad \lambda_i g_i(x^*) = 0 \quad \forall i = 1, \dots, l$$

$$\nabla f(x^*) + \sum_{i=1}^l \lambda_i \nabla g_i(x^*) = 0 \tag{KT}$$

Class 18

Nonlinear Programming

Let $\emptyset \neq D \subseteq \mathbb{R}^n$ be an open and convex set, $f : D \rightarrow \mathbb{R}$ be a continuous function and $g = (g_1, g_2, \dots, g_l) : D \rightarrow \mathbb{R}^l$ be all differentiable functions. We refer to f as the **objective function**. The functions g_i will describe the l inequality constraints of the optimization problem and will be named **constraint functions**. This is the canonical nonlinear programming problem:

$$\begin{aligned} \max f(x) & \tag{NLP} \\ \text{st : } g(x) & \geq 0 \\ x & \in D \end{aligned}$$

By the Weierstrass theorem, we already know that a function attains its maximum in a compact set. It follows immediately that if the constraint set $C := \{x \in D : g(x) \geq 0\}$ is compact the problem (NLP) has a solution. Moreover, if f is strictly quasi-concave and C is a convex set, this solution is unique. By Proposition (12), we know that C will be a convex set under the assumption of $(g_1, \dots, g_l)$ being quasi-concave functions.

Let $x^* \in C$ denote the solution of (NLP). At this point, some of the constraints may hold with equality, we call those constraints **binding** and some may hold with strict inequality which we call **slack**. Let $K \subset \{1, 2, \dots, l\}$ denote the set of binding constraints at x^* and for ease of notation we denote $|K| = k$. Without any loss of generality, we reorder the constraints to write:

$$\begin{aligned} g_i(x^*) &= 0, \quad i = 1, \dots, k \\ g_i(x^*) &> 0, \quad i = k + 1, \dots, l \end{aligned}$$

Analogously we can write the following definition.

Definition 189. We say that $x^* \in C$ satisfies the constraint qualification if x^* achieves the maximum of the following linear programming problem

$$\begin{aligned} \max \nabla f(x^*) \cdot x \\ \text{st : } \nabla g_i(x^*) \cdot (x - x^*) &\geq 0, \quad \forall i \in \{1, 2, \dots, k\} \\ x &\in \mathbb{R}^n \end{aligned}$$

Some references state the following condition as a definition for a point to satisfy constraint qualification:

Definition 190. Denote $g_K = (g_i)_{i \in K}$. We say that $x^* \in C$ satisfies constraint qualification if $\text{rank}(\nabla g_K(x^*)) = k$.

In the problem set, you will prove that Definition (190) actually implies Definition (189).

Theorem 26 (Kuhn-Tucker). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ be $\mathbf{C}^1$ functions for all $i \in \{1, 2, \dots, l\}$. Suppose x^* is a local maximum of f on

$$C = \{x \in \mathbb{R}^n : g_i(x) \geq 0, i = 1, \dots, l\}$$

And that x^* satisfies constraint qualification. Then, there exists a vector $\lambda \in \mathbb{R}^l$ such that the following conditions hold

$$\lambda_i \geq 0 \quad \text{and} \quad \lambda_i g_i(x^*) = 0 \quad \forall i = 1, \dots, l$$

$$\nabla f(x^*) + \sum_{i=1}^l \lambda_i \nabla g_i(x^*) = 0 \quad (\text{KT})$$

Proof. We rewrite the second expression in the Theorem's statement in matrix notation:

$$\nabla f(x^*) + \nabla g_i(x^*) \cdot \lambda^* = 0$$

Note that $\nabla g_i(x^*) = (\partial g_i / \partial x_1, \dots, \partial g_i / \partial x_n) \in \mathbb{R}^n$ and $\nabla f(x^*) = (\partial f / \partial x_1, \dots, \partial f / \partial x_n) \in \mathbb{R}^n$, the **Jacobian** is thus a $k \times n$ matrix such that:

$$\nabla g = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \dots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial g_k}{\partial x_1} & \dots & \dots & \frac{\partial g_k}{\partial x_n} \end{bmatrix}_{k \times n}$$

Note that since x^* satisfies constraint qualification:

$$\nabla f(x^*)(x^* - x) \geq 0 \quad \text{and} \quad \nabla g_i(x^*)(x - x^*) \geq 0$$

We can rewrite the above inequalities such as:

$$\underbrace{\nabla f(x^*)}_{v:=} \cdot \underbrace{(x - x^*)}_{\alpha:=} \leq 0 \quad \text{and} \quad \underbrace{-\nabla g_i(x^*)}_{A_i:=} \cdot (x - x^*) \leq 0$$

Therefore we have $v \cdot \alpha \leq 0$ and $A\alpha \leq 0$. By Farka's Lemma, it thus must be the case that $\exists \mu \in \mathbb{R}^k$ such that:

$$\nabla f(x^*) + \sum_{i=1}^k \mu_i \nabla g_i(x^*) = 0$$

Now define $\lambda \in \mathbb{R}^l$ such that:

$$\lambda_i := \begin{cases} \mu_i, & i = 1, \dots, k \\ 0, & i = k+1, \dots, l \end{cases}$$

We now show that the vector λ satisfies the properties stated in the theorem. First, note that since $i \in \{1, 2, \dots, k\}$ are the binding constraints we must have that $g_i(x^*) = 0$. Second, by definition of λ_i for $i \in \{k+1, \dots, l\}$ we have that $\lambda_i = 0$, then $\lambda_i g_i(x^*) = 0$. It remains to be shown that $\lambda_i \geq 0$ for any $i \in \{1, 2, \dots, k\}$. Let $z \in \mathbb{R}^n$ and $\eta \in \mathbb{R}$. Define the function $G = (G_1, \dots, G_k)$ from $\mathbb{R}^{n+1}$ to $\mathbb{R}^k$ by

$$G_1(y, \eta) = g_1(y) - \eta$$

$$G_i(y, \eta) = g_i(y), \quad \forall i = 1, \dots, k$$

Let $\nabla G_y(y, \eta)$ denote the $k \times n$ derivative matrix of G with respect to the y variables alone, and $\nabla G_\eta(y, \eta)$ denote the $k \times 1$ derivative vector of G with respect to η . Note that since we defined $g_K = (g_1, \dots, g_k)$ we have that $\nabla G_y(y, \eta) = \nabla g_K(y)$. We also have $\nabla G_\eta(y, \eta) = (-1, 0, \dots, 0)$ at any (y, η) . By the definition of G , we have that $G(x^*, 0) = 0$. Moreover, $\text{rank}(\nabla G(x^*, 0)) = \text{rank}(\nabla g_K(x^*)) = k$. Therefore, by the Implicit Function Theorem, there is a neighborhood N of zero, and a C^1 function $\xi : N \rightarrow \mathbb{R}^n$ such that $\xi(0) = x^*$, and

$$G(\xi(\eta), \eta) = 0$$

Differentiating this expression and using the fact that $\xi(0) = x^*$:

$$\nabla G_y(x^*, 0) \nabla \xi(0) + \nabla G_\eta(x^*, 0) = 0$$

Since $\nabla G_y(y, 0) = \nabla g_K(y)$ and $\nabla G_\eta(y, \eta) = (-1, 0, \dots, 0)$ the above expression turns into:

$$\nabla g_K(x^*) \nabla \xi(0) = (1, 0, \dots, 0)$$

Or equivalently:

$$\nabla g_1(x^*) \nabla \xi(0) = 1$$

$$\nabla g_i(x^*) \nabla \xi(0) = 0, \quad i = 2, \dots, k$$

Since $\lambda_i = 0$ for $i = k+1, \dots, l$ we now have

$$\begin{aligned} \nabla f(x^*) \nabla \xi(0) &= - \left(\sum_{i=1}^l \lambda_i \nabla g_i(x^*) \right) \nabla \xi(0) \\ &= - \left(\sum_{i=1}^k \lambda_i \nabla g_i(x^*) \nabla \xi(0) \right) \\ &= -\lambda_1 \end{aligned}$$

To complete the proof we will show that

$$\nabla f(x^*) \nabla \xi(0) \leq 0$$

To this end we first show that there exists $\eta^* > 0$ such that $\forall \eta \in [0, \eta^*)$ we must have $\xi(\eta) \in C$, where C is the constraint set of (NLP). If $\eta > 0$ then $G_i(\xi(\eta), \eta) = 0$ for $i = 1, \dots, k$ and from the definition of G_i this means:

$$g_1(\xi(\eta)) = \eta > 0$$

$$g_i(\xi(\eta)) = 0, \quad \forall i = 2, \dots, k$$

For $i = k+1, \dots, l$ we have that $g_i(\xi(0)) = g_i(x^*) > 0$. Since both $g_i(\cdot)$ and $\xi(\cdot)$ are continuous, it follows that by choosing η sufficiently small (say $\eta \in (0, \eta^*)$), we can also ensure that

$$g_i(\xi(\eta)) > 0 \quad \forall i = k+1, \dots, l$$

Now since $\xi(0) = x^*$ is a local maximum of f on D , and $\xi(\eta)$ is feasible for $\eta \in [0, \eta^*)$ it follows that for η sufficiently close to zero we must have

$$f(x^*) \geq f(\xi(\eta))$$

Therefore:

$$\frac{f(\xi(0)) - f(\xi(\eta))}{\eta} \leq 0$$

For η sufficiently small. Indeed, taking $\eta \rightarrow 0$ we get that

$$\nabla f(\xi(0)) \nabla \xi(0) \leq 0$$

Or similarly $\nabla f(x^*) \nabla \xi(0) \leq 0$ and the proof is complete. $\square$

This is a very important result. For any local maximum x^* that satisfies constraint qualification, there exists a vector of Lagrange multipliers λ such that (λ, x^*) satisfies the (KT) conditions. However, until now it is possible that there are some pairs (λ, x^*) that satisfy the (KT) conditions but for which x^* is **not** a local maximum.

We will derive another version of it when the constraint functions g_i are also concave. Recall that the constraint set generated by concave functions $g_i(x) \geq 0$ has an important property: it is convex! The following result states that the first-order conditions of the (KT) are both *necessary* and *sufficient* to identify optima of convex inequality-constrained optimization problems.

Theorem 27. Suppose that f and g_i are all concave and $\mathbf{C}^1$ functions. If $(\lambda, x^*) \in C \times \mathbb{R}_+^l$ solves (KT), then x^* is a solution to (NLP).

Proof. Suppose that x^* is **not** a solution to (NLP), that is, there exist $\tilde{x} \in C$ such that $f(\tilde{x}) > f(x^*)$. Since f is concave:

$$\nabla f(x^*)(\tilde{x} - x^*) \geq f(\tilde{x}) - f(x^*) > 0$$

We know that $g_i(x^*) = 0$ by construction and $g_i(\tilde{x}) \geq 0$ for all $i = 1, \dots, k$ since $\tilde{x} \in C$. Concavity of the constraint functions implies that:

$$\nabla g_i(x^*) \cdot (\tilde{x} - x^*) \geq 0$$

Let $\lambda \in \mathbb{R}_+^l$ be such that $\lambda_i g_i(x^*) = 0$ for any $i = 1, \dots, l$. Given that $\lambda_i = 0$ for $i > k$ and $\lambda_i \geq 0$ for $i = 1, \dots, k$:

$$\sum_{i=1}^l \lambda_i \nabla g_i(x^*) \cdot (\tilde{x} - x^*) \geq 0$$

Hence

$$\left(\nabla f(x^*) + \sum_{i=1}^l \lambda_i \nabla g_i(x^*) \right) \cdot (\tilde{x} - x^*) > 0$$

Since $\tilde{x} - x^* \neq 0$, we conclude that

$$\nabla f(x^*) + \sum_{i=1}^l \lambda_i \nabla g_i(x^*) \neq 0$$

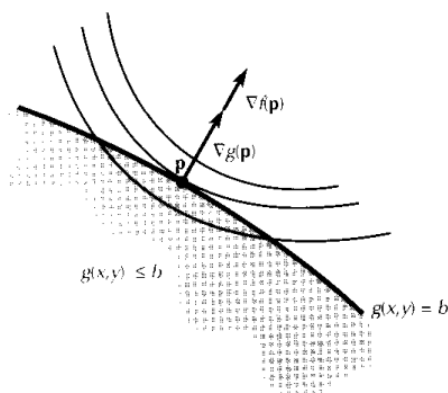
Thus (λ, x^*) is not a solution to (KT) since it does not satisfy one of the requirements of the theorem. Taking the counter-positive we get the result. $\square$

Geometric Interpretation

In this subsection, we use level curves to provide a geometrical intuition for the Lagrange multipliers, and the constrained optimization problem itself. For exposition, we let $n = 2$ so we can draw level curves, and solve the following problem:

$$\begin{array}{ll} \text{maximize} & f(x, y) \\ \text{subject to} & g(x, y) \leq b. \end{array}$$

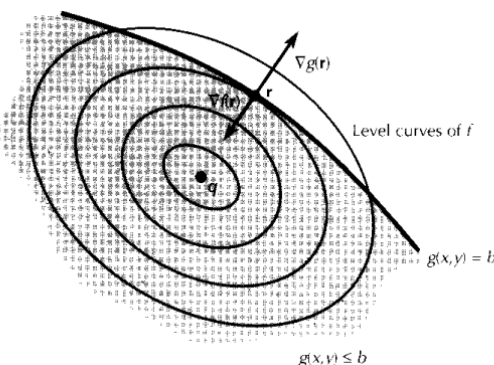
In the next figure, the thicker curve is the curve $g(x, y) = b$; the region to the left and below this curve is the constraint set $g(x, y) \leq b$. The thinner lines are the level sets of the objective function f .



Note that the highest level curve of f which meets the constraint set meets it at the point $\mathbf{p}$. Since $\mathbf{p}$ lies on the boundary of the constraint set where $g(x, y) = b$, we say that the constraint is binding at $\mathbf{p}$. Since the level curves are tangent at $\mathbf{p}$, their gradients ∇f and ∇g point in the same direction (the direction that would **increase** both functions, f and g), so ∇f is a multiple of ∇g , or:

$$\nabla f(\mathbf{p}) - \lambda \cdot \nabla g(\mathbf{p}) = 0$$

We now turn to the analysis of the other case. Suppose that the maximum of f on the constraint set $g(x, y) \leq b$ occurs not where $g(x, y) = b$ but at a point where $g(x, y) < b$.



The full constraint set $g(x, y) \leq b$ lies to the left and below this curve. However, this time the maximum of f occurs at the point $\mathbf{q}$ in the interior of the constraint set. There is a point $\mathbf{r}$ on the level set $g(x, y) = b$ where this level set is tangent to a level set of f , but ∇f and ∇g point in opposite directions at $\mathbf{r}$. In fact, one can increase the value of f by moving further into the constraint set from $\mathbf{r}$ until one reaches $\mathbf{q}$. Since $g(x, y)$ is strictly less than b at $\mathbf{q}$, in other words, $\mathbf{q}$ is in the interior of the constraint set, we say that the constraint is not binding at $\mathbf{q}$, and $\lambda = 0$. Note that the point $\mathbf{q}$ must be a local max of f , that is, a local unconstrained max.

Therefore, we will summarize our criterion that either $g(x, y) - b = 0$ or $\lambda = 0$ by requiring that

$$\lambda \cdot [g(x, y) - b] = 0.$$

Lagrange Multipliers

Theorem 28. Let $\mathbf{a} = (a_1, a_2, \dots, a_l) \in \mathbb{R}^l$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

$$\max f(\mathbf{x}) \tag{3}$$

$$\begin{aligned} \text{st: } & g_1(\mathbf{x}) \leq a_1 \\ & \vdots \\ & g_l(\mathbf{x}) \leq a_l \end{aligned}$$

Let $\mathbf{x}^*(\mathbf{a}) = (x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a}))$ denote the solution of problem and let $\lambda = (\lambda_1(\mathbf{a}), \dots, \lambda_l(\mathbf{a}))$ be the corresponding Lagrange multipliers. Suppose x_i and λ_j are differentiable functions of $(a_1, \dots, a_k)$ for any $i = 1, \dots, n$ and $j = 1, \dots, l$, and that the $\mathbf{x}^*$ satisfies constraint qualification. Then, for each $j = 1, \dots, l$,

$$\lambda_j(\mathbf{a}) = \frac{\partial}{\partial a_j} f(x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a}))$$

The above theorem states that the multiplier measures how the objective function evaluated at the solution varies with the level of the constraints. Imagine f is the profit of a firm, $\mathbf{x}$ are the n inputs used in production, and $\mathbf{a}$ is the availability of each input in the world. In this situation, the multiplier represents the change in the optimal profit resulting from the availability of one more unit of input j .

Alternatively, λ_j tells the maximum amount the firm would be willing to pay to acquire an additional unit of this input. For this reason, λ_j is often called **shadow price** of input j .

Parametric Optimization

Maximum Theorem In this subsection, we will deal with optimization problems in which the constraint set is indexed by a parameter (θ) in a determined parameter space ($\Theta \subseteq \mathbb{R}^l$). The problem will thus be:

$$\begin{aligned} \max f(x) \\ \text{st } x \in D(\theta) \end{aligned}$$

The solution to this problem will be a pair (f^*, D^*) defined by the following sets:

$$f^*(\theta) = \max\{f(x, \theta) : x \in D(\theta)\}$$

$$D^*(\theta) = \arg \max\{f(x, \theta) : x \in D(\theta)\}$$

We denote by $f^*(\theta)$ the *maximized value function* and $D^*(\theta)$ the *set of maximizers*. The main result of this subsection is The Maximum Theorem, which says basically that if primitives of a parametric optimization problem possess a sufficient degree of continuity, then the solutions will also be continuous albeit not to the same degree.

Theorem 29 (The Maximum Theorem). *Let $S \subseteq \mathbb{R}^n$ and $f : S \times \Theta \rightarrow \mathbb{R}$ be a continuous function, and $D : \Theta \rightarrow P(S)$ be a compact-valued, continuous correspondence. Let $f^* : \Theta \rightarrow \mathbb{R}$ and $D^* \rightarrow P(S)$ be defined by:*

$$f^*(\theta) = \max\{f(x, \theta) : x \in D(\theta)\}$$

$$D^*(\theta) = \{x \in D(\theta) : f(x, \theta) = f^*(\theta)\}$$

Then

1. f^* is continuous on Θ , and D^* is a USC correspondence.
2. If $f(\cdot, \theta)$ is concave in x for each θ , and $D(\theta)$ is convex for each θ . Then, $D^*(\theta)$ is also convex for each θ .⁹
3. If f is concave on $S \times \Theta$, and D has a convex graph, then f^* is a concave function, and D^* is a convex-valued USC correspondence¹⁰

⁹ When “concave” is substituted by “strictly concave”, the $D^*(\theta)$ is a single-valued USC correspondence, that is, a continuous function.

¹⁰ If “concave” is substituted for “strictly concave”, then f^* is also strictly concave and D^* is single-valued everywhere, therefore, it is a continuous function.

Proof

1. Pick $x_m \in D^*(\theta_m)$. By **Proposition 9.8**, there exists a strictly increasing function $\varphi \in \mathbb{N}^{\mathbb{N}}$ such that $x_{\varphi(m)} \rightarrow x \in D(\theta)$. We want to show that $f^*(\theta) = f(x, \theta)$.

Since f is continuous on $\Theta \times S$ it is the case that $f(x_{\varphi(m)}, \theta_{\varphi(m)}) \rightarrow f(x, \theta)$. Suppose there were $x^* \in D(\theta)$ such that $f(x^*, \theta) > f(x, \theta)$. Since D is LSC at θ , $x^* \in D(\theta)$ and $x_{\varphi(m)} \rightarrow x$ there is, by **Proposition 9.9**, a strictly increasing $\psi \in \mathbb{N}^{\mathbb{N}}$ such that $x_{\psi(\varphi(m))}^* \in D(\theta_{\psi(\varphi(m))})$ such that $x_{\psi(\varphi(m))}^* \rightarrow x^*$. It follows by continuity of f that:

$$\lim_{m \rightarrow \infty} f(x_{\psi(\varphi(m))}^*, \theta_{\psi(\varphi(m))}) = f(x^*, \theta) > f(x, \theta) = \lim_{m \rightarrow \infty} f(x_{\varphi(m)}, \theta_{\varphi(m)})$$

But this means that for m large enough we must have that $f(x_{\psi(\varphi(m))}^*, \theta_{\psi(\varphi(m))}) > f(x_{\varphi(m)}, \theta_{\varphi(m)})$ and therefore $x_{\psi(\varphi(m))}^* \notin D^*(\theta_{\psi(\varphi(m))})$, which is a contradiction.

2. Let $x, \bar{x} \in D^*(\theta)$ and $\lambda \in (0, 1)$. Define $x' := \lambda x + (1 - \lambda)\bar{x}$. Since $D(\theta)$ is convex for any θ we must have that $x' \in D(\theta)$. Then,

$$\begin{aligned} f(x', \theta) &\geq \lambda f(x, \theta) + (1 - \lambda)f(\bar{x}, \theta) \\ &= \lambda f^*(\theta) + (1 - \lambda)f^*(\theta) \\ &= f^*(\theta) \end{aligned}$$

We thus conclude that $x' \in D^*(\theta)$.

3. Let $\theta, \bar{\theta} \in \Theta$ and $\lambda \in (0, 1)$. Define $\theta' := \lambda\theta + (1 - \lambda)\bar{\theta}$. Pick any $x \in D^*(\theta)$ and $\bar{x} \in D(\bar{\theta})$. Let $x' := \lambda x + (1 - \lambda)\bar{x}$. Then, since $x \in D(\theta)$ and $\bar{x} \in D(\bar{\theta})$ and $D(\cdot)$ has a convex graph, we must have that $x' \in D(\theta')$. Since x' is feasible, but not necessarily optimal at θ' , we have by concavity of f :

$$\begin{aligned} f^*(\theta') &\geq f(x', \theta') \\ &= f(\lambda x + (1 - \lambda)\bar{x}, \lambda\theta + (1 - \lambda)\bar{\theta}) \\ &\geq \lambda f(x, \theta) + (1 - \lambda)f(\bar{x}, \bar{\theta}) \\ &= \lambda f^*(\theta) + (1 - \lambda)f^*(\bar{\theta}) \end{aligned}$$

This establishes the concavity of f^* . ■

Economic Application - Consumer Theory

Suppose a consumer has n goods available to be bought. A bundle is thus $x = (x_1, \dots, x_n) \in \mathbb{R}_+^n$. This consumer has some income $I \geq 0$ and faces strictly positive prices $p = (p_1, \dots, p_n) \in \mathbb{R}_{++}^n$. Define the **budget set** for a given income and vector of prices as follows:

$$\mathcal{B}(p, I) := \{x \in \mathbb{R}_+^n : p \cdot x \leq I\}$$

Claim. $\mathcal{B}(p, I)$ is continuous, compact, and convex valued.

Compactness will follow from the fact that $p \in \mathbb{R}_{++}^n$ as we saw in one of the previous chapters. Continuity and convexity will follow from the linear function defining the correspondence.

Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous and strictly concave function that we call *utility function*. Using $v(\cdot)$ to denote the maximized function, or in an economic language the *indirect utility* and $x(\cdot)$ to denote the set of maximizers (or *demand function*). You will see the following theorem at some point this year:

Proposition 46. *The indirect utility function $v(\cdot)$ and the demand correspondence $x(\cdot)$ have the following properties*

1. $v(\cdot)$ is a continuous function on $\mathbb{R} \times \mathbb{R}_{++}^n$, and $x(\cdot)$ is a compact-valued correspondence on $\mathbb{R} \times \mathbb{R}_{++}^n$
2. v is nondecreasing in I for a fixed p , and nonincreasing in p for a fixed I .
3. If u is concave, then v is quasi-convex in p for a fixed I , and concave in I for a fixed p ; further, x is a convex-valued correspondence.
4. If u is strictly concave, then x is a continuous function.

Class 19

Problem Set 4

Exercise 1. Suppose $f \in \mathbb{R}^D$ is differentiable and concave. Then, $x^* \in \text{int}(D)$ is a local maximum of f on D if, and only if, $\nabla f(x^*) = 0$

Proof. One of the directions was proved before. We focus here on the additional assumption that f is concave. Using one of our characterizations for concave functions

$$\forall y \in D, \quad f(y) - f(x^*) \leq \underbrace{\nabla f(x^*)(y - x^*)}_{=0}$$

$$\forall y \in D, \quad f(y) \leq f(x^*)$$

□

Exercise 2. Let $D \subseteq \mathbb{R}^n$ be an open, convex set, and $f : D \rightarrow \mathbb{R}$. Prove that f is quasiconcave if, and only if

$$\nabla f(x) \cdot (y - x) \geq 0, \quad \forall x, y \in D \quad \text{such that:} \quad f(y) \geq f(x)$$

Proof. We start by ($\implies$). Suppose f is quasi-concave and let $x, y \in D$ be such that $f(y) \geq f(x)$. Then, for all $\lambda \in [0, 1]$

$$f(x + \lambda(y - x)) = f((1 - \lambda)x + \lambda y) \geq \min\{f(x), f(y)\} = f(x).$$

Hence,

$$\frac{f(x + \lambda(y - x)) - f(x)}{\lambda} \geq 0.$$

Passing to the limit when $\lambda \rightarrow 0$:

$$\lim_{\lambda \rightarrow 0} \frac{f(x + \lambda(y - x)) - f(x)}{\lambda} = \nabla f(x) \cdot (y - x) \geq 0$$

To prove ($\impliedby$) assume that:

$$\nabla f(x) \cdot (y - x) \geq 0, \quad \forall x, y \in D \quad \text{such that:} \quad f(y) \geq f(x).$$

For each $\lambda \in [0, 1]$, let $z_\lambda = (1 - \lambda)x + \lambda y$ and define $g : [0, 1] \rightarrow \mathbb{R}$ to be such that $g(\lambda) = f(z_\lambda)$. Notice that $g(0) = f(x) \leq f(y) = g(1)$. By the chain rule:

$$g'(\lambda) = \nabla f(z_\lambda) \cdot (y - x), \quad \forall \lambda \in (0, 1)$$

We claim that if $\lambda \in (0, 1)$ is such that $f(z_\lambda) \leq f(x)$, then $g'(\lambda) = 0$. Indeed, $f(z_\lambda) \leq f(x) \leq f(y)$ implies that:

$$\nabla f(z_\lambda) \cdot (x - z_\lambda) = -\lambda \nabla f(z_\lambda) \cdot (y - x) \geq 0$$

and

$$\nabla f(z_\lambda) \cdot (y - z_\lambda) = (1 - \lambda) \nabla f(z_\lambda) \cdot (y - x) \geq 0$$

Hence, $\nabla f(z_\lambda) \cdot (y - x) = 0$ since $\lambda \in (0, 1)$ which is the desired result. We now claim that $g(\lambda) \geq g(0)$ for all $\lambda \in (0, 1)$ so that f is quasi-concave. Suppose not, that there exists $\alpha \in (0, 1)$ such that $g(\alpha) < g(0)$. Given that g is continuous there must exist $\theta \in (0, \alpha)$ such that $g(\alpha) < g(\theta) < g(0)$ and $g(\lambda) < g(0)$ for all $\lambda \in (\theta, \alpha)$. Hence, by MVT, there exists $\lambda^* \in (\theta, \alpha)$ with $g(\alpha) - g(\theta) = g'(\lambda^*)(\alpha - \theta)$, so $g'(\lambda^*) < 0$, a contradiction. $\square$

Exercise 3. Prove the following Proposition.

Proposition 47. Let S be an open and convex subset of $\mathbb{R}^n$ and $f \in \mathbb{R}^S$ be twice continuously differentiable. Fix any $x \neq y \in S$ and define $F \in \mathbb{R}^{(0,1)}$ by:

$$\forall t \in (0, 1), \quad F(t) := f(\underbrace{tx + (1-t)y}_{z:=})$$

The function F is twice differentiable on $(0, 1)$ and

$$\forall t \in (0, 1), \quad F''(t) = (x - y)^T [H_{f,z}](x - y)$$

Proof. Let $\theta : [0, 1] \rightarrow \text{co}\{x, y\}$ be defined by $\theta(t) := tx + (1 - t)y$. We have already proved that

$$\forall t \in (0, 1), \quad F'(t) = L_{f, \theta(t)}(x - y)$$

We then deduce that

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [F'(t + \epsilon) - F'(t)] = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [L_{f, \theta(t+\epsilon)}(x - y) - L_{f, \theta(t)}(x - y)]$$

This is the directional derivative of L_f at $\theta(t)$ in the direction $x - y$, that is

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [F'(t + \epsilon) - F'(t)] = [L_{f, \theta(t)}(x - y)](x - y) = (x - y)^T [\mathbf{H}_{f, \theta(t)}](x - y)$$

$\square$

Exercise 4. Prove the Second Mean Value Theorem

Proof. Let $F : [0, 1] \rightarrow \mathbb{R}$ be defined by $F(t) := f(tx + (1 - t)y)$. Note that F is twice continuously differentiable on $(0, 1)$. Applying the “Baby” Second MVT, there exists $c \in (0, 1)$ such that

$$F(1) = F(0) + F'(0) + \frac{1}{2}F''(c)$$

But this is the same as

$$f(x) - f(y) = \nabla f(y) \cdot (x - y) + \frac{1}{2}(x - y)^T [\mathbf{H}_{f,z}](x - y)$$

Where $z := cx + (1 - c)y$ belongs to $co\{x, y\} \setminus \{x, y\}$ □

Exercise 5. Let x^* be a solution for the canonical NLP problem. Prove that if the full rank condition holds, then x^* satisfies constraint qualification.

Proof. Suppose that x^* does not solve the linear programming problem that defines constraint qualification, that is, there exists $\tilde{x} \in \mathbb{R}^n$ such that

$$\nabla g_j(x^*) \cdot (\tilde{x} - x^*) \geq 0 \quad j = 1, \dots, k$$

$$\nabla f(x^*) \cdot (\tilde{x} - x^*) > 0$$

Since $\nabla g(x^*)$ has full rank, the matrix equation $\nabla g(x^*)h = \mathbf{1}$, where $\mathbf{1}$ is a vector of ones, has a solution h . Let $x(t) = \tilde{x} + th$, with $t \in [0, 1]$. If $t > 0$:

$$\nabla g_j(x^*) \cdot (x(t) - x^*) = \underbrace{(1 - t)\nabla g_j(x^*) \cdot (\tilde{x} - x^*)}_{\geq 0} + \underbrace{t\nabla g_j(x^*) \cdot h}_{> 0} > 0$$

for all $j = 1, \dots, k$. Now note that $t \in (0, 1]$ implies

$$\nabla f(x^*) \cdot (x(t) - x^*) = \underbrace{(1 - t)\nabla f(x^*) \cdot (\tilde{x} - x^*)}_{> 0} + t\nabla f(x^*) \cdot h,$$

and so $\nabla f(x^*) \cdot (x(t) - x^*) > 0$ if t is sufficiently close to zero. Let t_1 be one such value of t . In this case, by the Chain rule,

$$\left. \frac{dg_j}{d\alpha}(x^* + \alpha(x(t_1) - x^*)) \right|_{\alpha=0} = \nabla g_j(x^*) \cdot (x(t_1) - x^*) > 0$$

for $j = 1, \dots, k$ and

$$\left. \frac{df}{d\alpha}(x^* + \alpha(x(t_1) - x^*)) \right|_{\alpha=0} = \nabla f(x^*) \cdot (x(t_1) - x^*) > 0$$

Thus, $g(x^* + \alpha(x(t_1) - x^*)) > 0$ and $f(x^* + \alpha(x(t_1) - x^*)) > f(x^*)$ if α is sufficiently small. To finish, observe that X is open, $g_j(x^*) > 0$ for $j > k$, and g is continuous. Reducing α even further if necessary, we then have that $x^* + \alpha(x(t_1) - x^*) \in X$ and $g(x^* + \alpha(x(t_1) - x^*)) > 0$, but $f(x^* + \alpha(x(t_1) - x^*)) > f(x^*)$. Hence, x^* is not a solution to (NLP). □

Exercise 6.

1. Let f and g be both $C^2(\mathbb{R}^2)$ defined by $f(x, y) = -(x^2 + y^2)$, and $g(x, y) = (x - 1)^3 - y^2$. Is it possible to find:

$$\begin{aligned} &\max f(x, y) \\ \text{st: } &g(x, y) \geq 0 \end{aligned}$$

using the Kuhn-Tucker characterization? Explain.

2. Let $f(x) = 2x^3 - 3x^2$ and $g(x) = (3 - x)^3$. Consider the problem:

$$\begin{aligned} &\max f(x) \\ \text{st: } &g(x) \geq 0 \end{aligned}$$

Is it possible to find the maximum using KT characterization? Explain.

3. Let $f(x) = x^2 - x$ and $g(x) = x$. Consider the problem:

$$\begin{aligned} &\max f(x) \\ \text{st: } &g(x) \geq 0 \end{aligned}$$

Is it possible to find the maximum using KT characterization? Explain.

4. Let $f(x, y) = x + y$ and $g(x, y) = xy - 1$. Consider the problem:

$$\begin{aligned} &\max f(x) \\ \text{st: } &g(x) \geq 0 \end{aligned}$$

Is it possible to find the maximum using KT characterization? Explain.

Proof. See Sundaram's book (Chapter 6). More specifically examples 6.7-6.9

□

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